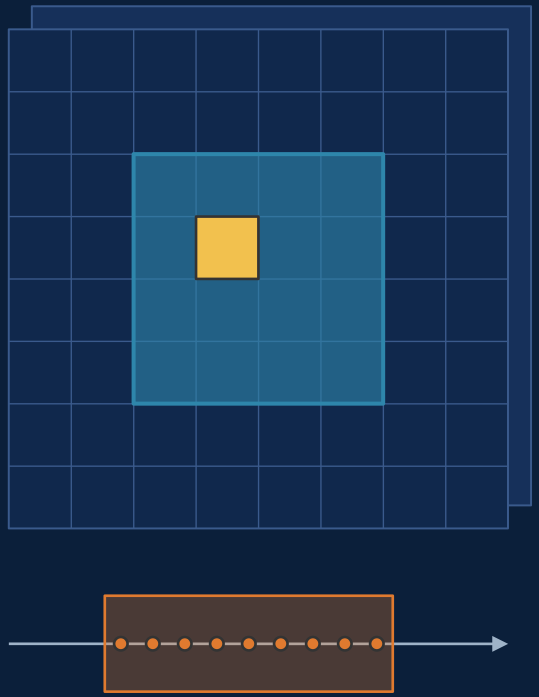
Geospatial representation models

A stencil-level overview of AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 and IBM's Prithvi-based ocean model (Granite-Geospatial-Ocean)

What area in space and what span of time goes into each embedding — and what each model actually hands you



Earth 2 representation-model research · 31 Aug – 2 Sep 2026

Geospatial representation models

This deck compares six "representation models" for Earth observation, and then uses them to sharpen the design of our own model for the ocean.

A representation model (also called an embedding model or foundation model) is a neural network that is trained on huge amounts of raw data without labels, and whose job is to turn each place on Earth into a short list of numbers — a vector — that summarises what is going on there. That vector is called an embedding. Once you have good embeddings, many downstream tasks (mapping crops, detecting floods, estimating an ocean current) become much easier because a small model can be trained on top of the embeddings with very few labels.

The word "stencil" runs through the whole deck. It is our term for the piece of the world that feeds one embedding: how large an area around a pixel the model looks at, and how long a stretch of time it summarises. The little drawing on the right shows the idea — a grid of pixels with one output pixel (yellow) and its spatial context (blue), and a timeline with the observations that go into it (orange).

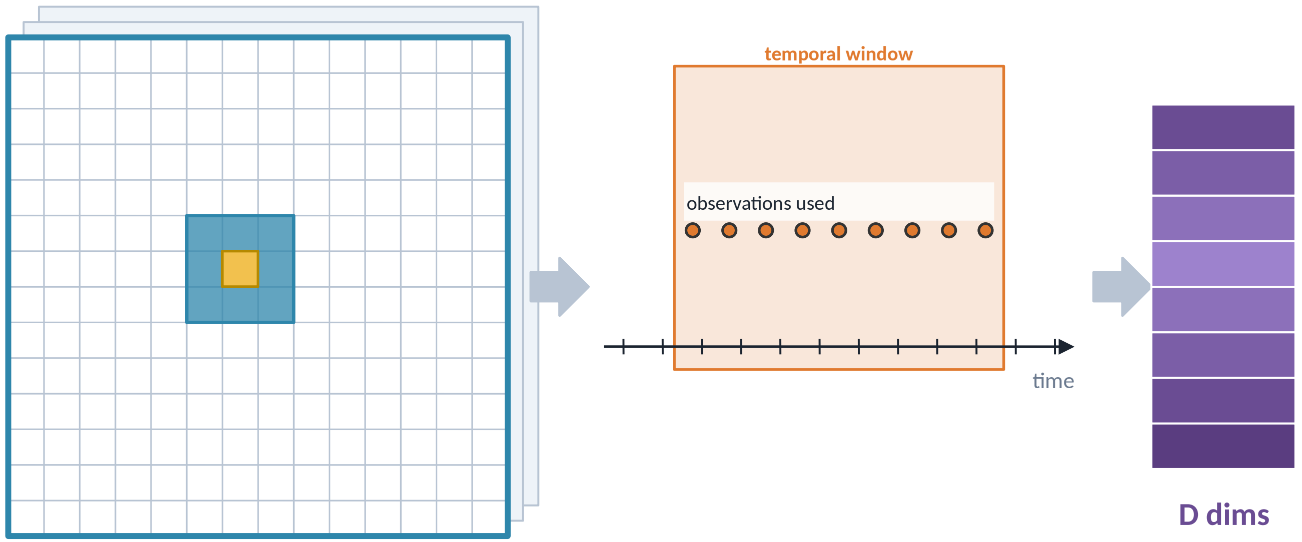
The six models are: AlphaEarth (Google DeepMind), TESSERA (University of Cambridge), OlmoEarth (Allen Institute for AI), TerraMind (IBM and ESA), Prithvi-EO 2.0 (IBM and NASA), and IBM's ocean model, Granite-Geospatial-Ocean. The last third of the deck applies what we learn from them to "Earth 2", our own ocean project.

How to read the stencil diagrams

Every model slide uses the same three-panel convention

Stencil — what feeds one embedding

SPACE | TIME | OUTPUT



Grid = input tile the encoder sees at once (blue outline). Blue square = one token / receptive field. Yellow = the output pixel.
Stacked sheets = co-registered modalities / channels.

Shaded band = window summarised by one embedding.
Dots = observations sampled inside it. A single tall bar = one snapshot acquisition.

one vector per pixel, per token, or per tile

Four questions we ask of each model

- Per-pixel or patch? Does the embedding of a pixel see its neighbours (patch/ViT) or only its own time series (pixel-wise)?
- Snapshot or period summary? One acquisition, a handful of dates, or a whole year folded into one vector?
- Product or model? Do we get precomputed global embeddings, open weights we can run on our own grid, or both?
- Does it touch the ocean? Land-only masking is the norm; only one of the six is trained on sea pixels.

How to read the stencil diagrams

Every model slide uses the same three-panel picture, so this slide is the legend.

SPACE panel (left). The grid is the tile — the square block of pixels the model processes at once. The blue outline marks the tile edge. The small blue square is one "token": a token is the unit the neural network actually reasons about, usually a small patch of pixels (for example 16 by 16) squashed into one vector. The yellow square is the single output pixel we are asking about. The stacked sheets behind the grid stand for the different data sources — for example optical imagery, radar and elevation — that are laid on top of each other ("co-registered") so that every pixel has a value from each source.

TIME panel (middle). The horizontal axis is time. The shaded band is the "temporal window": the span of time that one embedding summarises. The dots are the individual satellite observations inside that window. If instead there is a single tall bar, the model looks at one moment only — a snapshot.

OUTPUT panel (right). The purple stack represents the embedding vector. "D" is its dimension, the number of entries in the vector — 64, 128, 768 and so on. The caption says whether you get one vector per pixel, per token, or per whole tile.

The four questions on the right are the ones we ask of every model. "Per-pixel or patch" asks whether the model uses neighbouring pixels at all. "Snapshot or period summary" asks whether time is folded into the vector. "Product or model" asks whether the makers hand you pre-computed embeddings, the trained network ("weights"), or both. And "does it touch the ocean" is our own concern: almost all of these models deliberately throw away sea pixels.

AlphaEarth Foundations (Google DeepMind)

An 'embedding field': every 10 m land pixel → 64-D unit vector per year, 2017–2024

Who / when DeepMind + Google Earth Engine; arXiv 2507.22291 (Jul 2025, v2 Sep 2025).

Inputs Sentinel-2, Landsat 8/9, Sentinel-1, PALSAR-2, GEDI lidar, ERA5-Land, GRACE, GLO-30 DEM, NLCD, plus text (Wikipedia, GBIF) — >3 B observations.

Architecture Space-Time-Precision (STP) encoder — 15 blocks, each with three operators at three resolutions (spatial attention @ L/16, time-axial attention @ L/8, 3×3 convs @ L/2; widths 1024 / 512 / 128) exchanging state through a learned Laplacian pyramid (next slide). Teacher/student video embedders + text alignment; ~480 M deployed (1 B trained). Losses: reconstruction, batch-uniformity on S^{63}, consistency, text-contrastive.

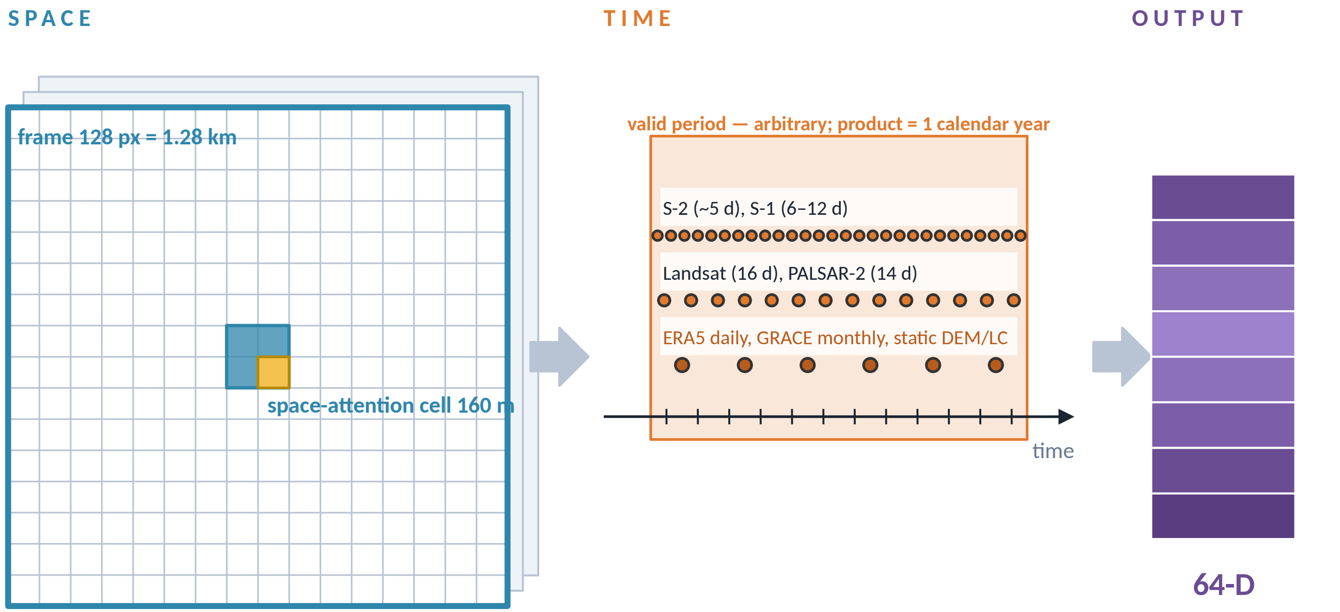
Stencil Output 10 m pixel. Input frame 1.28 km × 1.28 km (128 × 128 px; supplement S2.1/S8.2): inference runs on 960 m tiles buffered by 160 m, so a pixel's context is at most the 1.28 km frame, with spatial attention global inside it. Continuous 'valid period' $[t_0, t_1)$ — arbitrary windows in principle; released product = calendar-year summaries.

Output 64-D, unit-norm, int8 in the public dataset; ~1.4 T footprints / yr. Land + coastal/inland water; no open ocean.

Availability Dataset only (CC-BY 4.0) in Earth Engine / GCS. Weights and code not released.

Results ~23.9 % average error reduction vs. prior featurizations over 15 evaluations (10.4 % at 10-shot, 4.2 % at 1-shot); evapotranspiration regression R^2 0.58. Per-dataset scores are charts only.

Stencil — what feeds one embedding



Output pixel: 10 m × 10 m (drawn 8× too large). Context: the 1.28 km frame. Spatial attention is global inside it on an 8 × 8 grid of 160 m cells; the outer 80 m is trimmed at inference, so no pixel sees further than ~640 m.
≈10 co-registered sources at 10 m … 300 km native resolution.

All observations in the window are summarised — a period embedding, not a snapshot.
Model supports interpolation/extrapolation in time; only annual layers are public.

64-D

per 10 m pixel per year
int8, unit sphere

AlphaEarth Foundations (Google DeepMind)

AlphaEarth Foundations is Google DeepMind's model, published in July 2025. Its central idea is the "embedding field": every 10-metre pixel of land on Earth gets a 64-number vector for every year from 2017 to 2024, and Google publishes those vectors as a dataset inside Google Earth Engine.

Inputs: a very wide mix — optical imagery (Sentinel-2, Landsat), radar (Sentinel-1, PALSAR-2), laser altimetry (GEDI), climate reanalysis (ERA5-Land), gravity (GRACE), elevation, land cover and even text descriptions. Over three billion individual observations went in.

Architecture: the encoder is called Space-Time-Precision, STP, and has 15 blocks. The next four slides walk through it. The training uses several losses at once: reconstructing masked-out inputs, keeping the embeddings spread evenly over a sphere ("batch uniformity"), consistency between a teacher and a student network, and matching to text. The deployed network has about 480 million parameters.

The stencil is the key point for us. Output pixel: 10 m. Spatial context: the model uses attention over the whole input frame at coarse resolution, so a pixel's embedding is influenced by everything in its frame — and the supplement tells us how large the frame is: 1.28 km by 1.28 km, 128 pixels a side, for both training and the global inference run (which tiles the world in 960-m squares padded by 160 m on each side). So an AlphaEarth pixel never sees further than about 640 m in any direction. Time: the model takes a "valid period" — a start and end date — and produces a summary of everything observed in it. In principle any window; in practice the published product uses calendar years.

Output: 64 dimensions, normalised to length one (a "unit vector"), stored as 64 bytes. Coverage is land plus coastal and inland water; no open ocean.

Availability is the big caveat: only the dataset is released. The trained network is not, so nobody can run it on new data, new time windows, or the ocean.

"Error reduction ~23.9 %" means: across their 15 evaluation tasks, using AlphaEarth embeddings instead of the next-best method cut the error by about a quarter on average (about 10 % when only ten labels per class are available, about 4 % with one). The per-task numbers are published only as charts, so we do not quote them.

AlphaEarth — inside the STP encoder

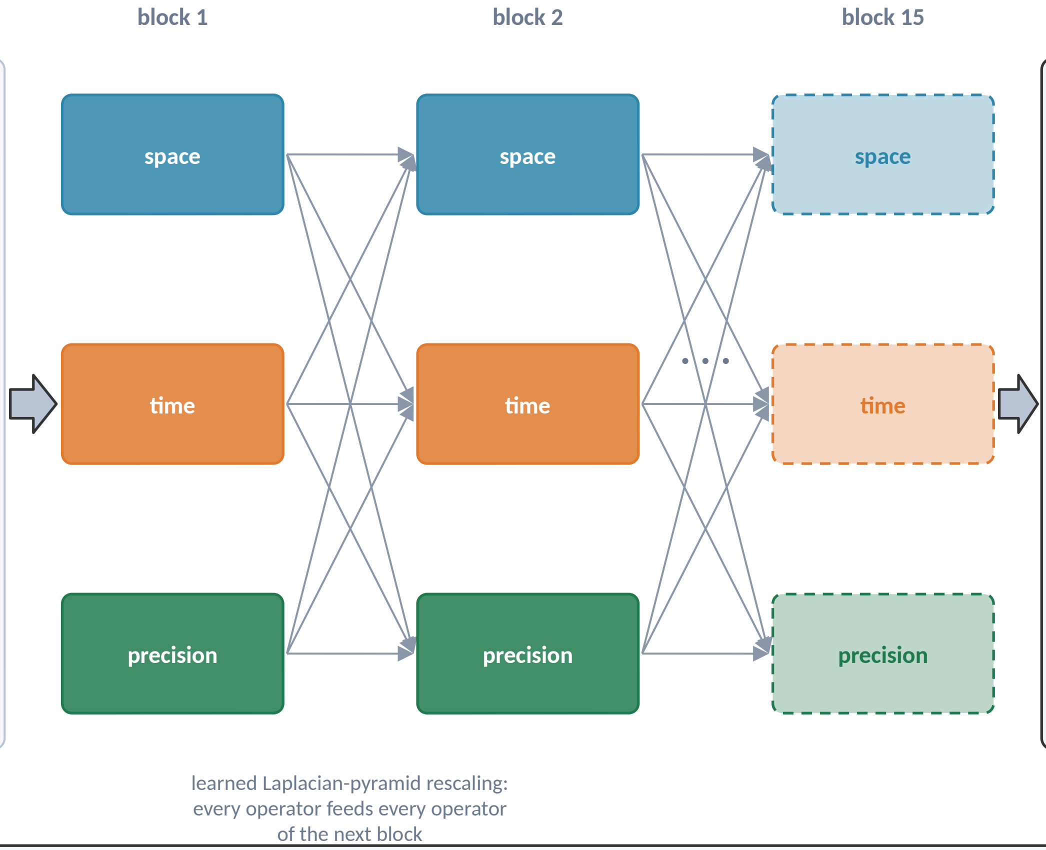
L = side length of the square input frame, in pixels — L = 128 (1.28 km at 10 m) for training and inference (supplement S2.1, S8.2). Three operators at L/16, L/8 and L/2 exchange state through a learned Laplacian pyramid after every one of 15 blocks.

SPACE operator
grid of L/16 × L/16 cells
ViT-like spatial self-attention over the whole (down-scaled) frame — global context, coarse grid

TIME operator
grid of L/8 × L/8 cells
time-axial self-attention along each pixel's sequence; every element conditioned on its sinusoidal timecode

PRECISION operator
grid of L/2 × L/2 cells
3×3 convolutions — keeps fine spatial detail near output resolution

Input projectors
one per source; frames + timestamps → feature maps at 1/2 L. Σ N_i frames in, Σ N_i maps out.



Output head
final learned spatial resampling up to the precision resolution, then to 10 m

64-D per pixel
unit-norm (S^{63}), quantized to 64 bytes (int8) in the released layers

Conditioning
a 'valid period' (t_s, t_e) is injected as timecodes, so the same inputs can be summarised over any date range — interpolating or extrapolating past the observations

Training trio teacher video embedder with implicit decoders · student with identical architecture (consistency + batch-uniformity on the sphere) · text-alignment model with a frozen language model (CLIP-style). ~1 B and ~480 M variants trained; the 480 M one is deployed. **In the supplement (S2.4, S8.2):** 15 STP blocks; widths D_S = 1024 (space), D_T = 512 (time), D_P = 128 (precision); implicit decoders = 2-hidden-layer MLPs of width 512; vMF bottleneck κ = $8{\cdot}10^3$; inference on 960 m tiles buffered to 1.28 km, outer 80 m trimmed. So the spatial receptive field is at most 1.28 km — the whole frame. **Still not stated:** the exact pyramid up/down-sampling kernels and how the three states are merged.

AlphaEarth — inside the STP encoder

This slide opens up the encoder. First the symbol: L is the side length of the square input frame, measured in pixels. The main text keeps it symbolic — "given a square input of L pixels a side" — but the supplement fixes it: frames are 128 by 128 pixels, 1.28 km at 10 m, so the three grids below are 8 by 8, 16 by 16 and 64 by 64 cells.

There are three "operators" — three different kinds of neural-network layer — that run in parallel inside every block:

The SPACE operator works on a coarse grid of L/16 by L/16 cells and uses self-attention, the mechanism from transformers in which every cell can look at every other cell. This gives global context across the frame at low resolution.

The TIME operator works on an L/8 grid and uses "time-axial" attention: each location looks only at its own history across all the dated frames. Every frame carries a sinusoidal "timecode" (a numeric encoding of its date), which is how the model knows when things happened.

The PRECISION operator works on the finest grid, L/2, with ordinary 3-by-3 convolutions to keep sharp spatial detail.

A "block" is one round of all three operators. After each block, a learned "Laplacian pyramid" rescaling step lets every operator hand its result to every operator of the next block: coarse results are up-sampled, fine results are down-sampled, so that information flows between scales. Repeat for 15 blocks (supplement S2.4, which also gives the widths: 1024 numbers per cell on the space path, 512 on the time path, 128 on the precision path).

Input projectors bring each source's frames down to the L/2 grid first. The output head resamples back up to the precision resolution, then to 10 m, and produces the 64-number vector per pixel.

"Teacher/student" refers to the training recipe: one network (teacher) produces targets that a second, identical network (student) learns to match; a third model aligns embeddings with text descriptions in the style of CLIP.

Bottom line: because the space operator uses attention over the whole frame, the effective receptive field is the frame — 1.28 km, so at most about 640 m from the pixel in any direction. What the paper still does not spell out is the exact form of the pyramid resampling and how the three states are merged.

STP, step 1 of 4 — input projectors bring every source to ½ L

One frame of the same 1.28 km footprint from each sensor, at its own native resolution, becomes a 64 × 64 feature map at 20 m

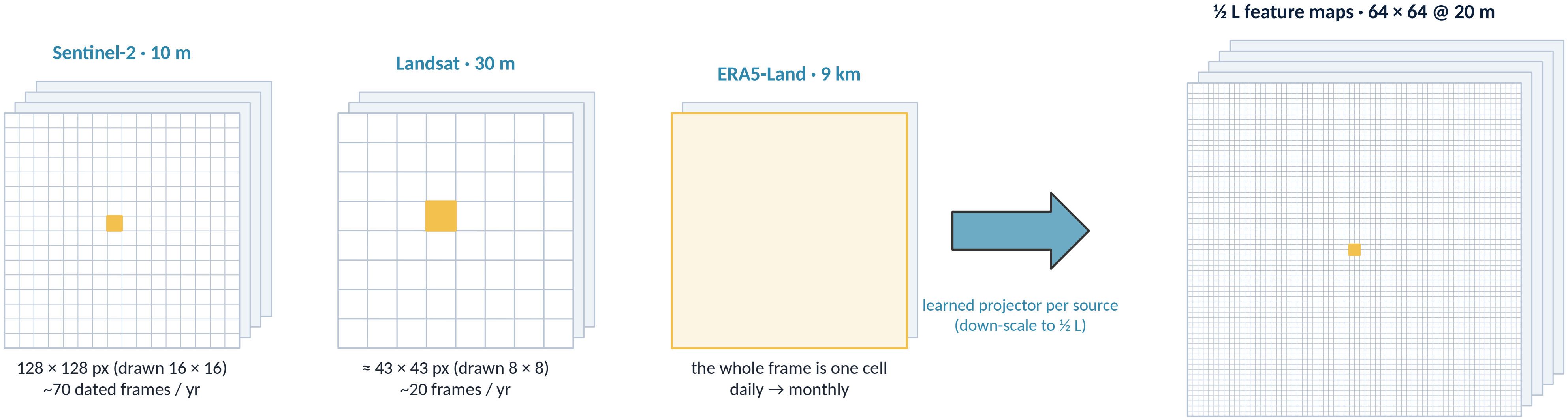
Inputs
many sources, native resolutions, dated frames

Input projectors
→ ½ L feature maps (64 px @ 20 m)

STP block
space 1/16 L · time 1/8 L · precision ½ L

Pyramid exchange
every operator → every operator, × N

Output head
→ L (10 m), 64-D unit vector



ΣN_i frames in → ΣN_i maps out. Every frame keeps its acquisition timestamp as a sinusoidal timecode, so the block that follows knows when each map was observed. The yellow 10 m pixel is now a 20 m feature cell — the first (and only) fixed loss of resolution before the output head restores it.

... + S-1, PALSAR-2, GEDI, GRACE, DEM, land-cover, text

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 1 of 4 — input projectors bring every source to ½ L

Steps 1 to 4 animate the previous slide with the paper's own frame: L = 128 pixels at 10 m, so the frame is 1.28 km across (supplement S2.1 and S8.2). Two things in these slides are simplified and the footers say so: we draw 4 blocks where the real network has 15, and the per-year frame counts are typical values rather than the paper's.

Step 1 is about the input projectors. Each data source arrives at its own native resolution over the same 1.28-km footprint: Sentinel-2 at 10 m gives a 128 by 128 grid (drawn coarser here); Landsat at 30 m gives about 43 by 43; ERA5-Land, a weather reanalysis at 9 km, gives just one value for the whole frame. Each source also arrives many times a year — Sentinel-2 roughly every five days, Landsat every 16.

A learned projector per source converts each frame into a "feature map" — a grid of learned numbers rather than raw pixel values — at half the input resolution: 64 by 64 cells of 20 m. This is the only place where resolution is permanently reduced before the output head restores it.

The important detail is that each frame keeps its timestamp: it is carried along as a sinusoidal timecode so the time operator later knows when each observation was made. "ΣN_i frames in, ΣN_i maps out" simply means: however many frames come in from all sources combined, that many feature maps come out — nothing is merged yet.

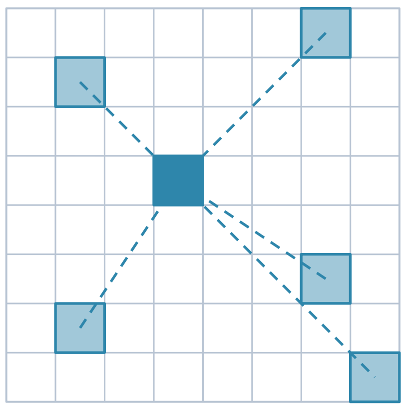
The yellow pixel is our 10-m output pixel; after this step it lives inside a 20-m feature cell.

STP, step 2 of 4 — three operators, three resolutions

Same 1.28 km footprint, three grids: what one cell can 'see' in each operator

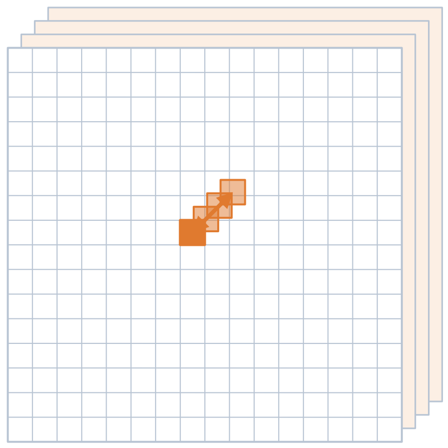
Inputs — many sources, native resolutions, dated frames → **Input projectors** — → $\frac{1}{2}$ L feature maps (64 px @ 20 m) → **STP block** — space $\frac{1}{16}$ L · time $\frac{1}{8}$ L · precision $\frac{1}{2}$ L → **Pyramid exchange** — every operator → every operator, × N → **Output head** — → L (10 m), 64-D unit vector

SPACE operator · $\frac{1}{16}$ L = 8 × 8 cells of 160 m



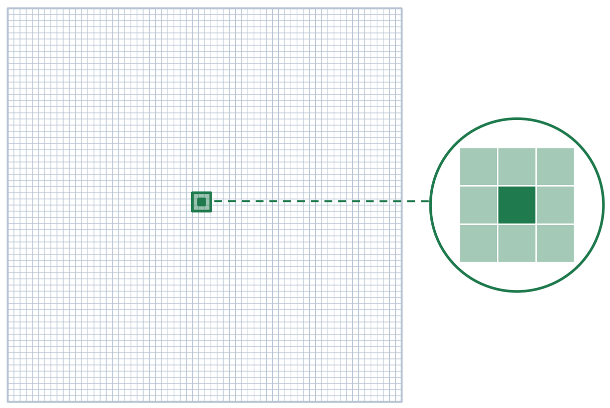
ViT-like self-attention: the cell attends to every other cell of the frame. Receptive field = whole frame, at coarse grain.

TIME operator · $\frac{1}{8}$ L = 16 × 16 cells of 80 m



Time-axial self-attention: the cell attends only to itself across all dated frames (stacked sheets), each tagged with its timecode.

PRECISION operator · $\frac{1}{2}$ L = 64 × 64 cells of 20 m



3 × 3 convolutions: the cell sees only its immediate neighbours (60 m), but at the finest grain the block carries.

Three views of one place, computed simultaneously. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture — none of them alone is the stencil; the exchange in step 3 is.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 2 of 4 — three operators, three resolutions

Step 2 shows what one cell can "see" inside each of the three operators, on the same 1.28-km footprint.

SPACE (left): the grid has 8 by 8 cells of 160 m (that is L/16 with L = 128). The highlighted cell attends to every other cell — the dashed lines. So the space operator gives every cell a view of the whole frame, but only at coarse grain.

TIME (middle): 16 by 16 cells of 80 m. The stacked sheets are the dated frames. The highlighted cell attends only to itself across all the sheets — this is "time-axial" attention. It knows the whole history of one spot, but nothing about the neighbours.

PRECISION (right): 64 by 64 cells of 20 m. The operator is a 3-by-3 convolution, so a cell sees only its eight immediate neighbours — 60 m across — but at the finest grain the block carries. The magnifier shows that 3-by-3 neighbourhood.

The point of running three views at once: none alone is the stencil. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture. Step 3 is where they are combined.

STP, step 3 of 4 — the pyramid exchange between blocks

End of block k: each of the three maps is resampled to all three resolutions and summed into block k+1's inputs — repeat for N blocks

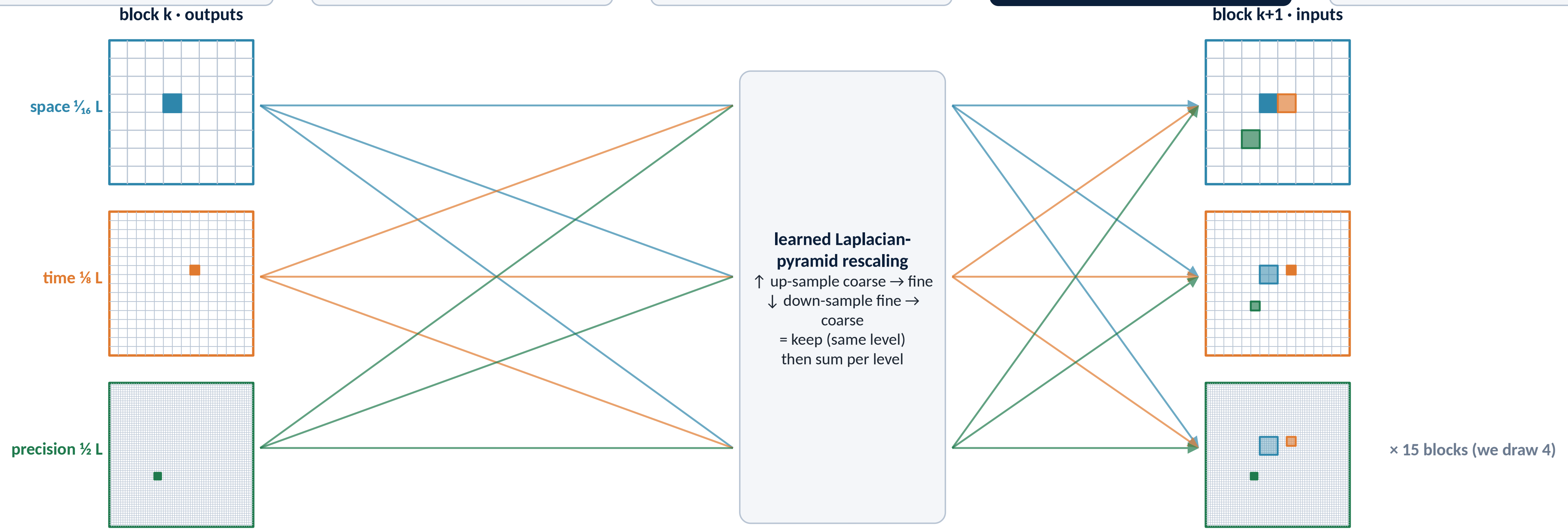
Inputs
many sources, native resolutions, dated frames

Input projectors
→ ½ L feature maps (64 px @ 20 m)

STP block
space 1/16 L · time ⅛ L · precision ½ L

Pyramid exchange
every operator → every operator, × N

Output head
→ L (10 m), 64-D unit vector



Follow the colours. The blue 160 m space cell arrives in the fine map as an 8 × 8 block of 20 m cells; the green 3 × 3 precision patch (60 m) arrives in the coarse map as a fraction of one 160 m cell; the orange time cell lands as 2 × 2 / 4 × 4. After the sum, every operator in block k+1 starts from all three views. Repeat N times — the receptive field of the fine path grows to the whole frame while it keeps 20 m grain.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 3 of 4 — the pyramid exchange between blocks

This is the sentence the walk-through was built to explain: "after each block, learned Laplacian-pyramid rescaling lets every operator feed every operator of the next block".

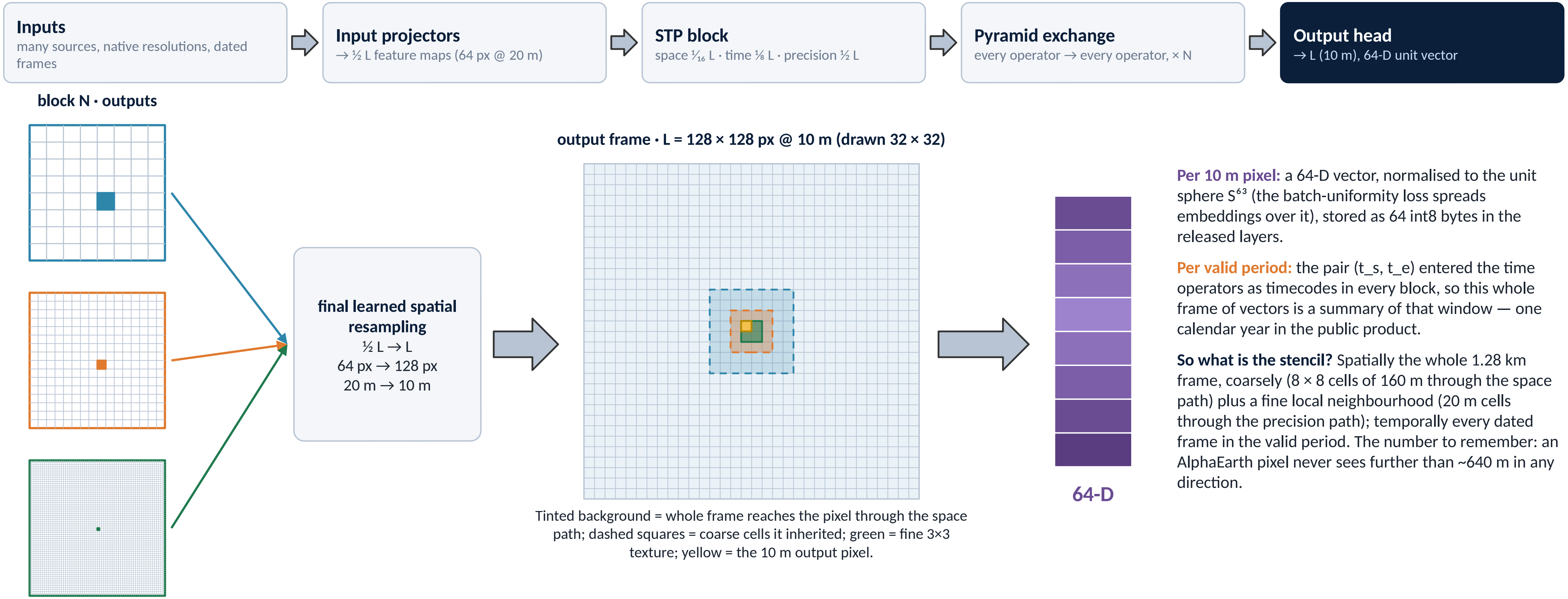
On the left are the three outputs of block k — the coarse space map, the medium time map, the fine precision map — each with one feature highlighted. In the middle is the rescaling stage. A Laplacian pyramid is a classic image-processing idea: an image is represented at several resolutions at once, and there are learned operations to move information up (coarse to fine, up-sampling) and down (fine to coarse, down-sampling). Here it is used to translate every map to every resolution.

On the right are the three inputs of block k+1. Each now contains all three colours: the blue space feature arrived in the fine map as an 8-by-8 block of 20-m cells, the green precision patch arrived in the coarse map as a fraction of one 160-m cell, the orange time cell landed as 2-by-2 or 4-by-4 blocks. Nine paths in total (three sources times three destinations), then a sum per level — the sum is our reading; the paper says "pass its state to each of the operators" without spelling out the merge.

Repeat 15 times (we draw 4). The consequence for our stencil question: after a few blocks, even the fine 20-m path has a receptive field that spans the whole frame, because it has received the space operator's global view several times — while keeping its own fine grain.

STP, step 4 of 4 — back to 10 m, 64-D per pixel

After block N: the ½ L maps are resampled to the precision resolution and then to L; each output pixel is a unit vector on S^{63}



Tinted background = whole frame reaches the pixel through the space path; dashed squares = coarse cells it inherited; green = fine 3×3 texture; yellow = the 10 m output pixel.

Per 10 m pixel: a 64-D vector, normalised to the unit sphere S^{63} (the batch-uniformity loss spreads embeddings over it), stored as 64 int8 bytes in the released layers.

Per valid period: the pair (t_s, t_e) entered the time operators as timecodes in every block, so this whole frame of vectors is a summary of that window — one calendar year in the public product.

So what is the stencil? Spatially the whole 1.28 km frame, coarsely (8 × 8 cells of 160 m through the space path) plus a fine local neighbourhood (20 m cells through the precision path); temporally every dated frame in the valid period. The number to remember: an AlphaEarth pixel never sees further than ~640 m in any direction.

L = side length of the square input frame in pixels. L = 128 px @ 10 m (1.28 km) is the paper's own frame (supplement S2.1, S8.2); the real depth is N = 15 blocks, we draw 4. Frame counts per year are typical values, not from the paper.

STP, step 4 of 4 — back to 10 m, 64-D per pixel

Step 4 closes the loop. After the last block, the three maps are merged and a final learned spatial resampling takes the result from the L/2 grid (64 by 64 at 20 m) back to L (128 by 128 at 10 m).

The output frame in the middle shows, for one yellow 10-m pixel, what it has inherited: the tinted background stands for the whole frame reaching it through the space path; the dashed squares are the coarse cells it inherited from the space and time operators; the green patch is the fine local texture from the precision path.

Each output pixel becomes a 64-number vector. "Unit vector on S^{63}" is the mathematical way of saying the 64 numbers are scaled so their total length is exactly one — they sit on the surface of a 64-dimensional sphere. The batch-uniformity loss during training spreads the embeddings evenly over that sphere. In the released dataset each vector is stored as 64 bytes (one "int8", a whole number between −128 and 127, per dimension).

"Per valid period": the pair of dates (t_s, t_e) — start and end of the window — was fed into the time operators in every block as timecodes, so the whole frame of vectors is a summary of that window. In the public product the window is one calendar year.

So what is AlphaEarth's stencil? Spatially, the whole 1.28-km frame coarsely (8 by 8 cells of 160 m through the space path) plus a fine local neighbourhood (20-m cells through the precision path); temporally, every dated observation in the window. The number to remember: an AlphaEarth pixel never sees further than about 640 m.

TESSERA (University of Cambridge)

Pixel-wise 10 m annual embeddings from S-1 + S-2 time series — fully open (CC0), CVPR 2026

Who / when Cambridge Centre for Earth Observation; arXiv 2506.20380 (Jun 2025), CVPR 2026. Code github.com/ucam-eo/tessera, access via geotessera.

Inputs Sentinel-2 L2A (10 bands, 60 m bands dropped, resampled to 10 m) + Sentinel-1 RTC (VV, VH). Two modality-specific branches, fused late.

Architecture Per-pixel temporal transformer (4 layers) + GRU pooling per modality; encoder ~46 M params (1.34 B projector used only in training). Barlow Twins self-supervision on ~800 M pixel-series from 3,012 MGRS tiles; ~6,200 GPU-h on MI300X.

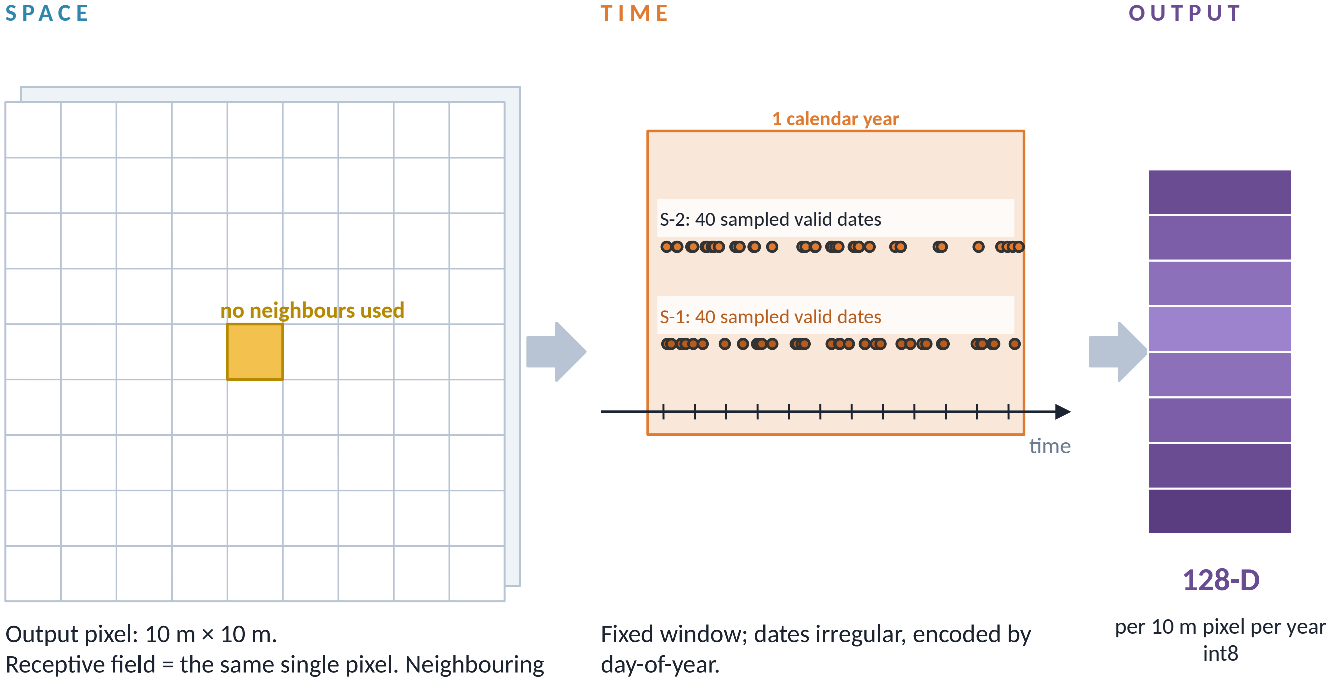
Stencil Strictly one 10 m pixel — no spatial context at all. One calendar year; 40 valid dates sampled per modality with learnable day-of-year embeddings (irregular cadence tolerated).

Output 128-D per pixel-year, int8 (QAT). Years 2017–2025, global land (water masked); tile-sampled coverage, gaps on request.

Availability Weights (CC0), code (MIT), precomputed embeddings via GeoTessera CDN. Self-run: ~10 h per 110 km tile-year (128-core CPU + A30).

Results Frozen + MLP head: PASTIS-R mIoU 50.7, TreeSatAI-TS F1 78.0, canopy height RMSE 12.2 m; holds up until <~20 valid S-2 obs/yr.

Stencil — what feeds one embedding



Output pixel: 10 m × 10 m.
Receptive field = the same single pixel. Neighbouring pixels never enter.
Two modality sheets: S-2 (10 bands) and S-1 (2 bands).

Fixed window; dates irregular, encoded by day-of-year.
Sampling with replacement pads pixel-years with <40 clear observations.

TESSERA (University of Cambridge)

TESSERA comes from the University of Cambridge (published June 2025, accepted at CVPR 2026) and is the most radical design in the deck: it uses no spatial context at all.

Inputs: Sentinel-2 optical (10 bands) and Sentinel-1 radar (2 polarisations, VV and VH). Sentinel-1 is radar, so it sees through clouds; Sentinel-2 is optical.

Architecture: for each 10-m pixel separately, a small transformer reads the pixel's own time series — 40 dates sampled from one calendar year for each sensor — and a GRU (a type of recurrent network) pools them into one vector. The two sensors have their own branches which are fused at the end. The encoder is only about 46 million parameters; a much larger "projector" is used only during training and then discarded. Training uses Barlow Twins, a self-supervised objective that asks two differently-sampled views of the same pixel-year to produce the same embedding while keeping the embedding's dimensions non-redundant — no reconstruction of inputs is involved.

Stencil: strictly one 10-m pixel; neighbours never enter. One calendar year; dates are irregular, and the model is told the day of year of each observation. If a pixel has fewer than 40 clear observations, dates are sampled with replacement to fill the 40 slots.

Output: 128 numbers per pixel per year, stored as int8. Years 2017–2025, land only (water masked). Everything is released: weights (CC0, the most permissive licence), code (MIT), and precomputed embeddings through the GeoTessera service.

Results quoted are with the embeddings frozen and only a small head trained on top: PASTIS-R crop segmentation mIoU 50.7, TreeSatAI tree-species F1 78.0, canopy height error 12.2 m. mIoU is "mean intersection over union", the standard segmentation score (100 = perfect overlap). F1 is the balanced average of precision and recall for classification.

Why it matters for us: TESSERA proves that a per-pixel temporal encoder with zero spatial context is competitive on land. That is the opposite extreme from our 3-by-3 patch, and a useful reference point.

OlmoEarth (Ai2)

Open multimodal space-time ViT family (Nano→Large); embeddings computed on demand, not precomputed

Who / when Allen Institute for AI (+ UW, ASU, UBC). v1 Nov 2025 (arXiv 2511.13655), v1.1 May 2026, v1.2 later 2026. Weights on Hugging Face; custom 'OlmoEarth Artifact License' (no military/extractive use).

Inputs Sentinel-1, Sentinel-2, Landsat-8 monthly time series, all resampled to 10 m; static layers (OSM, WorldCover, CDL, SRTM, canopy height, WorldCereal) as inputs/targets.

Architecture ViT over space × time × band-group tokens (FlexiViT patches 1–8 px in v1; single-bandset token merging in v1.1, RoPE in v1.2). 'Latent MIM Lite': masked modelling with patch-discrimination + instance-contrastive losses. Sizes: Nano 1.4 M · Tiny 6.2 M · Base 89 M · Large 308 M (encoder).

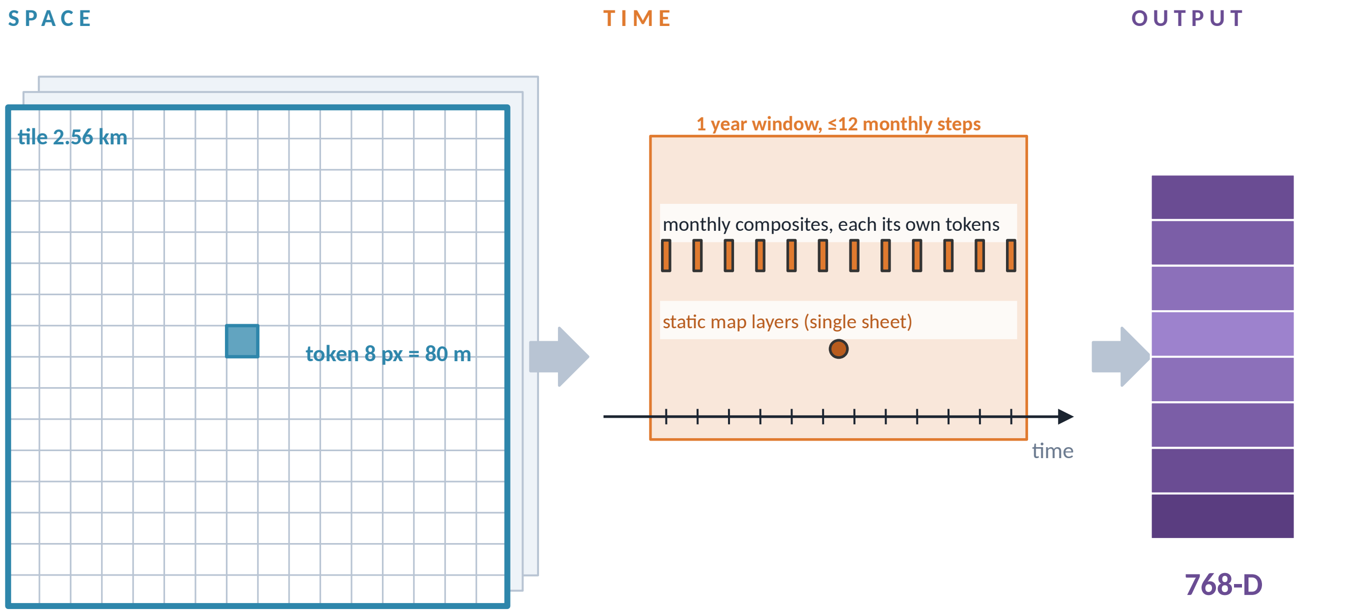
Stencil Tile 2.56 km × 2.56 km = 256 × 256 px @ 10 m; token = 1–8 px patch (≤ 80 m). Up to 12 monthly steps over one year, missing months masked.

Output Per-patch, per-timestep tokens (Base 768-D, Large 1024-D) + pooled vector. No standing global product; Studio exports embeddings for any AOI at 10–80 m, 1–12 monthly steps (int8 COGs).

Results Best on 15/24 tasks (kNN/linear) and 19/29 (fine-tuned) vs. Prithvi v2, TerraMind, Galileo, DINOv3; matches/exceeds AlphaEarth after fine-tuning on 5 tasks.

Pretraining data 285 k samples × 2.56 km × 1 yr, stratified over 120 OSM classes (~10 TB) — land/infrastructure-biased.

Stencil — what feeds one embedding



Tile: 256 × 256 px @ 10 m = 2.56 km (drawn 16 × 16 for legibility).
Token: flexible 1–8 px patch → 10–80 m per token, full attention across the tile.
Sheets: S-1, S-2, Landsat + static maps.

Each month is tokenised separately; attention runs across space, time and modality. Temporal masking (v1.2) trains the model to fill missing months.

per patch per month (Base)
+ pooled tile vector

OlmoEarth (Ai2)

OlmoEarth is the Allen Institute for AI's family of open models, first released November 2025; version 1.1 (May 2026) merged Sentinel-2's band groups into a single token to cut compute, and version 1.2 added rotary position encodings. It is a "space-time ViT": a Vision Transformer that tokenises not just space but also time and sensor band groups.

Inputs: Sentinel-1, Sentinel-2 and Landsat-8 monthly time series, all resampled to 10 m, plus static map layers (OpenStreetMap, land cover, crop data, elevation, canopy height) used as extra inputs and as prediction targets.

Architecture: the tile is 2.56 km square (256 by 256 pixels at 10 m). Tokens are flexible patches of 1 to 8 pixels — so 10 to 80 m — and up to 12 monthly steps are tokenised separately. Attention runs across space, time and modality together. The training objective, "Latent MIM Lite", is masked modelling: hide some tokens, predict them in the embedding space rather than as pixels, plus contrastive terms. Four sizes: Nano (1.4 M parameters), Tiny (6 M), Base (89 M), Large (308 M).

Output: one vector per patch per month (768 dimensions for Base, 1024 for Large) plus a pooled vector for the tile. There is no global precomputed product; instead the "OlmoEarth Studio" computes embeddings on demand for any region you draw.

Results: in the authors' own benchmark the model is best on 15 of 24 tasks with frozen embeddings and 19 of 29 when fine-tuned; on five tasks a fine-tuned OlmoEarth matches or beats AlphaEarth.

Licence: open weights, but under a custom licence that forbids some uses (military, extractive industries).

For us: OlmoEarth is the closest analogue to a general space-time transformer we could actually retrain — open weights, monthly tokens, a masked latent objective — but its pretraining sample is land-biased.

TerraMind (IBM Research + ESA Φ-lab)

Any-to-any generative multimodal EO model; 'Thinking in Modalities' — ICCV 2025

Who / when IBM Research Europe, ESA Φ-lab, ETH Zürich, FZ Jülich, NASA IMPACT, Univ. of Iceland. arXiv 2504.11171 (Apr 2025); TerraMesh dataset 2504.11172. Apache 2.0, weights on Hugging Face (ibm-esa-geospatial), TerraTorch integration. Tiny/Small edge variants Jun–Jul 2025.

Inputs Pixel-level: S-2 L2A & L1C, S-1 GRD & RTC, DEM, RGB, NDVI. Token-level: LULC classes, captions, geolocation. All co-registered at 10 m in 224 px tiles.

Architecture Dual-scale encoder–decoder: token path (cross-modal correlation) + pixel path (fine detail). Per-modality FSQ-VAE tokenizers (16 k codebook). Objective: masked cross-modal token reconstruction over 500 B tokens from ~9 M TerraMesh samples. Base: 6 d, Large: 10 d on 32 A100s.

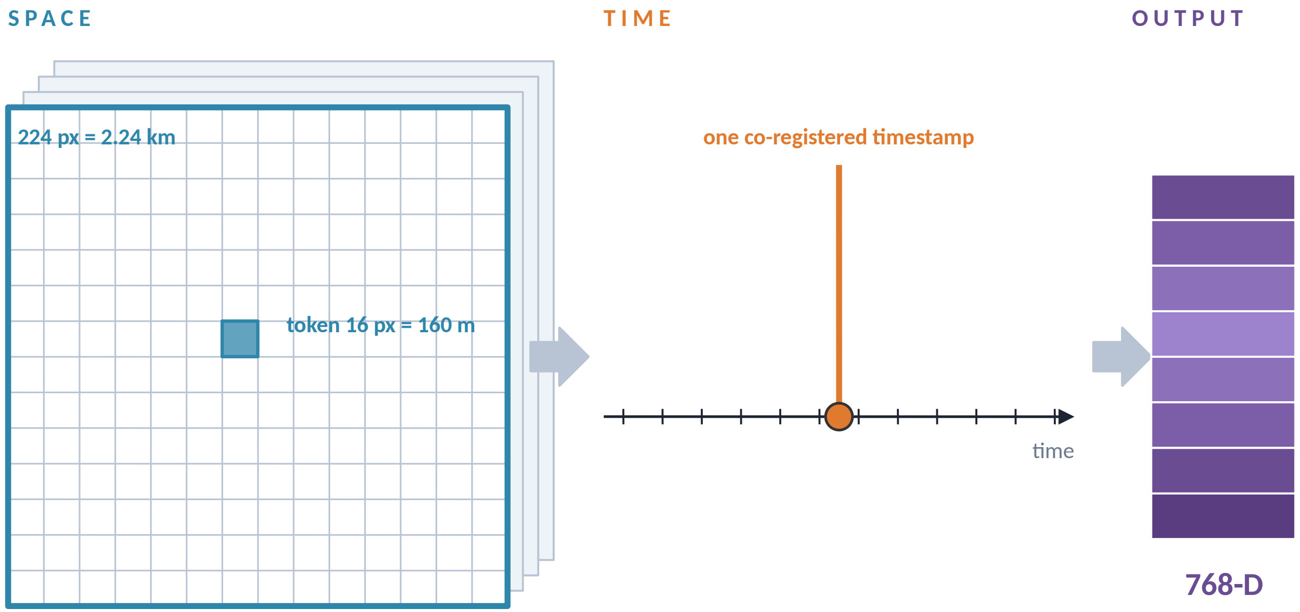
Stencil Tile 224 × 224 px @ 10 m = 2.24 km; token 16 × 16 px = 160 m → 14 × 14 = 196 tokens per modality. Single co-registered timestamp per sample — no multi-temporal fusion.

Output 196 tokens per modality of 768-D (Base) / 1024-D (Large), mergeable to one vector; plus generated rasters of missing modalities (TiM). No precomputed embedding archive.

Results PANGAEA avg mIoU: Large 59.6, Base 58.4 (≥3 pp over CROMA, DOFA and the other GeoFMs in that table); only GeoFM beating task-specific U-Nets across the benchmark.

Ocean TerraMesh is land/ecoregion-stratified; no marine modalities. Open-ocean behaviour untested.

Stencil — what feeds one embedding



Tile: 224 × 224 px @ 10 m = 2.24 km × 2.24 km.
Token: 16 × 16 px = 160 m; 14 × 14 grid per modality.
Up to ~9 modality sheets, each with its own tokenizer.

All modalities come from the same date and place.
A sample is a snapshot: every modality is aligned to one acquisition.
Temporal change is not part of the pretraining stencil.

per 160 m token, per modality
(Base; Large 1024-D)
or merged per tile

TerraMind (IBM Research + ESA Φ-lab)

TerraMind is a joint IBM Research and ESA Φ-lab model (April 2025, ICCV 2025). Its distinguishing feature is that it is generative and "any-to-any": given some modalities it can produce the missing ones.

Inputs come in two kinds. Pixel-level: Sentinel-2 optical, Sentinel-1 radar, elevation, RGB, vegetation index (NDVI). Token-level: land-use class maps, text captions, geographic coordinates. All are aligned on 10-m, 224-pixel tiles.

Architecture: each modality has its own "tokenizer" — a small network that turns its raster into discrete codes from a 16,000-entry codebook (an FSQ-VAE, a kind of quantising autoencoder). The main encoder–decoder then learns to predict masked codes of one modality from the others. "Thinking in Modalities" (TiM) is the trick of generating a missing modality — say land cover from optical — and feeding it back in as extra input. Trained on 500 billion tokens from about 9 million TerraMesh samples.

Stencil: tile 2.24 km (224 pixels at 10 m); token 16 pixels = 160 m, so 14 by 14 tokens per modality. One co-registered timestamp per sample — every modality comes from the same date and place; there is no multi-temporal fusion in pretraining.

Output: 196 tokens per modality, 768 dimensions for the Base model and 1024 for Large, or one merged vector per tile, plus generated rasters. No precomputed embedding archive; weights are Apache 2.0 (fully open) on Hugging Face.

Results: on the PANGAEA benchmark (a standard suite of segmentation tasks) TerraMind Large scores 59.6 mean IoU, the only foundation model to beat U-Nets trained from scratch on that suite.

Ocean: the training data is stratified by land ecoregion; there are no marine modalities and open-ocean behaviour is untested.

Prithvi-EO 2.0 (IBM + NASA)

Multi-temporal HLS masked autoencoder at 30 m; 300 M and 600 M, with temporal-location (TL) variants

Who / when IBM Research + NASA IMPACT, trained at JSC; arXiv 2412.02732 (Dec 2024, v3 Mar 2026). Weights on Hugging Face (ibm-nasa-geospatial); fine-tuning via TerraTorch. Succeeds Prithvi-EO 1.0 (100 M).

Inputs HLS v2 (Harmonized Landsat–Sentinel-2) at 30 m, 6 bands (B02–B07: blue, green, red, NIR-narrow, SWIR1, SWIR2). 4.2 M training samples, 2014–2023; sea-only tiles removed with Fmask, Greenland excluded.

Architecture ViT-L (300 M: 1024-D, 24 layers) / ViT-H (600 M: 1280-D, 32 layers) masked autoencoder, mask ratio 0.75, 3D sin-cos positions. TL variants add lat/lon + date encodings with metadata dropout.

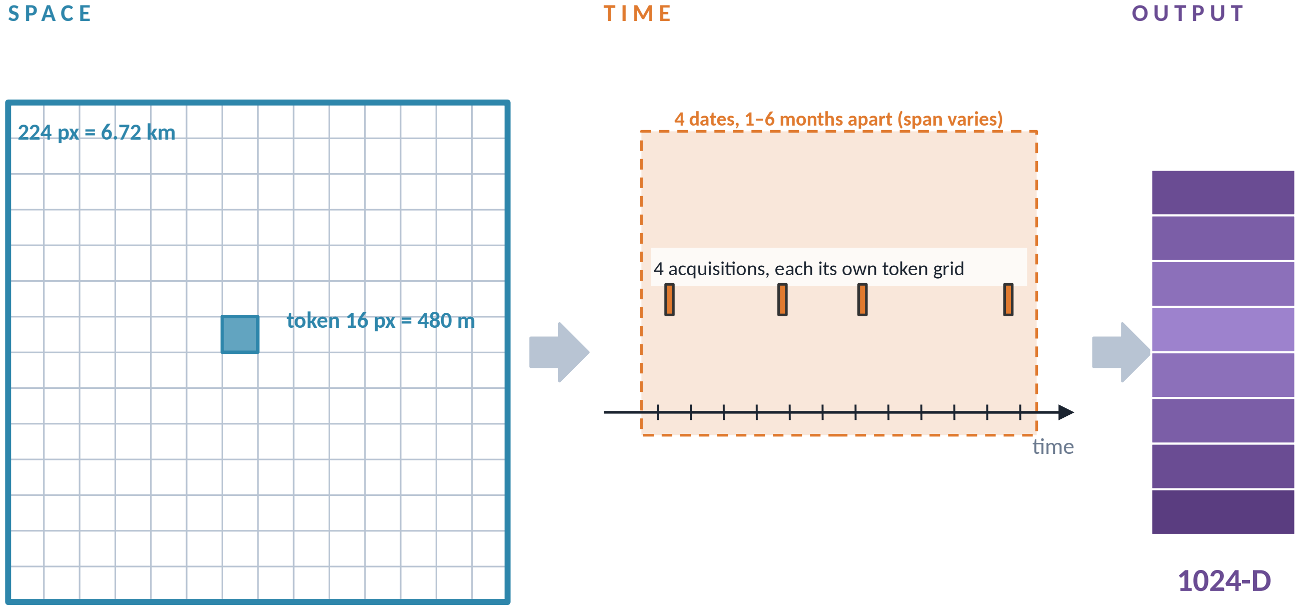
Stencil Tile 224 × 224 px @ 30 m = 6.72 km. Token 16 px = 480 m (300 M) or 14 px = 420 m (600 M); tubelet depth 1, so each of 4 timestamps gets its own token grid. Frames 1–6 months apart (seasonal), not a fixed window; single-frame inference supported.

Output 4 × 196 (or 4 × 256) tokens of 1024/1280-D. No precomputed embedding product; typical use is full fine-tuning of encoder + light decoder.

Results GEO-Bench: 600 M-TL best on average, ~8 % over v1.0 and ahead of six other GFMs incl. DOFA and Scale-MAE (per-set numbers are charts); tasks: floods, burn scars, crops, landslides, PASTIS, Sen4Map.

In orbit May 2026: deployed on Kanyini satellite and ISS IMAGIN-e for onboard flood/cloud detection (still the 2.0 architecture).

Stencil — what feeds one embedding



Tile: 224 × 224 px @ 30 m = 6.72 km × 6.72 km.
Token: 16 × 16 px = 480 m (300 M) · 14 × 14 px = 420 m (600 M).
One sheet: 6 HLS bands.

Time is a stack of 4 snapshots with 3D position codes — not a period summary.
TL variants inject acquisition date + lat/lon as extra encodings.

per 480 m token, per date
300 M: 1024-D · 600 M: 1280-D

Prithvi-EO 2.0 (IBM + NASA)

Prithvi-EO 2.0 is IBM and NASA's second-generation Earth observation model (December 2024). It matters here for two reasons: it is a widely used open baseline, and IBM's ocean model on the next slide reuses its recipe.

Inputs: HLS, the Harmonized Landsat–Sentinel-2 product, at 30 m with six spectral bands (blue, green, red, near-infrared, and two shortwave-infrared bands). 4.2 million training samples from 2014–2023. Importantly for us, sea-only tiles were explicitly removed using a cloud/water mask, and Greenland was excluded.

Architecture: a masked autoencoder (MAE) — the network sees a tile with 75 % of its patches hidden and learns to reconstruct them. Two sizes: 300 million parameters (a "ViT-Large", 1024-dimensional embeddings, 24 layers) and 600 million (ViT-Huge, 1280 dimensions, 32 layers). The "TL" variants additionally receive the acquisition date and latitude/longitude as extra encodings.

Stencil: tile 224 pixels at 30 m = 6.72 km. Token 16 pixels = 480 m for the 300 M model, 14 pixels = 420 m for the 600 M model (we checked the published configuration files). Time: four frames, each 1–6 months apart to capture seasons; the "tubelet depth" is 1, meaning each frame gets its own grid of tokens rather than frames being merged into one token — time is fused only inside attention. Single-frame use is also supported.

Output: one vector per token per frame; no precomputed product. The typical way to use it is full fine-tuning — retraining the encoder together with a small decoder for your task — through IBM's TerraTorch toolkit.

Results: on GEO-Bench (a 12-task standard benchmark) the 600 M-TL model is the best on average, about 8 % above version 1.0 and ahead of six other foundation models; the per-dataset scores are published as charts, so we quote no single number. In May 2026 the model was deployed on a satellite for onboard flood and cloud detection.

Granite-Geospatial-Ocean (IBM · STFC · PML · Exeter)

The Prithvi-EO recipe moved onto Sentinel-3 ocean colour + SST — github.com/ibm-granite/geospatial

Who / when IBM Research Europe (UK), STFC Hartree Centre, Plymouth Marine Lab, Univ. of Exeter (UK HNCDI). Released 6 Oct 2025; arXiv 2509.21273 'A Sentinel-3 foundation model for ocean colour'. Apache 2.0; weights on Hugging Face, code + TerraTorch notebook at github.com/ibm-granite/geospatial.

Inputs Sentinel-3 OLCI Level-2 reflectance, 16 bands (Oa01–Oa12, Oa16–18, Oa21) + SLSTR sea-surface temperature = 17 channels at 300 m. ~512 k tiles (the paper also gives 470 k train + 50 k val), 2017–2021, balanced over 83 Longhurst provinces and months.

Architecture Prithvi-EO ViT-MAE with the tile shrunk for the coarser sensor: embed 512-D, 12 layers, 8 heads, decoder 256-D × 4; ~50 M params (larger gave no gain). RMSE reconstruction loss on masked patches.

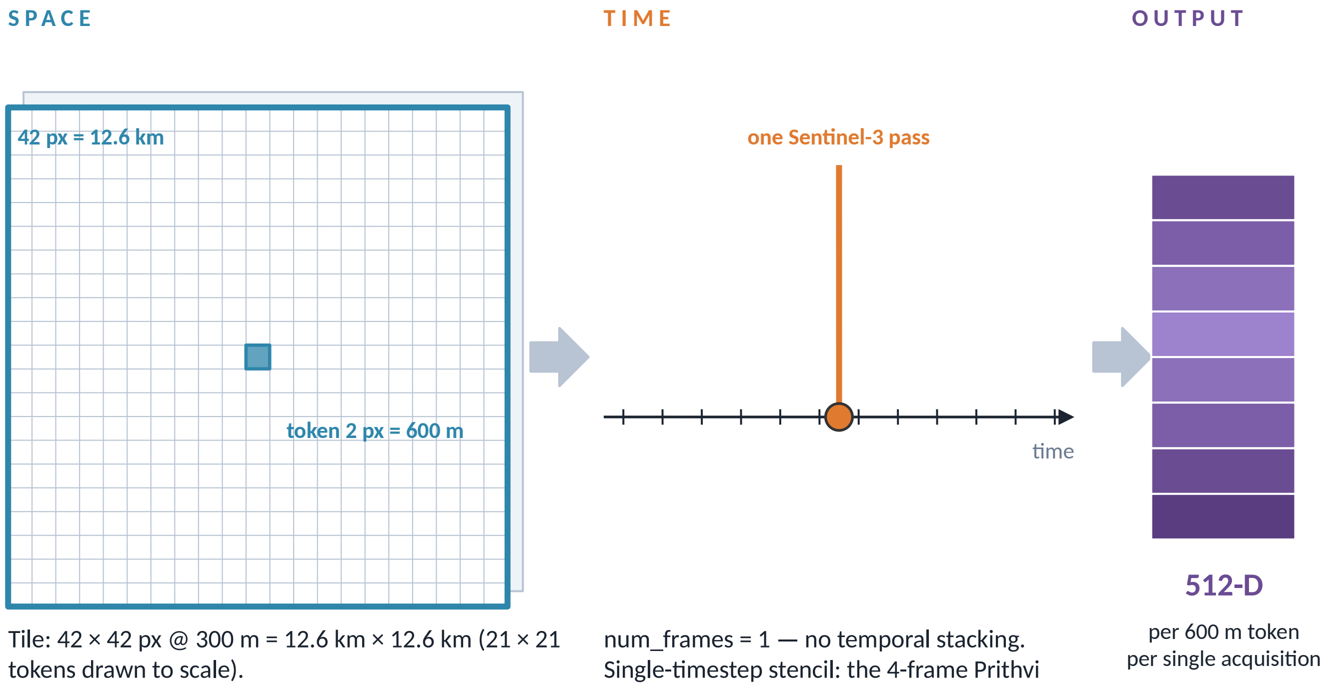
Stencil Tile 42 × 42 px @ 300 m = 12.6 km × 12.6 km; token 2 × 2 px = 600 m → 21 × 21 = 441 tokens. num_frames = 1: a single acquisition, no temporal stacking (fine-tuning pairs each in-situ sample with cloud-free imagery from a 6-day window centred on it).

Output 441 × 512-D tokens per tile; downstream pixel-wise regression heads (TerraTorch) for chlorophyll-a and primary production.

Results Chl-a (188 in-situ points): RMSE 0.14 ± 0.05 vs 0.16 ± 0.10 $\log_{10}$ mg m^{-3} from scratch; primary production (103 points): 0.39 ± 0.04 vs 0.42 ± 0.07 $\log_{10}$ mgC m^{-2} d^{-1}. Gains grow as labels shrink (useful at 25 % of labels).

Caveats Surface-only, cloud-limited optical; single snapshot; authors say not yet competitive with operational algorithms; under-predicts bloom-level chl-a.

Stencil — what feeds one embedding



Tile: 42 × 42 px @ 300 m = 12.6 km × 12.6 km (21 × 21 tokens drawn to scale).
Token: 2 × 2 px = 600 m.
Two sheets: 16 OLCI bands + 1 SST band.

num_frames = 1 — no temporal stacking.
Single-timestep stencil: the 4-frame Prithvi option is switched off.
Fine-tuning pairs each in-situ measurement with imagery from a 6-day window centred on it.

Granite-Geospatial-Ocean (IBM · STFC · PML · Exeter)

This is IBM's Prithvi-based ocean model, developed with the UK's STFC Hartree Centre, Plymouth Marine Laboratory and the University of Exeter, released October 2025. The code lives in the ibm-granite GitHub organisation; we read its configuration files directly to get the numbers below.

Inputs: Sentinel-3 OLCI, the ocean-colour instrument, 16 spectral bands at 300 m, plus sea-surface temperature from the SLSTR instrument — 17 channels in total. About 512,000 tiles from 2017–2021 (the paper also quotes 470,000 for training plus 50,000 for validation), sampled evenly across 83 "Longhurst provinces" (a standard division of the ocean into biogeochemical regions) and across months.

Architecture: the Prithvi masked-autoencoder recipe with the network shrunk to fit the coarser sensor: embedding size 512, 12 layers, about 50 million parameters, trained to reconstruct masked patches under a root-mean-square-error loss. The authors report that larger networks gave no gain.

Stencil: tile 42 by 42 pixels at 300 m = 12.6 km square. Token 2 by 2 pixels = 600 m, so 21 by 21 = 441 tokens. num_frames = 1: a single acquisition, no temporal stacking — the four-frame option of Prithvi is switched off. For fine-tuning, each ship measurement is paired with cloud-free imagery from a six-day window centred on it.

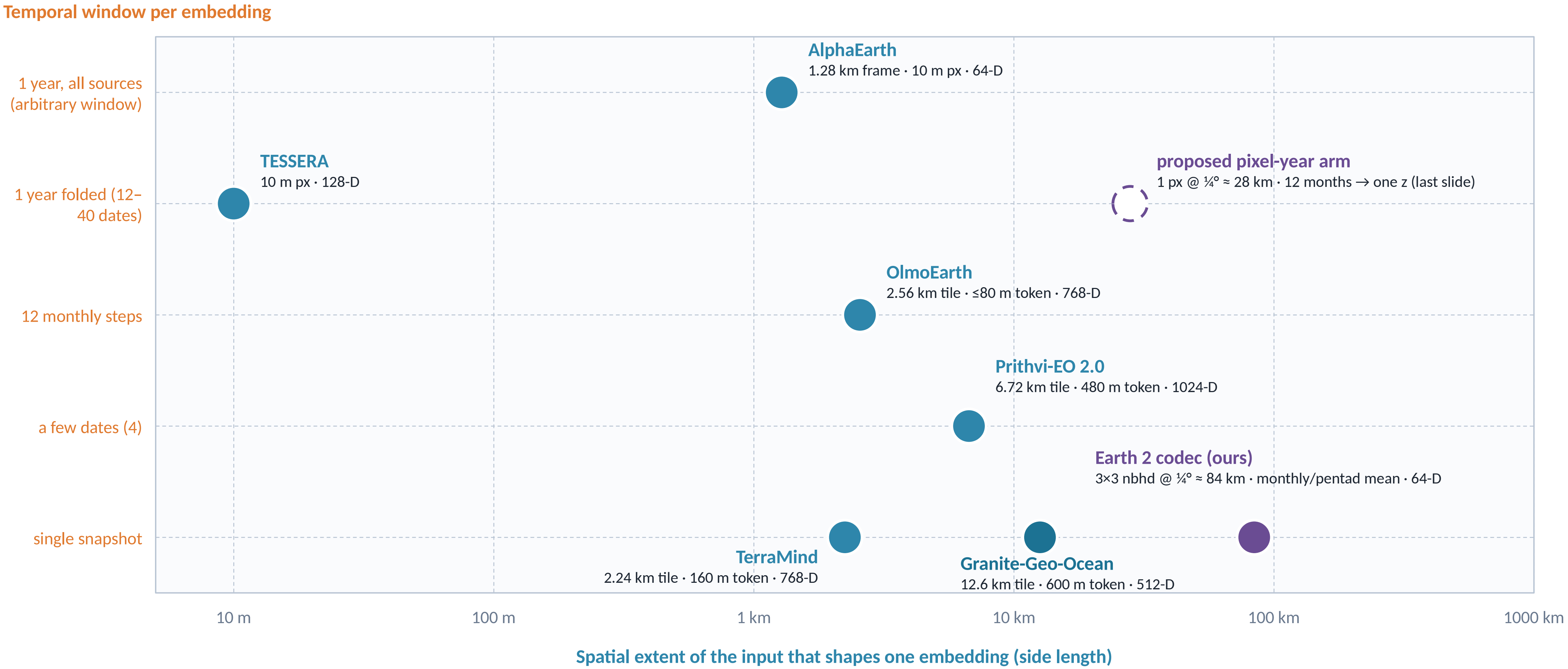
Output: 441 vectors of 512 dimensions per tile. Downstream, a small regression head is trained to predict chlorophyll-a (a proxy for phytoplankton) and primary production (how much carbon the phytoplankton fix).

Results: the numbers are RMSE — root-mean-square error — in log10 units, because these quantities vary over orders of magnitude. With the foundation model: 0.14 for chlorophyll, versus 0.16 training the same network from scratch; 0.39 versus 0.42 for primary production. The gains are small and within the fold-to-fold spread; the authors' stronger claim is that the advantage grows when labels are scarce.

Caveats: surface-only, optical (so cloud-limited), one snapshot in time. It is a model of ocean colour — biology at the surface — not of the ocean's physical state.

Stencils side by side

Spatial footprint that feeds one embedding (log scale) vs. how much time it summarises



Earth 2 shown for reference: the codec's channel token sees a 3 × 3 neighbourhood of ¼° (or 1°) cells at one monthly or 5-day mean — a snapshot stencil ~7× wider than the widest EO tile here (Granite, 12.6 km).

Stencils side by side

This chart puts all six models — and our own — on two axes that summarise the whole deck.

Horizontal axis (logarithmic): the spatial extent of the input that shapes one embedding, from 10 m to 1000 km. Vertical axis (categorical): how much time one embedding summarises, from a single snapshot at the bottom to a full year of all sources at the top.

Reading the points: TESSERA sits at 10 m on the far left — a per-pixel output with no spatial context at all — and AlphaEarth at 1.28 km, the frame each of its 10-m pixels is computed from; both sit at the top because both fold a whole year into one vector. TerraMind and Granite are single snapshots at 2–13 km tile scale. Prithvi stacks four dates; OlmoEarth stacks twelve monthly steps.

Earth 2, our codec, is the purple dot at about 84 km: each channel token sees a 3-by-3 neighbourhood of quarter-degree cells (a quarter degree is about 28 km) at one monthly or five-day mean. It is a snapshot stencil roughly seven times wider than the widest Earth-observation tile here (Granite, 12.6 km). The hollow dot is the proposed "pixel-year" experiment from a later slide, which would move us up to the "one year folded" row.

Two design axes separate all these models: per-pixel versus patch, and snapshot versus period summary. Every later recommendation in the deck comes back to these two axes.

Summary I — what goes into each embedding

Inputs, spatial stencil, temporal stencil, output vector

Model	Inputs (sensors / modalities)	Spatial stencil	Temporal stencil	Output per embedding
AlphaEarth (DeepMind)	S-2, Landsat 8/9, S-1, PALSAR-2, GEDI, ERA5-Land, GRACE, DEM, NLCD, text	10 m output pixel inside a 1.28 km (128 px) frame; attention global within the frame	Period summary; arbitrary valid window in the model, calendar year in the product	64-D unit vector, int8; per pixel per year
TESSERA (Cambridge)	S-2 L2A (10 bands) + S-1 RTC (VV, VH)	Single 10 m pixel — no spatial context	1 calendar year; 40 sampled dates per modality, day-of-year encoded	128-D int8; per pixel per year
OlmoEarth (Ai2)	S-1, S-2, Landsat-8 monthly + static maps (OSM, WorldCover, CDL, SRTM, canopy, WorldCereal)	2.56 km tile (256 px @ 10 m); 1–8 px tokens (10–80 m)	≤12 monthly steps over 1 year, each step tokenised; temporal masking	768-D (Base) / 1024-D (Large) per patch per step + pooled vector
TerraMind (IBM/ESA)	S-2 L2A/L1C, S-1 GRD/RTC, DEM, RGB, NDVI, LULC, captions, geolocation	2.24 km tile (224 px @ 10 m); 16 px = 160 m tokens, 14×14 grid	Single co-registered timestamp	196 × 768-D tokens per modality; merged 768-D; generated modalities
Prithvi-EO 2.0 (IBM/NASA)	HLS v2 at 30 m, 6 bands (blue, green, red, NIR, SWIR1, SWIR2)	6.72 km tile (224 px @ 30 m); 480 m (300 M) or 420 m (600 M) tokens	4 dated frames, 1–6 months apart, tubelet depth 1; single frame allowed	1024-D (300 M) / 1280-D (600 M) per token per frame
Granite-Geo-Ocean (IBM/STFC/PML)	Sentinel-3 OLCI L2, 16 bands + SLSTR SST (17 ch) at 300 m	12.6 km tile (42 px @ 300 m); 2 px = 600 m tokens, 21×21 grid	Single acquisition (num_frames = 1)	512-D per token; regression heads for chl-a, primary production

Summary I — what goes into each embedding

A reference table of what goes into each embedding, condensing the model slides. Column by column:

Inputs: the sensors and data sources. S-1 and S-2 are Sentinel-1 (radar) and Sentinel-2 (optical); HLS is the harmonised Landsat–Sentinel product; OLCI and SLSTR are the Sentinel-3 ocean-colour and temperature instruments.

Spatial stencil: what area feeds one embedding — the output pixel size, the token size (the patch the network reasons about) and the tile size (the block it processes at once).

Temporal stencil: whether the model summarises a period (AlphaEarth, TESSERA), stacks dated frames (OlmoEarth, Prithvi), or takes a single snapshot (TerraMind, Granite).

Output: the vector dimension and what one vector describes — a pixel-year, a token, a tile.

The table is designed to be read across a row to reconstruct a model's stencil, or down a column to compare one design choice.

Summary II — what each project hands you

Weights, precomputed embeddings, licence, ocean coverage, and the intended way to use it

Model	Open weights / code	Precomputed embeddings	Licence	Ocean?	Intended use / evaluation style
AlphaEarth	**No — model and code unreleased**	**Yes — global 10 m annual 2017–2024 in Earth Engine + GCS**	CC-BY 4.0 (data)	**Coastal/inland water only**	Frozen embeddings + linear/shallow heads on sparse labels; 'featurization' replacement
TESSERA	**Yes — weights CC0, code MIT, full pipeline**	**Yes — 10 m annual 2017–2025 via GeoTessera (tile-sampled)**	CC0 / MIT	**Land only (water masked)**	Frozen per-pixel embeddings + MLP/conv heads; label-efficient mapping
OlmoEarth	**Yes — 4 sizes on Hugging Face**	On demand — Studio exports for any AOI; no global archive	OlmoEarth Artifact License (use restrictions)	**Not in pretraining sample**	Both: kNN/linear probes and full fine-tuning; platform for NGOs/agencies
TerraMind	**Yes — tiny/small/base/large + tokenizers**	**No**	Apache 2.0	**Land-stratified TerraMesh**	Fine-tuning via TerraTorch; any-to-any generation; edge/onboard variants
Prithvi-EO 2.0	**Yes — 300 M / 600 M, ±TL**	**No**	Apache 2.0 (HF) / MIT (GitHub) — verify	**Sea-only tiles removed**	Full fine-tuning of encoder + decoder (TerraTorch); GEO-Bench
Granite-Geo-Ocean	**Yes — HF weights + GitHub notebook**	**No**	Apache 2.0	**Yes — open ocean, surface optical + SST**	Fine-tune regression heads on ~100–200 in-situ points; ocean-colour biogeochemistry

Summary II — what each project hands you

The companion table: what each project actually hands you, which is often as important as the architecture.

Open weights: whether the trained network is downloadable, so you can run it on your own data or fine-tune it. AlphaEarth is the exception — no weights, no code.

Precomputed embeddings: whether a ready-made global product exists. AlphaEarth and TESSERA yes; OlmoEarth on demand through its Studio; the others no.

Licence: CC-BY and CC0 are Creative Commons licences (attribution required, or no restrictions); Apache 2.0 and MIT are permissive software licences; OlmoEarth's custom licence has use restrictions. Prithvi's licence is stated differently on Hugging Face and GitHub, hence "verify".

Ocean: green means open ocean is in the training data, red means it is masked out or not targeted. Only Granite is green among the six.

Intended use: "frozen embeddings" means you keep the network fixed and train only a small head on top; "fine-tuning" means you retrain the network for your task; kNN is k-nearest-neighbours, the simplest possible classifier, used to test how good raw embeddings are.

Who embeds the ocean — and how deep?

Representation models whose pretraining contains sea pixels: the published ones stop at the surface; Earth 2 is the one with the interior (checked Aug 2026)

Model (org, year)	Ocean data in pretraining	Open ocean? depth?	Stencil	Embedding / availability	What it is for
Earth 2 codec (ours, 2026)	Gridded ocean state at ¼° / 1°: T and S at depth from Argo (plus pre-Argo gaps marked 'never measured'), SSH, currents, wind and surface forcing; 14–25 channels, monthly or pentad means	**Yes — open ocean, surface to ~2000 dbar interior**	3 × 3 patch per channel token; one pixel-month snapshot; 24-month history in stage 2 with a 222–4444 km stencil	64-D per pixel-month (128-D variant training); weights + embeddings ours to release	Ocean-state field prediction; frozen-embedding probes to AMOC transport (RAPID, MOVE)
Granite-Geospatial-Ocean (IBM · STFC · PML · Exeter, 2025)	Sentinel-3 OLCI 16 bands + SLSTR SST, 300 m; 512 k tiles 2017–2021, balanced over 83 Longhurst provinces	**Yes — surface optical only**	42 px = 12.6 km tile; 600 m tokens; single acquisition	512-D per token; Apache 2.0 weights on HF + TerraTorch notebook	Ocean-colour biogeochemistry: chl-a, primary production (surface optical)
OceanSAR-1 / -2 (Galeio, Apr 2025 / Jan 2026)	Sentinel-1 Wave-Mode SAR vignettes over open sea, ~12 M images 2015–2024, VV, downsampled to 50 m	**Yes — surface roughness only**	256 × 256 px @ 50 m = 12.8 km; single acquisition; DINO / DINOv2 with dynamic dataset curation	ResNet50 2048-D · ViT-S/B; v2 ViT 384-D; v1 Apache 2.0 on HF	Sea-state: TenGeoP geophysical classes, wave height (RMSE 0.63–0.72 m), wind speed (RMSE ~1.4 m s^{-1}), icebergs
Copernicus-FM / Copernicus-Pretrain (TUM, 2025)	S-1, S-2, S-3 OLCI (300 m), S-5P (1 km), DEM on a 0.25° ERA5 grid; 18.7 M images, ~312 k grid cells	**Partial — 'whole land surface and near-land ocean'**	per-modality patches inside 0.25° cells; dynamic hypernetwork accepts any spectral/non-spectral input + metadata	ViT encoders; code, data and weights released (github zhu-xlab); Copernicus-Bench, 15 tasks	Unified multi-sensor EO backbone; coastal seas incidental, not a target
Weather/climate FMs (Prithvi WxC, Aurora, ClimaX, AtmoRep)	Reanalysis grids (MERRA-2, ERA5) whose surface fields exist over ocean cells; Aurora has an ocean-wave fine-tune (HRES-WAM)	**Ocean cells present, but as reanalysis background — SST/ocean state not the learning target**	0.25°–0.5° global grids; 1–2 lead-time steps	Open weights (varies); no ocean embedding product	Atmospheric forecasting; wave forecasting (Aurora)
Location encoders (SatCLIP, GeoCLIP)	Coordinates ↔ imagery/photos; SatCLIP samples S-2 globally, GeoCLIP uses geotagged photos	**Effectively no — sea coordinates unverified or absent**	point location → vector	Open	Geographic priors, not ocean state

Verdict. Two purpose-built ocean encoders exist besides ours — both surface, both single-snapshot, one optical (colour + SST) and one radar (sea state). Everything else masks the sea out or carries it as reanalysis background; forecast-only systems (GLONET, Xihe, WenHai, AI-GOMS, Samudra) learn ocean dynamics but publish forecasts, not reusable embeddings. Earth 2 is, to our knowledge, the only embedding model whose inputs include the ocean interior through time.

Who embeds the ocean — and how deep?

This slide answers a question we kept returning to: which representation models have ever seen the sea in their training data, and how deep do they go?

Earth 2 (ours) is in the first row because it is the only entry whose inputs include the ocean interior — temperature and salinity at depth from Argo floats (the global fleet of autonomous profiling instruments), plus sea-surface height, currents and wind at the surface. Our stencil is a 3-by-3 patch of quarter-degree cells, one monthly or five-day snapshot, with 24 months of history in the second-stage forecasting model. Output: 64 numbers per pixel-month.

Granite-Geospatial-Ocean (previous slide) and OceanSAR are the two published encoders trained deliberately on sea pixels — but both are surface-only. Granite sees ocean colour and temperature; OceanSAR (from the company Galeio) sees Sentinel-1 radar "wave mode" images of the sea surface, about 12 million of them, and predicts sea state, wave height and wind speed.

Copernicus-FM (TU Munich) covers "the whole land surface and near-land ocean" on a quarter-degree grid — coastal seas are incidental. Weather and climate models (Prithvi WxC, Aurora, ClimaX) contain ocean grid cells only because reanalysis fields are global; the sea is background, not the learning target. Location encoders such as SatCLIP work on coordinates, not ocean state.

Verdict: nothing else embeds the ocean interior through time. Forecast systems such as GLONET or Samudra learn ocean dynamics but publish forecasts, not reusable embeddings.

Four interacting spheres — who embeds which

Atmospheric weather · ocean weather (incl. the interior) · ocean biosphere · land biosphere. Nobody represents two of them jointly today.

Sphere	Who embeds it today	Native cadence · memory	Observing geometry	Couples to the others through
a) Atmospheric weather	Forecast FMs: Prithvi WxC, Aurora, ClimaX, GraphCast/AIFS/Pangu (states, not reusable embeddings). AlphaEarth ingests ERA5-Land only as an input.	hours–days · ~10 d predictability; slow modes (ENSO, NAO) months	dense gridded reanalysis; radiances and stations upstream	wind stress, heat & freshwater flux → ocean; precipitation, radiation → both biospheres
b) Ocean weather	Surface only: OceanSAR (sea state), Granite (SST as one band), AIFS-ocean 2026 (SST/SSS/SSH/currents, forecast). Interior: Earth 2 — the only embedding with T, S at depth.	days (surface) to decades (interior) · mixed layer months, interior years; top 2.5 m hold the heat of the whole atmospheric column	gridded SST/SSH from satellites; sparse Argo profiles below; nothing under ice	SST → atmosphere; mixed-layer depth, nutrients, light → ocean biosphere; carbon uptake → land via CO_2 (slow)
c) Ocean biosphere	Granite-Geospatial-Ocean (colour → chl-a, primary production); carbon-sink mappers SOM-FFN, CSIR-ML6; CANYON-B for BGC-Argo — all task models, one FM	days–seasons · blooms weeks, seasonal cycle, decadal sink trends	cloud-gapped optical swaths (Sentinel-3, MODIS); BGC-Argo sparse; pCO_2 ship tracks	phytoplankton ← MLD, SST, wind (physics); CO_2 flux → atmosphere; export → interior carbon
d) Land biosphere / surface	The mainstream: AlphaEarth, TESSERA, OlmoEarth, TerraMind, Prithvi-EO 2.0 — all fold a year, none touch the sea	seasonal–annual · phenology, multi-year land-use memory	dense 10–30 m tiles, frequent revisits; LiDAR, SAR	runoff, dust, nutrients → coastal ocean; evapotranspiration, albedo → atmosphere

1 Ocean interior
the anchor: longest memory, least covered by anyone else — what we already have

→

2 + Ocean colour
same grid, Sentinel-3 recipe exists (Granite); closes physics ↔ biology via MLD, light, SST

→

3 + Atmosphere as forcing
already in our channels as wind/flux; at monthly cadence it is a wide, shallow forcing (cone slide), not a state to roll

→

4 + Land biosphere
last: the mainstream does it well, and the coupling to the ocean is slow (rivers, dust, CO_2)

Three design consequences. (i) Memories differ by orders of magnitude, so each sphere needs its own (v, τ) cone and the coupling variables (SST, fluxes, chlorophyll–light–MLD) are the L-shaped unions. (ii) Biospheres run on the calendar, physics on the clock: a joint model needs period tokens (pixel-year, slide on the TESSERA arm) for c) and d) and dated-state tokens for a) and b). (iii) The observing systems differ in geometry — grids, sparse profiles, cloud-gapped swaths, 10 m tiles — and our masked-vs-never-measured token distinction is the primitive that lets one encoder take all of them.

Four interacting spheres — who embeds which

Yannick's framing of the long-term goal: the Earth system has four interacting "spheres", and eventually we want one representation that spans them all. This slide maps who covers which today and proposes an order for joining them.

a) Atmospheric weather: covered by forecast models (Prithvi WxC, Aurora, GraphCast and others) that produce states, not reusable embeddings. Native timescale hours to days; the atmosphere is predictable for about ten days.

b) Ocean weather: the surface is covered by OceanSAR, Granite and ECMWF's 2026 AIFS ocean extension; the interior only by Earth 2. Timescales run from days at the surface to decades in the deep. A physical fact drives everything: the top two and a half metres of ocean hold as much heat as the entire atmosphere above it, so the ocean is the slow, remembering partner.

c) Ocean biosphere: phytoplankton, ocean colour, carbon uptake. Granite plus task-specific carbon-flux models (SOM-FFN, CSIR-ML6, CANYON-B). Timescales: blooms in weeks, seasons, decadal trends.

d) Land biosphere: the entire mainstream of Earth observation models, all of which fold a year and none of which touch the sea.

Nobody represents two spheres jointly. The proposed sequence: start from the ocean interior (longest memory, least covered by anyone else — what we already have), add ocean colour (same grid, Granite's recipe exists, and it closes the physics-to-biology loop through mixed-layer depth, light and temperature), then treat the atmosphere as a forcing (already in our channels as wind and fluxes), and add land last.

Three design consequences follow: each sphere needs its own speed-and-memory "cone" (introduced on slide 26); biospheres run on the calendar so they want year-summary tokens while physics wants dated snapshots; and the observing systems differ in shape (grids, sparse profiles, cloud-gapped swaths, 10-m tiles), which our distinction between "hidden from the model" and "never measured" tokens is built to handle.

Head-to-head I — frozen embeddings, one protocol

OlmoEarth paper, Table 2: kNN (k = 20, cosine) for single-image classification, linear probe for segmentation and multi-temporal tasks. Five of our six appear.

Model (frozen encoder)	BigEarthNet µF1	So2Sat acc	EuroSAT acc	CropHarvest-Togo S1+S2 acc	m-cashew mIoU	SA-crop mIoU	PASTIS (S2) mIoU	MADOS mIoU	Sen1Floods11 mIoU
OlmoEarth Base (89 M)	62.4	67.7	94.7	**82.0**	32.3	28.9	50.6	67.2	79.2
OlmoEarth Large (308 M)	62.0	**68.2**	**96.3**	79.7	30.9	28.5	**51.8**	66.4	**79.8**
TerraMind Base	63.9	46.7	85.6	77.5	46.0	30.4	40.9	66.0	78.7
TerraMind Large	63.9	47.4	90.0	75.2	50.4	31.2	41.3	67.5	78.4
Prithvi-EO 2.0 300M	51.6	34.7	82.2	–	46.8	24.7	37.2	50.9	–
Prithvi-EO 2.0 600M	51.0	34.0	81.2	–	45.8	26.6	37.5	52.2	–
TESSERA (precomputed)	–	–	–	81.0	–	–	–	–	–
Galileo Base	58.3	55.7	92.8	77.5	28.9	25.3	39.6	**68.4**	79.4
DINOv3-Sat 7B	61.6	50.1	91.3	–	**54.1**	**31.7**	26.3	59.7	–
Panopticon Base	**64.9**	60.5	95.2	76.5	32.7	27.3	30.2	66.1	78.0
Copernicus-FM Base	64.6	50.3	84.7	70.6	32.2	28.4	32.1	63.9	77.6
CROMA Large	59.2	48.2	85.8	81.0	27.0	30.4	42.7	66.4	78.8

Reading the table. Green = best in column (all rows). Shaded rows are the models in this deck; grey rows are context. On frozen features OlmoEarth leads the classification columns (So2Sat, EuroSAT, Sen1Floods11, PASTIS), TerraMind leads BigEarthNet and the cashew/SA-crop segmentation, and Prithvi-EO 2.0 trails on every classification column but is competitive on m-cashew. TESSERA appears once (CropHarvest-Togo, 81.0 — within 1 pp of OlmoEarth Base). AlphaEarth is not in this table (its embeddings are annual and 10 m, so the authors compared it separately — next slides). Caveat: authored by the OlmoEarth team.

Head-to-head I — frozen embeddings, one protocol

The first of four slides with actual benchmark numbers, transcribed from the papers. This one uses the widest comparison available: Table 2 of the OlmoEarth paper, which evaluates many models with a single frozen-embedding protocol.

"Frozen" means the encoder is not retrained; only a trivial classifier sits on top. kNN (k-nearest neighbours) assigns a test image the majority label of its 20 nearest training embeddings — the purest test of embedding quality. "Linear probe" fits a single linear layer, used for segmentation and multi-temporal tasks.

Columns are benchmark datasets: BigEarthNet and So2Sat (land-cover classification), EuroSAT (scene classification), CropHarvest-Togo (crop vs non-crop), m-cashew and SA-crop (crop segmentation), PASTIS (crop parcels through time), MADOS (marine debris and oil), Sen1Floods11 (flood mapping from radar). Scores are accuracy or micro-F1 for classification and mIoU for segmentation — all 0 to 100, higher is better.

Shaded rows are the models in this deck; grey rows are other models for context. Green is the best in each column across all rows.

What it says: on frozen features OlmoEarth leads the classification columns by large margins (So2Sat 68 vs 47 for TerraMind vs 35 for Prithvi), TerraMind leads BigEarthNet and the cashew and SA-crop segmentation, and Prithvi trails on every classification column. TESSERA appears once, on CropHarvest-Togo, within one point of OlmoEarth. AlphaEarth is absent from this table because its embeddings are annual and could not be run under this protocol.

Caveat, which applies to every table in this section: it was authored by one of the contestants.

Head-to-head II — fine-tuning and PANGAEA

Top: OlmoEarth Table 3 (full fine-tuning, encoder unfrozen after 20 % of epochs). Bottom: TerraMind Table 6 (PANGAEA, frozen encoder + UPerNet, mIoU).

Fine-tuned — accuracy / mIoU (OlmoEarth paper, Table 3)

Model (fine-tuned)	BigEarthNet	So2Sat	EuroSAT	m-cashew	SA-crop	PASTIS	MADOS	Sen1Floods11
OlmoEarth Base	72.0	**68.6**	**98.7**	79.8	39.6	64.3	77.8	**79.8**
OlmoEarth Large	72.4	68.1	98.5	80.6	40.8	**66.3**	**81.8**	**79.8**
TerraMind Base	72.6	66.1	97.6	80.9	39.2	59.9	73.2	79.5
TerraMind Large	**74.0**	65.4	97.8	**81.3**	**41.1**	60.9	71.5	79.5
Prithvi-EO 2.0 600M	70.6	64.7	96.8	81.1	38.8	58.6	69.3	–
Galileo Base	69.2	64.7	97.8	78.8	35.7	61.2	71.9	79.7
Copernicus-FM	71.3	66.8	98.5	78.7	33.6	54.6	66.0	78.6
OlmoEarth Base, random init (no pretraining)	61.0	48.9	80.3	43.0	27.5	43.9	45.6	77.0

PANGAEA — mIoU, frozen encoder + UPerNet (TerraMind paper, Table 6; Prithvi appears only as v1.0)

Model	HLS Burn Scars	MADOS	PASTIS-R	Sen1Floods11	FiveBillionPx	DynamicEarthNet	CropType S.Sudan	SpaceNet 7	AI4SmallFarms	Average	Avg rank
TerraMind Large	82.93	**75.57**	**43.13**	90.78	**63.38**	37.89	55.04	59.98	27.47	**59.57**	**3.44**
TerraMind Base	82.42	69.52	40.51	90.62	59.72	37.87	**55.80**	60.61	28.12	58.35	3.94
Prithvi-EO 1.0 100M	83.62	49.98	33.93	90.37	46.81	27.86	43.07	56.54	26.86	51.00	11.00
CROMA	82.42	67.55	32.32	90.89	51.83	38.29	49.38	59.28	25.65	55.29	6.61
DOFA	80.63	59.58	30.02	89.37	43.18	39.29	51.33	61.84	27.07	53.59	8.22
U-Net trained from scratch	**84.51**	54.79	31.60	**91.42**	60.47	**39.46**	47.57	**62.09**	**46.34**	57.58	4.89

Reading. Once fine-tuned, the gap closes: TerraMind Large and OlmoEarth Large split the columns (TerraMind: BigEarthNet, cashew, SA-crop; OlmoEarth: So2Sat, EuroSAT, PASTIS, MADOS) and Prithvi-EO 2.0 600M sits 1–3 pp behind on the classification columns but 8–13 pp behind on PASTIS and MADOS. The random-init row is the size of the pretraining effect itself (+10 to +37 pp). On PANGAEA, TerraMind is the only foundation model that beats a U-Net trained from scratch on average (59.6 vs 57.6); Prithvi-EO 1.0 loses to it. Prithvi-EO 2.0's own GEO-Bench table (600M-TL best on 6 of 12 sets vs DOFA / Scale-MAE / DeCUR / Satlas) contains none of the other models here.

Head-to-head II — fine-tuning and PANGAEA

Two more tables. The top one is OlmoEarth's Table 3: the same models, now fully fine-tuned — the whole network is retrained for each task (the encoder is unfrozen after the first 20 % of training). The bottom one is TerraMind's Table 6 on the PANGAEA benchmark, a community suite of about eleven segmentation, change-detection and regression tasks, evaluated with a frozen encoder plus a standard segmentation decoder (UPerNet).

Top table: once fine-tuned, the gap between models closes. TerraMind Large and OlmoEarth Large split the columns roughly evenly, and Prithvi-EO 2.0 (600 M) sits one to three points behind on the classification columns but eight to thirteen points behind on the PASTIS and MADOS segmentation columns. The last row is the same OlmoEarth architecture trained from random initialisation — no pretraining at all. The distance between that row and the others (10 to 37 points) is the value of pretraining itself.

Bottom table: mIoU per dataset plus an average and an average rank. TerraMind Large is the only foundation model whose average (59.6) beats a U-Net trained from scratch (57.6). Prithvi appears only as version 1.0 and comes last. HLS Burn Scars, FiveBillionPixels, DynamicEarthNet, SpaceNet 7 and AI4SmallFarms are the other PANGAEA datasets.

Not shown: Prithvi-EO 2.0's own GEO-Bench table, because none of the other models in this deck appear in it.

Head-to-head III — the two embedding products

TESSERA paper, Table 1 (frozen embeddings + light head, 100 % labels unless noted) · OlmoEarth paper, Table 7 · an independent Local-Climate-Zone study (Bern)

TESSERA vs AlphaEarth vs frozen RSFMs (TESSERA paper) — F1 / mIoU ↑, RMSE ↓

Model	TreeSatAI-TS F1	Austrian crop F1 (30 % labels)	PASTIS-R mIoU	Austrian crop seg. mIoU	BioMassters RMSE (t/ha)	Borneo canopy RMSE (m)
TESSERA (128-D, frozen)	**77.96**	**82.09**	50.68	**53.12**	**27.43**	**12.21**
AlphaEarth (64-D, frozen)	76.90	56.36	**51.08**	25.70	29.59	16.11
Prithvi-EO (frozen)	65.86	–	34.09	16.25	41.12	19.71
Presto (frozen)	67.81	57.89	–	34.04	–	17.88
Galileo (frozen)	69.44	–	27.92	22.80	38.52	20.48
SkySense (frozen)	69.35	–	32.87	26.82	32.52	16.56
U-Net baseline (frozen features)	71.39	–	30.16	27.00	36.41	17.79

Reading. TESSERA beats AlphaEarth on 5 of these 6 tasks in TESSERA's own table (AlphaEarth edges PASTIS-R by 0.4 mIoU), with the largest margins on pixel-wise crop tasks where per-pixel time series matter most; both precomputed products beat every frozen ViT-style RSFM including Prithvi by wide margins. Against OlmoEarth, AlphaEarth wins only where OlmoEarth is also frozen (Nandi frozen+decoder); once OlmoEarth is fine-tuned it wins all five — and AlphaEarth cannot be fine-tuned. The independent Bern study puts the two products within 1 pp of each other and 2–5 pp above raw composites. AlphaEarth's own paper reports a 23.9 % average error reduction vs the next-best method over 15 evaluations (per-dataset values are charts only), losing on just one (LCMAP land-use change).

AlphaEarth vs OlmoEarth Base (OlmoEarth paper, Table 7)

Model · protocol	Nandi acc	AWF acc	Ecosystem acc	LFMC L1 ↓	Solar farm mIoU
AlphaEarth · kNN	55.6	81.0	60.6	–	–
AlphaEarth · frozen + decoder	66.0	75.9	61.2	23.1	77.5
AlphaEarth · full fine-tune	n/a	n/a	n/a	n/a	n/a
OlmoEarth · kNN	66.2	82.0	59.3	–	–
OlmoEarth · frozen + decoder	62.9	84.0	61.1	19.9	84.8
OlmoEarth · full fine-tune	**82.2**	**86.0**	**62.4**	**17.9**	**86.7**

Independent: LCZ mapping, Bern (arXiv 2606.20034)

Input to Attention U-Net	Accuracy	IoU
Sentinel-1 + 2 composites	0.87	0.77
AlphaEarth 64-D	0.89	0.81
TESSERA 128-D	**0.90**	**0.82**

Head-to-head III — the two embedding products

This slide compares the two "products" — the models whose main deliverable is a precomputed global embedding layer: AlphaEarth and TESSERA.

Left table, from the TESSERA paper: frozen embeddings plus a light head, on six tasks. F1 and mIoU are higher-is-better (arrows up); RMSE — root-mean-square error, in tonnes per hectare for biomass and metres for canopy height — is lower-is-better (arrows down). TESSERA beats AlphaEarth on five of six, with the largest margins on pixel-wise crop tasks where a per-pixel time series matters most (Austrian crop F1 82 vs 56). AlphaEarth edges PASTIS-R by 0.4 points. Both products beat every frozen ViT-style model by wide margins.

Top-right table, from the OlmoEarth paper: AlphaEarth versus OlmoEarth under three protocols. With both frozen, results are mixed. Fine-tuned OlmoEarth wins all five tasks — and the "n/a" row is the point: AlphaEarth cannot be fine-tuned, because only the embeddings, not the network, are released. LFMC is live fuel moisture content, an L1 (absolute) error, lower is better.

Bottom-right, an independent study: mapping Local Climate Zones in Bern with an Attention U-Net fed either raw Sentinel composites, AlphaEarth, or TESSERA. The two products are within one point of each other and two to five points above raw imagery.

AlphaEarth's own paper reports a 23.9 % average error reduction versus the next-best method across 15 evaluations, losing only on land-use change; its per-task values are published as charts without numbers, which is why they are not tabulated here.

Head-to-head IV — the ocean encoders

Granite-Geospatial-Ocean, Table 1 (5-fold CV RMSE, $\log_{10}$ units) · OceanSAR-1, Tables 2–3 (Sentinel-1 wave-mode tasks)

Granite-Geospatial-Ocean — fine-tuned FM vs from scratch vs random forest (RMSE ↓)

Model	Chl-a RMSE $\log_{10}$ mg m^{-3}	Primary production RMSE $\log_{10}$ mgC m^{-2} d^{-1}
OLCI + SST · foundation model, fine-tuned	**0.14**	**0.39**
OLCI only · foundation model, fine-tuned	0.16	**0.39**
OLCI + SST · same net from scratch	0.16	0.42
OLCI only · same net from scratch	0.16	0.43
Random forest on pixel values	0.16	0.40

± spreads: chl-a 0.03–0.10, PP 0.04–0.07 across folds — the FM gain (0.02 / 0.03–0.04) is inside the fold spread; the authors' stronger claim is label-efficiency (gain grows as labels shrink to 25 %) and spatial pattern (SSIM vs the operational L2 product: FM 0.88, scratch 0.82, decision tree 0.68). No comparison to OC4Me-type algorithms is given.

OceanSAR-1 — kNN on frozen features vs other SAR FMs (acc ↑, RMSE ↓)

Model	TenGeoP acc %	Wave height RMSE (m)	Wind speed RMSE (m s^{-1})
OceanSAR-1 ViT-B/8	**83.6**	**0.63**	**1.37**
OceanSAR-1 ViT-S/8	82.1	0.64	1.38
OceanSAR-1 ResNet50	75.5	0.75	1.62
SoftCon ViT-B/14	74.8	1.01	2.08
CROMA ViT-B/8	65.4	0.78	1.95
DOFA ViT-L/16	63.4	0.95	2.43
MoCo ResNet50	60.9	0.83	1.98

With linear-probe / LoRA fine-tuning OceanSAR-1 ViT-B/8 reaches 88.3 % / 0.54 m / 1.29 m s^{-1} (MoCo-FT 86.5 / 0.77 / 1.80). Ocean-only pretraining beats land-trained SAR FMs by 9–23 pp on sea-state classification and cuts wind-speed error by 30–45 %.

Why no cross-model ocean table. The two ocean encoders measure different physics (optical colour vs radar roughness) against different in-situ truths, and neither has been run on a land benchmark or on each other's tasks. The only common denominator with the land models is the protocol — frozen features + small head, and 'FM vs same net from scratch' as the pretraining control — which is also the protocol of our attribution matrix. Our numbers (RAPID r 0.617 pixel / 0.672 patch vs raw 0.613 / 0.659) are the same kind of statement as Granite's 0.14 vs 0.16: a pretraining margin measured against a matched from-scratch control, and in both cases a small one.

Head-to-head IV — the ocean encoders

The two ocean encoders cannot be compared with the land models — no shared benchmark exists — so this slide shows each against its own baselines.

Left: Granite-Geospatial-Ocean, Table 1. Five-fold cross-validated RMSE in log10 units for chlorophyll-a and primary production, comparing the fine-tuned foundation model with the same network trained from scratch and with a random forest on raw pixel values. The foundation model is best in both columns, but the margins (0.02 and 0.03) are smaller than the spread between folds. The authors' stronger claims are label efficiency (the advantage grows as training labels shrink) and better spatial pattern (SSIM — structural similarity — of 0.88 versus 0.82 against the operational product).

Right: OceanSAR-1, kNN on frozen features against other SAR foundation models. TenGeoP is a ten-class geophysical-phenomena dataset (accuracy, higher is better); wave height and wind speed are RMSE in metres and metres per second (lower is better). Ocean-only pretraining beats land-trained SAR models by 9 to 23 points on classification and cuts wind-speed error by 30 to 45 %.

Bottom box: the only thing the ocean and land models share is the protocol — frozen features plus a small head, and "foundation model versus same network from scratch" as the control for the value of pretraining. That is also exactly what our attribution matrix does: our pixel and patch codecs versus raw data at matched receptive field. Our pretraining margin (0.617 vs 0.613 correlation for a single pixel; 0.672 vs 0.659 for the 3-by-3 patch) is the same kind of statement as Granite's 0.14 versus 0.16 — a small one.

Who wins where — a digest of the tables

Rankings restricted to the models in this deck, per protocol; every source table was written by one of the contestants

Protocol · metric family	Ranking among our models	Margin / evidence	Source & caveat
Frozen features · classification accuracy / µF1	OlmoEarth > TerraMind > Prithvi-EO 2.0	So2Sat 68 vs 47 vs 35; EuroSAT 96 vs 90 vs 82; BigEarthNet TerraMind 63.9 edges OlmoEarth 62.4	OlmoEarth Table 2 (own paper)
Frozen features · segmentation mIoU	split: TerraMind (cashew 50.4, SA-crop 31.2) · OlmoEarth (PASTIS 51.8, MADOS 67.2, floods 79.8) · Prithvi last	Prithvi-EO 2.0 competitive only on m-cashew (46.8)	OlmoEarth Table 2
Fine-tuned · accuracy / mIoU	TerraMind-L ≈ OlmoEarth-L > Prithvi-EO 2.0 600M	columns split 3/4 between the two leaders; Prithvi 1–3 pp behind on classification, 8–13 pp on PASTIS/MADOS; pretraining itself worth +10 to +37 pp (random-init row)	OlmoEarth Table 3
PANGAEA · mIoU, frozen + UPerNet	TerraMind-L 59.6 > TerraMind-B 58.4 > U-Net scratch 57.6 > … > Prithvi-EO 1.0 51.0	only GFM above the from-scratch U-Net; Prithvi 2.0 not evaluated	TerraMind Table 6
GEO-Bench-2 · capability ranks (of 14)	Prithvi 600M-TL core rank 5 · TerraMind-L 6 · Prithvi 300M-TL 9 · TerraMind-B 11; TerraMind-L rank 1 on multi-temporal and multi-spectral	ranks only, no raw scores	GEO-Bench-2, arXiv 2511.15658 (independent); no AlphaEarth / TESSERA / OlmoEarth / Granite
Precomputed products · frozen + head	TESSERA > AlphaEarth on 5/6 TESSERA tasks; ≈ tie on Bern LCZ (0.82 vs 0.81 IoU); both ≫ frozen Prithvi	Austrian crop F1 82 vs 56; canopy RMSE 12.2 vs 16.1 m; PASTIS AlphaEarth +0.4	TESSERA Table 1 (own) · LCZ study (independent)
Product vs fine-tunable model	fine-tuned OlmoEarth > AlphaEarth on 5/5; frozen-vs-frozen mixed (AlphaEarth wins Nandi 66.0 vs 62.9)	Nandi 82.2 vs 66.0; solar farm 86.7 vs 77.5	OlmoEarth Table 7 (own)
Ocean · RMSE vs in-situ, FM vs scratch	Granite: 0.14 vs 0.16 (chl-a), 0.39 vs 0.42 (PP) · OceanSAR-1: 83.6 % vs 74.8 % best land-trained SAR FM	Granite gain inside fold spread; OceanSAR gain large	own papers; no shared task with land models

Two structural caveats. No benchmark contains all six models — the widest single table (OlmoEarth's) has five and omits AlphaEarth; the only independent multi-model source (GEO-Bench-2) has two. And no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem; the transferable lesson is the protocol — frozen probe vs fine-tune, and a from-scratch control at matched receptive field, which is what our attribution matrix already does.

Who wins where — a digest of the tables

A one-page digest of the four head-to-head slides: for each evaluation protocol and metric family, how the models in this deck rank, with the margin and the source.

Reading down: with frozen features OlmoEarth leads classification and shares segmentation with TerraMind; fine-tuned, TerraMind Large and OlmoEarth Large tie with Prithvi behind; on PANGAEA TerraMind is the only model above a from-scratch U-Net; GEO-Bench-2 (an independent 2025 benchmark that publishes ranks, not scores) places Prithvi 600M-TL fifth and TerraMind Large sixth of fourteen; among the two products TESSERA beats AlphaEarth on five of six tasks in TESSERA's own table but they tie in an independent study; a fine-tuned OlmoEarth beats AlphaEarth everywhere because AlphaEarth cannot be fine-tuned; and the ocean models are measured only against themselves.

The two structural caveats at the bottom are the honest summary. First, no benchmark contains all six models — the widest single table has five and omits AlphaEarth, and the only independent multi-model source has two. Second, no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem. What transfers is the protocol: frozen probe versus fine-tune, and a from-scratch control at matched receptive field, which our attribution matrix already implements.

What this means for our ocean representation work

Six models, two design axes, one gap

1 Axis 1 — pixel vs. patch

TESSERA proves a per-pixel encoder with zero spatial context is competitive on land; every other model buys context with a ViT patch grid. We have already priced this axis: the attribution matrix reads single-pixel raw 0.613 vs codec 0.617, and 3×3 raw 0.659 vs codec 0.672 (two seeds) — the neighbourhood is worth ≈ +0.05, pretraining ≈ +0.01.

2 Axis 2 — snapshot vs. period

TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our stage-1 codec is a snapshot of a monthly (or 5-day) mean and the month-block codec folds only ~6 pentads. The year-folded design is the untested axis — hence the proposed arm on the last slide.

3 Product vs. model

AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both weights and embeddings. For the ocean nothing precomputed exists — whatever we release is the first embedding-field product for the ocean interior.

4 The gap

Only two published encoders are trained on sea pixels — Granite-Geospatial-Ocean (colour + SST) and OceanSAR (sea-state radar) — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state (T, S, SSH, currents) through time — the territory our codec occupies.

What this means for our ocean representation work

Four takeaways for Earth 2, each tied to something measured.

Axis 1, pixel versus patch. TESSERA shows a per-pixel encoder with no spatial context is competitive on land; every other model buys context with a patch grid. We have already priced this axis in our attribution matrix: reading the RAPID transport with a single-pixel codec gives a correlation of 0.617 (raw data 0.613); with the 3-by-3 patch, 0.672 averaged over two seeds (raw 0.659). So the neighbourhood is worth about +0.05, and pretraining about +0.01. (RAPID is the mooring array at 26.5°N that measures the Atlantic overturning circulation; "correlation" is how well our read-out tracks it.) Two cautions that arrived on 2 September: these are probe numbers, whose seed-to-seed spread in our own record runs from 0.04 to 0.25, so the +0.01 for pretraining cannot be resolved and "consistent with parity" is the honest reading. And the paper was reset the same day, because its rolled forecast numbers came from second-stage heads that had seen the held-out years; the probe numbers quoted here are unaffected, but no forecast number from before 2 September may be quoted.

Axis 2, snapshot versus period. TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our first-stage codec is a snapshot of a monthly (or five-day) mean, and the month-block codec folds only about six five-day bins. The year-folded design is the axis we have not tested — hence the proposed experiment on the next slide.

Product versus model. AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both. For the ocean nothing precomputed exists at all.

The gap. Only two published encoders are trained on sea pixels — Granite and OceanSAR — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state through time. That is the territory our codec occupies.

Proposed arm: a TESSERA-style pixel-year codec

Fold the year, not the neighbourhood — the one TESSERA ingredient the programme has not yet tested

Why not a 1×1 spatial ablation Already done. The attribution matrix has p = 1 and 3×3 at both raw and codec: 0.613 / 0.617 vs 0.659 / 0.672 — single seed, inside the probe noise band (ml/CLAUDE.md §3b): read as parity. The temporal axis is unmeasured.

What TESSERA does that we don't One embedding per pixel-YEAR: a 4-layer temporal transformer + GRU pooling over ~40 dated observations of a single pixel, day-of-year embeddings, Barlow Twins objective (no reconstruction), int8 QAT. Our stage-1 z is one pixel-MONTH; stage 2 then runs over the monthly z's.

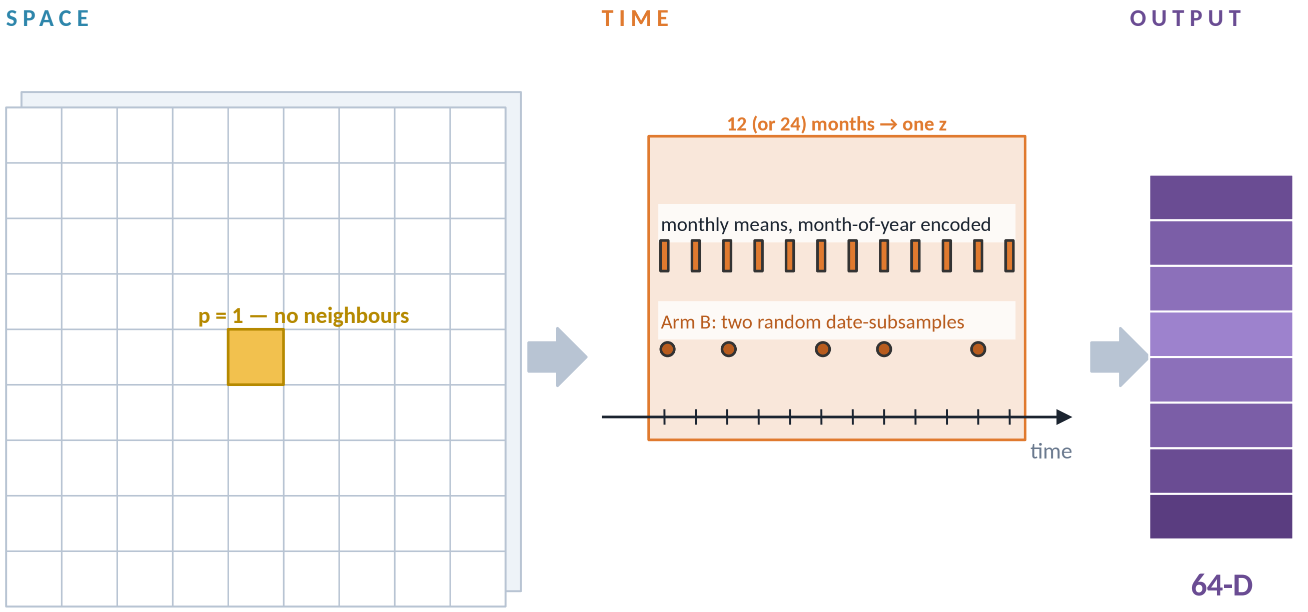
Arm A — pixel-year codec Encoder over the 12 (or 24) pixel-months of one ¼° pixel: channel tokens carry month-of-year; p = 1 to isolate the temporal fold; emit one $z \in \mathbb{R}^{64}$ per pixel-year (or rolling 12-month window ending at t). Decoder queries (channel, Δmonth) up to ±12. Token count ≈ 12 × (C+2) ≈ 310 at 24 channels — same order as the 282-token month-block codec, so it fits the existing recipe.

Arm B — objective swap Same encoder, Barlow Twins on two random date-subsamples of the same pixel-year (TESSERA's augmentation) instead of masked reconstruction. Tests whether a non-reconstructive objective keeps the deep-density detail the decoder had to be re-taught to read.

Controls (matched) Mean of the 12 monthly z's (pooling control) · stage-2 over monthly z's (current path) · raw 12-month stack into the same attention head (end-to-end control, as in the attribution matrix). Two codec seeds minimum — head numbers are never quoted from one.

Read-outs Probe ladder → RAPID r and RMSE (Sv); round-trip variance lost per channel; rolled skill under the corrected pool (MSSS per lead vs climatology and damped persistence; paper v8, 2 Sep 2026) with a year-folded z as the state. Success = beats the control beyond bootstrap intervals, n ≥ 3 seeds.

Stencil — what feeds one embedding



Output pixel: one ¼° cell (≈ 28 km).
Receptive field = the same cell; the 3×3 patch is deliberately switched off so only the time axis changes.
Sheets: the 24–25 ocean channels (T, S, SSH, wind, ...).

A period embedding of the ocean state, TESSERA-style, instead of a snapshot of one monthly mean.
Moves our point on the space × time map from 'single snapshot' to '1 year folded'.

per ¼° pixel per year
(or rolling 12-month window)

Proposed arm: a TESSERA-style pixel-year codec

A concrete experiment proposal, framed honestly. An earlier draft of this deck suggested a "1-by-1 pixel-only control"; it turned out our paper already contains it (the attribution matrix has both single-pixel and 3-by-3 versions of raw and codec inputs). So the spatial question is settled; what TESSERA adds that we have not tried is the temporal fold.

What TESSERA does that we don't: it produces one embedding per pixel-YEAR from about 40 dated observations of one pixel, using a temporal transformer and day-of-year encodings, trained with Barlow Twins (no reconstruction) and quantised to int8 during training. Our first-stage vector z describes one pixel-MONTH; a second-stage model then runs over the sequence of monthly vectors.

Arm A — the pixel-year codec: an encoder that reads the 12 (or 24) pixel-months of one quarter-degree cell at once, with month-of-year tokens, and emits one 64-number vector per pixel-year. We keep the patch size at 1 so that only the time axis changes. Token count about 12 times (channels + 2), roughly 310 at 24 channels — the same order as our existing 282-token month-block codec, so it fits the current training recipe.

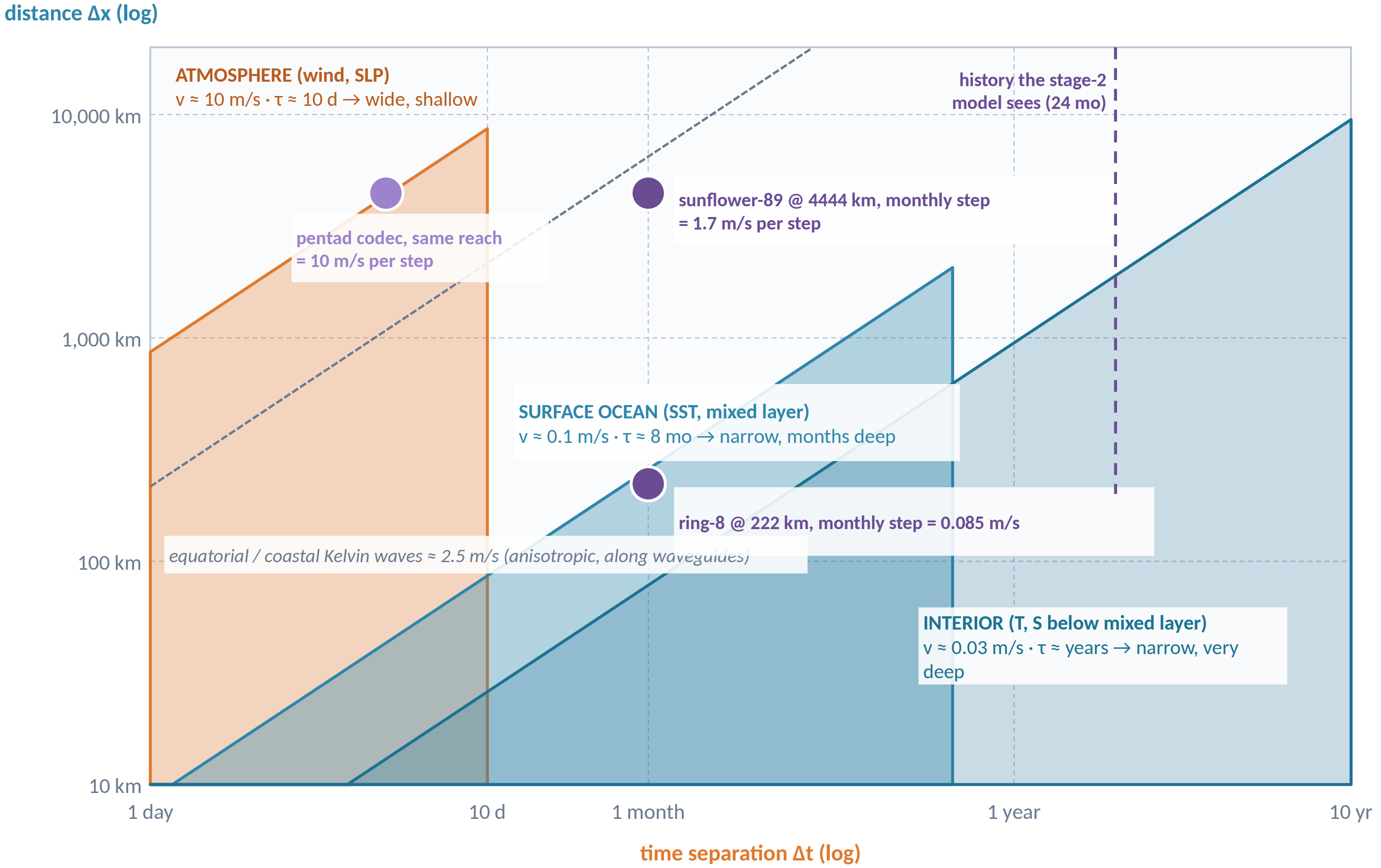
Arm B — swap the training objective: same encoder, but Barlow Twins on two random subsets of dates from the same pixel-year, instead of masked reconstruction.

Controls: the average of the 12 monthly vectors (does folding beat pooling?), the current stage-2 pathway, and the raw 12-month stack fed to the same attention head (the end-to-end control from the attribution matrix). At least two codec seeds, because head numbers from one seed have proven noisy.

Read-outs: our probe ladder to the RAPID transport (correlation and RMSE in Sverdrups), the fraction of variance lost on the round trip, and rolled skill under the corrected protocol — the mean skill against climatology and against damped persistence at each lead, with trained and held-out longitudes reported separately; the old "corridor AUC" name was retired on 2 September 2026 when the paper was reset. Success means the folded vector beats the control beyond block-bootstrap intervals, with at least three seeds.

A velocity-specific dependency cone per channel

What can influence one pixel Δt ahead is bounded by how fast each process propagates (v) and how long it remembers (τ): $\Delta x \le v\cdot\Delta t$, $\Delta t \le \tau$



The model

For a process p with signal speed v_p and memory τ_p, the state at (x, t+Δt) can depend on the past only inside

$$C_p(\Delta t) = \{ (\Delta x, \ell) : \Delta x \le v_p (\Delta t + \ell),\ \ \Delta t + \ell \le \tau_p \}$$

Δt is the forecast step (we predict t+Δt from data up to now, t); ℓ is the lag of an input — it was observed at t−ℓ, so it sits Δt+ℓ before the target (ℓ = 0 is the newest input). $\Delta x \le v\cdot(\Delta t+\ell)$ is the domain of dependence (the CFL / light-cone argument); $\Delta t+\ell \le \tau$ says that beyond the process's own predictability its history is noise, not signal.

The union rule

A channel c is forced by several processes, so its cone is the union: C_c = $\cup_{p\in P(c)} C_p$. Temperature is forced by the atmosphere (fluxes) and by currents (advection), so C_SST = C_atm ∪ C_ocean — wide-and-shallow plus narrow-and-deep, which is not a cone but an L-shape in (Δx, Δt).

Where our stencils sit

A stencil is a cylinder — the same reach at every lag. ring-8 @ 222 km per month is an ocean-advection speed (0.085 m/s); sunflower-89 @ 4444 km is 1.7 m/s, the speed of the fast ocean waveguides and inside the atmosphere's cone at Δt = 1 month. The rolled forecast rebuilds the cone recursively (dependency-cone note, §1), so reach-per-step × horizon is what the target finally sees.

Notation t = now (last observed step) · Δt = forecast step · ℓ = lag of an input (observed at t−ℓ; ℓ = 0 newest) · both in months here.

A velocity-specific dependency cone per channel

This slide introduces the physical idea Yannick proposed: what can influence a pixel Δt ahead is bounded by how fast each process propagates and how long it remembers.

Notation first. t is "now", the last observed step. Δt is the forecast step — we predict the target at $t+\Delta t$. ℓ (a script L) is the lag of an input: it was observed at $t-\ell$, so it sits $\Delta t+\ell$ before the target; $\ell = 0$ is the newest input. v is a process's signal speed; τ (tau) is its memory — how long it stays predictable.

The model: for a process p, an input at lag ℓ can matter only if it is close enough to have travelled to the target in time, $|\Delta x| \le v\cdot(\Delta t+\ell)$, and only if the process still remembers, $\Delta t+\ell \le \tau$. The first condition is the domain-of-dependence or "light-cone" argument from numerical physics (the CFL condition); the second says that beyond a process's predictability its history is noise.

The chart is log-log: time separation on the horizontal axis (1 day to 10 years), distance on the vertical (10 km to 20,000 km), so a constant speed is a straight diagonal. Each process is a region: the atmosphere (amber) is wide but shallow — 10 m/s but only 10 days of memory; the surface ocean (blue) is narrow but months deep — 0.1 m/s, 8 months; the interior (teal) is narrower still but years deep — 0.03 m/s. The dashed line is the Kelvin-wave speed along coasts and the equator, an exception that is fast but anisotropic.

The union rule: a channel forced by several processes has the union of their cones. Sea-surface temperature is forced by the atmosphere (heat flux) and by currents (advection), so its region is wide-and-shallow plus narrow-and-deep — an L-shape, not a cone.

Where our stencils sit: a fixed stencil is a cylinder — same reach at every lag. Our ring-8 at 222 km per month is an ocean-advection speed (0.085 m/s); our sunflower-89 at 4444 km is 1.7 m/s, inside the atmosphere's cone at a one-month step. The dashed vertical line marks the 24 months of history our second-stage model sees.

What the cone predicts — and how to test it on our stencil

Per channel group: dominant processes, the useful reach per monthly step, the useful history, and the falsifiable consequence

Channel group	Processes (v, τ)	Useful reach per monthly step	Useful history	Design consequence
10 m wind, SLP, fluxes	synoptic advection 10 m/s, $\tau \approx 5$–10 d; planetary modes (NAO, ENSO teleconnections) $\tau \approx$ months, forced by SST	causal reach in one month = the globe, but $\tau < \Delta t$: monthly wind is not predictable from its own history — only via slow ocean modes	1 step (its own); the ocean's history through SST	treat as wide, shallow forcing: large stencil, no depth. At pentad $\Delta t \approx \tau$ the wind becomes partly self-predictable — a pentad-only gain to look for
SST, mixed-layer T	surface currents 0.1–0.3 m/s (260–800 km/mo); air–sea forcing (atmospheric cone); mixed-layer memory $\tau \approx 3$–12 mo	advective 300–800 km, plus the atmospheric cone via forcing ≈ synoptic correlation length 1,000–3,000 km	several months (re-emergence up to a year)	union: needs BOTH the wide wind stencil and months of history — the one group that justifies the full cylinder
Interior T, S (below ML)	gyre advection 0.02–0.1 m/s (50–260 km/mo); midlatitude Rossby 0.02–0.05 m/s; $\tau \approx$ years–decades	50–300 km — a 3 × 3 patch at ¼° already covers it; 4444 km is 15–90× too wide for one step	24 months and more; skill should keep rising with history	narrow, deep: small reach, long history. Reach can be cut with no loss — the concrete answer to the open question in the cone note (§5)
SSH	baroclinic Rossby 0.03 m/s (mid-lat) → ~0.3 m/s (10°); Kelvin / coastal waves 1–3 m/s along waveguides; barotropic adjustment in hours (averaged out at monthly Δt)	interior: hundreds of km; along coasts and the equator: thousands of km in one step	months	anisotropic: an along-boundary / along-equator arm on the stencil, narrow elsewhere — a shape the sunflower approximates by brute force
Transport target (RAPID 26°N)	thermal wind = east–west density difference across the basin; set by interior Rossby adjustment (yrs) and by fast boundary waves (weeks)	the whole section, both boundaries	years for the interior arm; the current step for the boundary arm	why the pooled ridge failed and the unpooled attention head worked (paper §ladder): the target's cone is the section, not its mean

Four falsifiable predictions (each an ablation we can run)

1 Group-specific reach: cutting reach to ~500 km for interior T/S slots costs no rolled skill (MSSS per lead); cutting it for wind/SST slots does.
2 Group-specific history: wind slots beyond lag 1 add nothing; interior slots gain from 24 → 48 months.
3 A cone-shaped stencil (reach ∝ Δt + lag) matches the 4444 km cylinder at a fraction of the slots.
4 The pentad stage-2 shows one-step wind skill the monthly one cannot, because $\Delta t \approx \tau\_atm$.

Where the simple picture needs care

v is the fastest relevant signal speed, not the mean flow — waves (Rossby, Kelvin, barotropic) outrun advection. τ is predictability, not causality: the cone is truncated by memory, so wind's 'large area' is bounded by v·τ (~4,000–8,000 km), not by v·horizon. Cones are tilted, not symmetric: upstream along the flow, westward for Rossby waves, along boundaries for coastal waves. Time-averaging to monthly means turns anything with $\tau < \Delta t$ into spatially-correlated forcing — the atmosphere is a stochastic boundary condition at our cadence, not a dynamical state. Eddy diffusion adds only $\sqrt{(K \cdot t)} \approx 50$ km/month at $K \approx 10^3$ m² s^{-1} — negligible at ¼°.

What the cone predicts — and how to test it on our stencil

The cone model applied channel group by channel group, with the useful reach per monthly step, the useful history, and a design consequence for each.

Wind, pressure and fluxes: the atmosphere moves at 10 m/s and forgets in 5–10 days. Its causal reach in one month is the whole globe, but because its memory is shorter than our step, monthly wind is not predictable from its own history — only through slow ocean modes that force it. So treat it as a wide, shallow forcing: large stencil, no depth. At the five-day step the memory and the step are similar, so wind becomes partly self-predictable — a gain only the pentad model could show.

Sea-surface and mixed-layer temperature: advected at 0.1–0.3 m/s (260–800 km per month), forced by the atmosphere, remembered for 3–12 months (including "re-emergence" — anomalies that sink in autumn and reappear next spring). It needs both the wide wind stencil and months of history: the one group that justifies the full cylinder.

Interior temperature and salinity: 0.02–0.1 m/s, memory of years to decades. Per monthly step the useful reach is 50–300 km, which a 3-by-3 patch at quarter degree already covers; our 4444-km stencil is 15–90 times too wide for one step. Small reach, long history — the concrete answer to the open question "can reach come down?" from the project's dependency-cone note.

Sea-surface height: slow Rossby waves in the interior (0.03 m/s at mid-latitudes), fast Kelvin and coastal waves along boundaries (1–3 m/s). An anisotropic stencil — an arm along coasts, narrow elsewhere.

The transport target (RAPID): the overturning transport is set by the east-west density difference across the basin ("thermal wind"), which means its cone is the whole section — the reason our pooled read-out failed and the unpooled attention head worked.

Bottom left: four falsifiable predictions, each an ablation we can run with the existing evaluator. Bottom right: where the simple picture needs care — v is the fastest signal speed, not the mean flow; τ is predictability, not causality; cones are tilted, not symmetric; monthly averaging turns anything faster than the step into forcing.

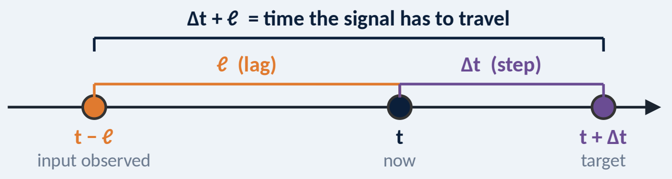
Worked example — the surface current at one pixel

Target: (u, v) at a ¼° pixel P, one month ahead. Which drivers, how each one moves the current, and what (v, τ) it carries. Live version: the Dependency-Cone Explorer artifact.

Driver	Mechanism → current	How it enters the stencil	v	τ	L_corr	Useful lags (Δt = 1 mo)	Reach: lag 0 → last useful lag
Wind stress	Ekman transport & surface drift — responds within ~1 day, locally	direct forcing; $\tau < \Delta t \Rightarrow$ lag 0 only, not itself predictable	10 m/s	10 d	1,500 km	lag 0 only	10,000 (cap) km
Air temperature / heat flux	net heat flux → mixed-layer T → density gradient → geostrophic current	fast driver integrated by a slow medium: inherits $\tau_{ML} \approx 6$ mo and, for old lags, ocean advection speed	10 m/s (atm.) 0.15 m/s (ocean)	6 mo	1,500 km	0 – 5 (6 of 24)	1,889 → 3,833 km
Ocean surface T (mixed layer)	horizontal density gradient → thermal wind	via the gradient ⇒ neighbours needed (≥ 3 × 3); advected at 0.15 m/s	0.15 m/s	8 mo	—	0 – 7 (8 of 24)	389 → 3,110 km
Interior T, S (below ML)	thermal wind integrated from depth; Rossby adjustment	via the gradient; 0.03 m/s ⇒ 78 km per month — the 3 × 3 patch covers lag 0	0.03 m/s	5 yr	—	0 – 23 (24 of 24)	78 → 1,866 km
SSH	pressure gradient → barotropic geostrophic current	via the gradient; Rossby 0.03 m/s westward + Kelvin arm 2.5 m/s along coasts (anisotropic)	0.03 m/s	1 yr	—	0 – 11 (12 of 24)	78 → 933 km
Own history (u, v at P)	momentum persistence, mesoscale eddies (westward drift ~0.03 m/s; eddies live 4+ months but pass through a fixed pixel in weeks–months)	direct; eddy memory ≈ 3 months	0.15 m/s	3 mo	—	0 – 2 (3 of 24)	389 → 1,166 km

Two rules do all the work. $\text{reach}(\ell) = \min(\max(v\cdot(\Delta t+\ell),\ L_{corr}),\ 10{,}000\text{ km})$ — a signal observed ℓ before now has $\Delta t+\ell$ to travel; a field is never less coherent than its own correlation length. A lag is useful iff $\Delta t+\ell \le \tau$ (lag 0 is always kept). One exception: a fast driver acting through a slow medium (air temperature → heat flux → mixed layer) adds instead of maxing — $\text{reach} = L_{corr} + v_{ocean}\cdot(\Delta t+\ell)$ — because the flux pattern is laid down over L_corr and then carried away. Wind is the extreme case: at monthly cadence its memory is shorter than the step, so it enters as one wide sheet of forcing at lag 0 and nothing older carries information about next month's wind. Air temperature is the subtle case: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it delivered for months and carries it along at 0.15 m/s — so the channel inherits the ocean's τ and v.

Notation: the lag ℓ



Worked example — the surface current at one pixel

A concrete walk-through of the cone idea for one target: the surface current (u, v) at a quarter-degree pixel, one month ahead. Each row is a driver — something that pushes the current around — with its mechanism and its two numbers, speed and memory, plus a correlation length.

Wind stress acts through Ekman transport (the wind drags the surface layer, which turns at right angles because of Earth's rotation), within about a day. Because its 10-day memory is shorter than the monthly step, only the newest input (lag 0) is useful, and only as forcing that is not itself predictable.

Air temperature acts through heat flux, which changes mixed-layer temperature, which changes density gradients, which drive geostrophic currents. This is the subtle row: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it received for months and carries it along at ocean speed — so the channel inherits the ocean's memory (6 months) and speed (0.15 m/s). Its reach is the 1,500-km atmospheric correlation length plus what the ocean advects.

Ocean surface temperature drives currents through horizontal density gradients ("thermal wind") — which is why neighbours are mandatory: a gradient does not exist inside one pixel.

Interior temperature and salinity: same mechanism, integrated from depth; at 0.03 m/s (78 km per month) the 3-by-3 patch already covers lag 0.

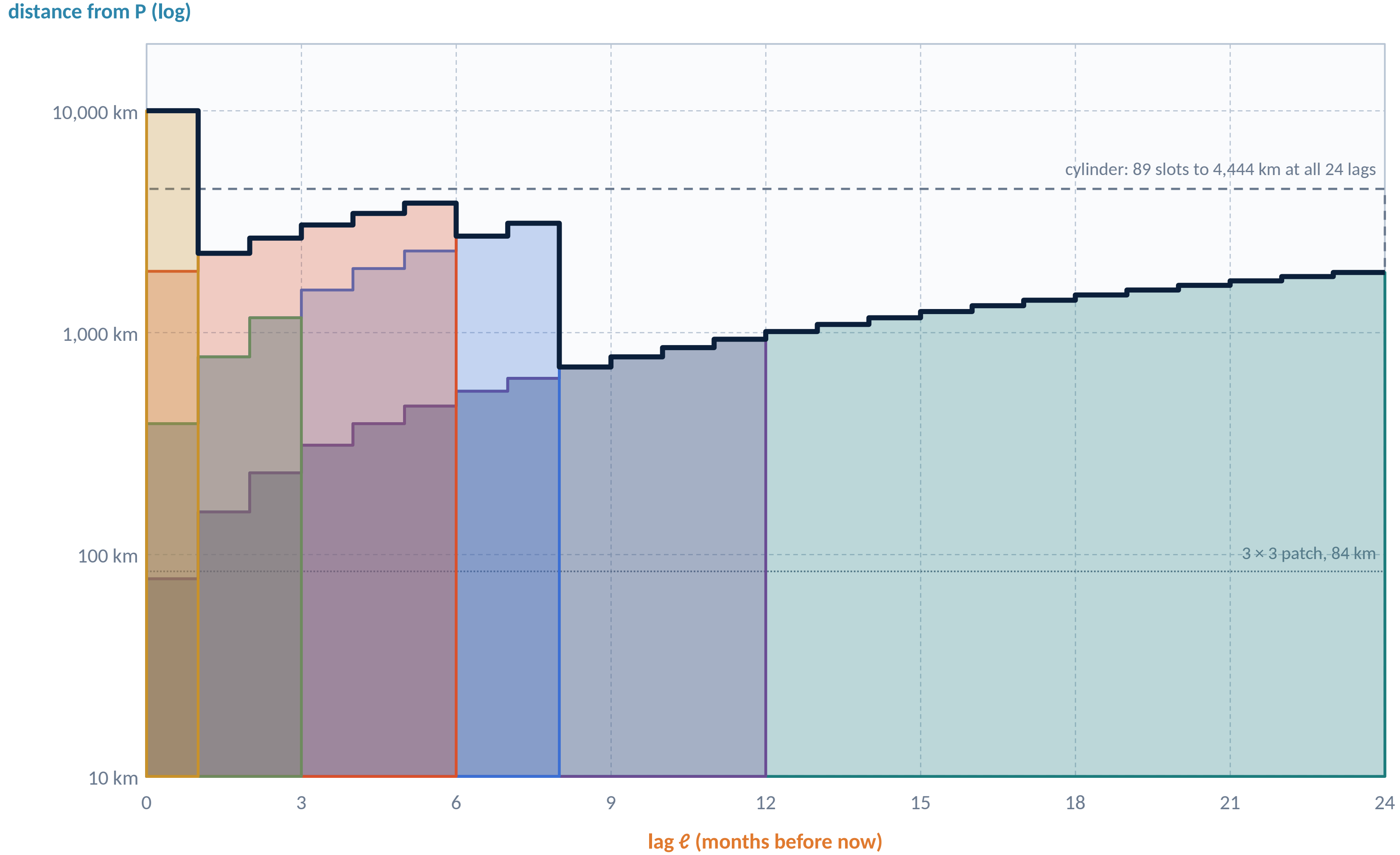
Sea-surface height sets the pressure gradient; slow Rossby waves in the interior plus a fast Kelvin arm along coasts.

Own history: momentum persistence and eddies. Individual eddies live four months and longer and drift westward at a few centimetres per second, but at a fixed pixel the current decorrelates in weeks to a few months because they pass through — that decorrelation time, about three months, is the memory that matters here.

The "two rules" box states the arithmetic used throughout: reach at lag ℓ is the larger of $v\cdot(\Delta t+\ell)$ and the correlation length, capped at 10,000 km; a lag is useful if $\Delta t+\ell \le \tau$. There is one deliberate exception, stated in the box: for a fast driver acting through a slow medium (air temperature → heat flux → mixed layer) the two lengths add rather than compete — the flux pattern is laid down over the atmospheric correlation length and then carried away by the ocean — which is why the air-temperature row reads 1,889 to 3,833 km. The notation panel defines ℓ with a timeline. All parameter values are order-of-magnitude illustrations and can be changed live in the Dependency-Cone Explorer page.

Step 1 — each driver becomes a wedge in (lag, distance)

x: how long before now the input was observed · y: how far from P it may sit and still matter (log). Bold outline = the union the stencil should follow.



Drivers (Δt = 1 month)

Wind stress
lag 0 only · reach capped at 10,000 km

Air temperature / heat flux
lags 0–5 · reach 1,889 → 3,833 km

Ocean surface T (mixed layer)
lags 0–7 · reach 389 → 3,110 km

Interior T, S (below ML)
lags 0–23 · reach 78 → 1,866 km

SSH
lags 0–11 · reach 78 → 933 km

Own history (u, v at P)
lags 0–2 · reach 389 → 1,166 km

Reading. The union is an L: a tall thin spike at lag 0 (the atmosphere, wide but instant), a shoulder out to ~6–8 months (mixed-layer T and the integrated heat flux, 2–4,000 km), then a long low tail to 24 months (interior T/S at ≤ 1,900 km). The cylinder over-covers the tail by 2–50× and cannot cover the lag-0 spike at all.

Step 1 — each driver becomes a wedge in (lag, distance)

Each driver from the previous slide becomes a wedge on this chart. Horizontal axis: lag ℓ, how many months before now the input was observed (0 to 24). Vertical axis (logarithmic): how far from the pixel the input may sit and still matter, 10 km to 10,000 km.

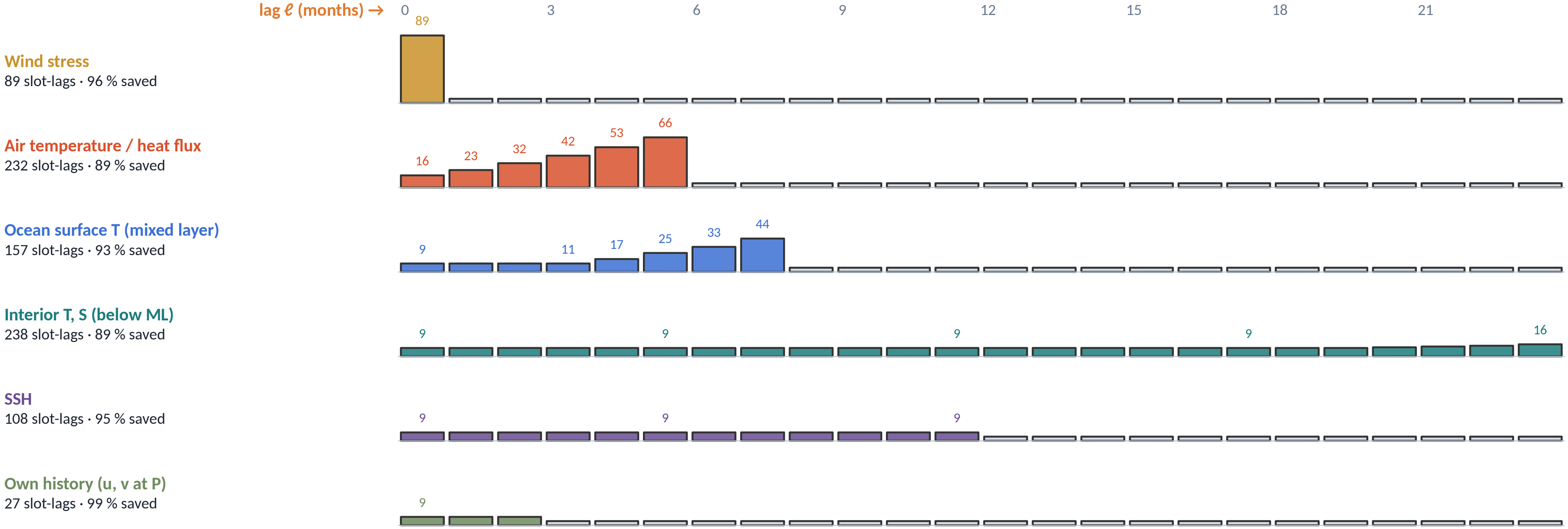
A wedge rises to the right because older inputs have had more time to travel — reach grows as $v\cdot(\Delta t+\ell)$ — and stops abruptly where the driver's memory τ runs out. Wind is a tall spike at lag 0 only. Air temperature is a wide band for six lags. Surface temperature is a true wedge for eight lags. Interior temperature and salinity and sea-surface height are low and long. Own history is short.

The bold black line is the union of all wedges: the region a stencil for this target should follow. Its shape is an L: a spike at lag 0 (the atmosphere, wide but instant), a shoulder out to 6–8 months (mixed-layer temperature and integrated heat flux, 2,000–4,000 km), then a long low tail to 24 months (the interior, under 1,900 km).

Two reference lines: the dashed rectangle is our current "cylinder" — 89 stencil slots reaching 4,444 km at every one of 24 lags; the dotted line at 84 km is the 3-by-3 patch. The cylinder over-covers the tail by a factor of 2 to 50 and cannot cover the lag-0 spike at all.

Step 2 — the union becomes a per-channel, lag-dependent stencil

Slots needed at each lag = 89·(reach / 4,444 km)^2, floored at the 9-slot 3 × 3 patch and capped at 89. Same 89-slot sunflower, different footprint per channel and lag.



851 cone-shaped slot-lags for the six channels vs **12,816** for the cylinder (89 × 24 × 6) → **93.4 % fewer**

Bar height = slots needed at that lag (max 89). Grey stubs = lags outside the driver's memory: dropped. Numbers label the first lag, every sixth, and the first few changes.

What the shapes say. Wind is one full-width sheet at lag 0 and nothing else. Air temperature is a wide sheet for six lags (the mixed layer's memory). Surface T is a true cone: 9 slots at lag 0 growing to 44 at lag 7. Interior T/S is a thin 24-lag column at the 9-slot floor for a year and a half before it widens. SSH is a thin 12-lag column (plus the coastal arm the isotropic count ignores). Own history is three short lags. The cylinder spends 2,136 on each of them; the budget frees ~93 % of the slots for the ones that matter — or lets us afford 48-month history for the interior channels at no extra cost.

Step 2 — the union becomes a per-channel, lag-dependent stencil

This slide turns the wedges into a stencil budget. Our production stencil, the "sunflower-89", spreads 89 slots over a disc of radius 4,444 km and is applied at all 24 lags: 89 × 24 = 2,136 "slot-lags" per channel — the cylinder.

The rule: slots needed for a given reach r scale with area, $89 \cdot (r/4{,}444\ \text{km})^2$, floored at 9 (the 3-by-3 patch) and capped at 89. Each row is a driver; each bar is a lag; bar height is the slots needed at that lag; grey stubs are lags outside the driver's memory, which are dropped.

Wind: one full-width sheet at lag 0, nothing else — 89 slot-lags. Air temperature: a wide sheet for six lags, 232. Surface temperature: a true cone from 9 to 44 slots over eight lags, 157. Interior temperature and salinity: a thin 24-lag column at the 9-slot floor for a year and a half before it widens, 238. Sea-surface height: a thin 12-lag column, 108 (its coastal arm is not counted by this isotropic formula). Own history: three short lags, 27.

Total: 851 cone-shaped slot-lags for the six channels against 12,816 for the cylinder — 93 % fewer. Equivalently, the saving could buy 48 months of history for the interior channels at no extra cost. These numbers follow directly from the illustrative speeds and memories; the Explorer page recomputes them if you change any parameter.

Step 3 — the physics behind each wedge

The stencil fixes the support; attention learns the weights. Four ablations on this target decide whether the support is right.

Ekman — wind → drift

Surface stress τ drives a transport ≈ $\tau/(\rho f)$ at 90° to the wind within ~1 inertial period (≈ 1 day at 30°). Local, fast, shallow (tens of m). At monthly cadence this is same-step forcing: the wind of month t+1 is unknown, the wind of month t is already in the ocean state.

Thermal wind — T, S gradients → shear

$\partial u/\partial z = (g/(f \rho_0))\, \partial \rho/\partial y$: horizontal density gradients set the vertical shear of the geostrophic current, integrated from depth. A gradient does not exist inside one pixel — this is why the 3 × 3 patch is the floor for T/S channels and why heat flux matters through its spatial pattern, not its local value.

Pressure gradient — SSH → surface geostrophy

$u_g = -(g/f)\, \partial \eta/\partial y$. SSH anomalies arrive as slow westward Rossby waves in the interior (0.03 m/s) and as fast Kelvin/coastal waves along boundaries (2.5 m/s): a thin column with one long arm — the one wedge the isotropic count under-states.

Advection & eddies — own history

Momentum persists and mesoscale eddies drift westward at ~0.03 m/s. Individual eddies live 4 months and more (Chelton et al. 2011), but at a fixed pixel the velocity decorrelates in weeks to a few months because they pass through — hence τ ≈ 3 months for own history. Upstream cells at 0.15 m/s × $(\Delta t+\ell)$ carry what will arrive. Short cone, tilted upstream.

What the stencil fixes vs what attention learns

The cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only two physical facts per driver: how fast its signal travels and how long it is predictable. Everything else — which upstream cell matters this month, how strongly the SSH gradient projects onto u, whether the heat-flux pattern is the one that tilts the density field — is left to the attention weights inside the support. A support that is too small cannot be learned around; one that is too large costs slots and lets noise in. The four ablations to the right test whether the support is right, run with the roll battery under the corrected window-scope pool (paper v8, 2 Sep 2026) — never with the retired 'corridor AUC' of the contaminated heads.

The four ablations, specialised to this target

1 Interior T/S slots at 500 km reach instead of 4,444 km: predicted no loss (their wedge never exceeds 1,900 km).
2 Wind and heat-flux slots restricted to lags 0–1: predicted no loss ($\tau_{atm} < \Delta t$); the same cut on surface T: predicted loss (its wedge runs to lag 7).
3 Cone-shaped stencil (851 slot-lags) vs cylinder (12,816): predicted equal rolled skill at 7 % of the slots — and headroom for 48-month interior history.
4 Pentad codec ($\Delta t = 5$ d ≈ τ_{atm}): predicted one-step wind skill that the monthly model cannot show, because wind's own wedge now has lags > 0.

Two codec seeds minimum; head numbers are never quoted from one. Explore the parameters live in the Dependency-Cone Explorer artifact (same formulas, editable v, τ, L_corr, Δt, anisotropy).

Step 3 — the physics behind each wedge

The physics behind each wedge, in four cards, and then what is fixed versus what is learned.

Ekman: wind stress τ drives a surface transport of about $\tau/(\rho f)$ at right angles to the wind — ρ is water density, f the Coriolis parameter set by latitude — within about one inertial period, roughly a day at 30°. Local, fast, shallow. At monthly cadence this is same-step forcing.

Thermal wind: the vertical shear of the geostrophic current is proportional to the horizontal density gradient, $\partial u/\partial z = (g/(f \rho_0))\, \partial\rho/\partial y$. A gradient does not exist inside one pixel, which is why the 3-by-3 patch is the floor for temperature and salinity channels, and why heat flux matters through its spatial pattern rather than its local value.

Pressure gradient: the surface geostrophic current is $u = -(g/f)\, \partial\eta/\partial y$, where η is sea-surface height. SSH anomalies arrive slowly as westward Rossby waves in the interior and fast as Kelvin or coastal waves along boundaries.

Advection and eddies: momentum persists and mesoscale eddies drift westward at a few centimetres per second. The eddies themselves live four months and more, but at a fixed pixel the current decorrelates in weeks to a few months as they pass through — that is the memory used for "own history". Upstream cells carry what will arrive.

What the stencil fixes versus what attention learns: the cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only speed and memory per driver. Everything else — which upstream cell matters this month, how strongly an SSH gradient projects onto the current — is learned by the attention weights inside the support. A support too small cannot be learned around; one too large costs slots and admits noise.

Right box: the four ablations specialised to this target, with the predicted outcomes. They are hypotheses; nothing here has been run. The Dependency-Cone Explorer page uses the same formulas with editable parameters.

Related work I — the dependency cone vs. the literature

Who has used 'speed × time' to size what a model sees, and who has not (survey Aug 2026)

Work	What it does	Shares with the cone idea	Does not
Courant, Friedrichs & Lewy 1928 — domain of dependence	Numerical PDE stability: the numerical stencil must contain the physical domain of dependence $\|\Delta x\| \le v\,\Delta t$.	The origin of the argument; our '$\Delta x \le v \cdot \Delta t$' is exactly this	No notion of memory τ; about stability, not information
OceanNet — Chattopadhyay et al. 2024, Sci. Rep. (arXiv 2310.00813)	Regional FNO for Gulf Stream SSH, 4–5-day mean lead, 80 M params; justifies dropping lateral boundary conditions: 'Rossby waves at 40°N ≈ 1 cm/s … eddies travel only 30–120 km from the open boundary' over the horizon.	Explicit wave-speed × time argument in an ocean ML model	Used to shrink a boundary, not to shape a per-channel stencil; single process
PARADIS 'Learning to Advect' — Pereira et al. 2026 (arXiv 2601.21151)	Neural semi-Lagrangian layer: gathers features at the departure point $x_d = x - \Delta t \cdot u(x)$; stability becomes a Lipschitz condition on u instead of CFL; a 20-parameter NSL layer moves a tracer a 3-layer U-Net's receptive field cannot.	Input support set by velocity × Δt, learned per point — the anisotropic, tilted cone made operational	One transport field for all variables; no τ truncation; atmosphere only
ClimODE — Verma et al., ICLR 2024	Neural ODE with the continuity/advection equation; learns a velocity field, but reads inputs with fixed 3×3 convs plus a global-attention branch.	Advection as inductive bias	Local/global mix fixed, not sized by $v \cdot \Delta t$ per variable
Limited-area ML weather: Neural-LAM (Oskarsson 2023/24), stretched grid (Nipen 2024), Diffusion-LAM 2025	Boundary forcing zone of a fixed 10 grid nodes (~100 km) at every step, or a smooth global→regional mesh.	A boundary halo is a cone cut by the domain edge — our 'window edge' problem	Halo width is static; not derived from wind speed × lead time
Global ML weather: Keisler 2022 (9 GNN hops / 6 h), GraphCast (multi-mesh, 16 layers), Pangu (1/3/6/24 h chain), FuXi (0–5/5–10/10–15 d cascade)	Per-step reach is set by hop count or window size; longer horizons handled by bigger steps or horizon-specific models.	Implicit speed bound per step; growing the cone with lead time by changing Δt	Same reach for every channel; no receptive-field-vs-lead-time ablation published
Per-variable tokenisation: ClimaX 2023, Aurora 2024/25, Prithvi WxC 2024, Stormer	Each variable gets its own embedding / patch tokens, aggregated by attention.	Variable-specific processing as a design pattern	Variable-specific embedding, not variable-specific spatial or temporal extent
Ocean emulators: Samudra 2024/25 (1°, 19 levels, 5-day step, ConvNeXt, atmosphere prescribed), ACE2-SOM (mirror: ML atmosphere + slab ocean), Xihe, GLONET	One medium is treated as prescribed forcing at a cadence set by the other's dynamics.	'Atmosphere as forcing' at slow cadence — our monthly-Δt inversion in practice	Not stated as a τ-vs-Δt rule; no channel-specific stencils
Earth 2 dependency-cone note (16 Aug) + this deck	Cone recursion proved bit-equal to a full roll with NaN poisoning; reach 222 vs 4444 km per month measured as 29.5 % vs 100 % of the world ocean after 12 months; now: per-process (v, τ) cones, union rule, channel-group stencils.	—	Open: the τ-truncation + union → per-channel stencil principle was not found stated anywhere in our search; it is a hypothesis to test (four ablations), not a result

Reading. The 'speed × time' half of the idea has two clear precedents — OceanNet's Rossby-speed boundary argument and PARADIS's semi-Lagrangian gather — and the 'one medium as forcing' half is standard practice in ocean emulators. What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil shape. Absence in a one-day search is not absence in the literature; treat it as 'not yet found', and cite OceanNet and PARADIS as the nearest neighbours.

Related work I — the dependency cone vs. the literature

Where does the dependency-cone idea stand against the literature? Row by row.

The origin is the Courant–Friedrichs–Lewy condition from 1928: a numerical scheme's stencil must contain the physical domain of dependence, $|\Delta x| \leq v \cdot \Delta t$. Our first rule is exactly this; the CFL condition has no notion of memory.

OceanNet (2024), a regional neural forecaster for the Gulf Stream, uses precisely a wave-speed-times-time argument to justify dropping lateral boundary conditions: Rossby waves at 40°N move about 1 cm/s, so eddies travel only 30–120 km from the open boundary over the forecast horizon. Same arithmetic, used to shrink a boundary rather than to shape a per-channel stencil.

PARADIS (2026) is the architectural version: a "neural semi-Lagrangian" layer gathers features at the departure point $x - \Delta t \cdot u$ — where the flow came from — turning the CFL limit into a smoothness condition on the velocity. A 20-parameter layer moves a tracer that a U-Net's fixed receptive field cannot. That is the tilted, anisotropic cone made operational, though with one transport field for all variables and no memory truncation.

ClimODE uses advection as an inductive bias but with fixed local and global branches. Limited-area weather models use fixed boundary halos, not sized by wind speed times lead time. Global models such as GraphCast, Pangu and FuXi set reach per step by hop count or window size and grow it with lead time by changing the step — the same reach for every channel. Per-variable tokenisation (ClimaX, Aurora, Prithvi WxC) gives each variable its own embedding, not its own spatial or temporal extent. Ocean emulators such as Samudra prescribe the atmosphere as forcing — our monthly-step inversion in practice.

What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil. A one-day search is not a literature review; treat it as "not yet found", and cite OceanNet and PARADIS as the nearest neighbours.

Related work II — the four-sphere plan vs. the literature

Coupled emulators, observation-native models, 'Earth-system' foundation models, biosphere ML, coupled data assimilation (survey Aug 2026)

Family · nearest works	What they do	Shares with our plan	Does not
Coupled emulators — Samudra → SamudrACE (Dheeshjith 2025; Duncan 2026), ACE2-SOM (2025), ACE2-NEMO (2026)	Samudra: 3-D global ocean emulator (T, S, currents at depth, 1°, 5-day step); SamudrACE couples it to Ai2's ML atmosphere ACE2. ACE2-SOM/NEMO couple an ML atmosphere to a slab / dynamical ocean.	SamudrACE is the closest structural analogue: an ocean-interior model first, atmosphere joined later	Emulators of a simulator, not embeddings of observations; no ocean colour, no land biosphere
AIFS surface ocean — Hahner, Zampieri et al., ECMWF 2026 (arXiv 2604.25559)	One encoder–processor–decoder forecasts atmosphere + SST, SSS, SSH, surface currents, waves, sea ice 'in a component-agnostic way'; slowly evolving ocean/ice fields get larger loss-scaling factors.	Joint multi-sphere state space; explicit handling of the timescale mismatch	Surface ocean only; forecast model, no reusable embedding; no biosphere
Aurora — Bodnar et al., Nature 2025 ('A foundation model for the Earth system'); Aurora 1.5	One pretrained 3-D Swin backbone, fine-tuned to air quality, ocean waves, cyclones.	One representation, many sphere-specific heads	Atmosphere-centric; no interior ocean, no two-way coupling, no biosphere. Note the title collision
Observation-native — Aardvark Weather (Allen et al., Nature 2025), GraphDOP (ECMWF 2024/25), 4DVarNet (Fablet et al. 2023/24)	Aardvark/GraphDOP learn forecasts or a latent state directly from raw observations, skipping reanalysis; 4DVarNet learns a variational solver that fuses sparse altimetry + SST into gap-free surface fields.	The recipe for taking Argo profiles, swaths and tracks natively — what 'combine all relevant data' needs	Atmosphere (Aardvark, GraphDOP) or 2-D surface ocean (4DVarNet); no Argo-depth encoder found
'Earth-system' FMs — ESFM (Ozdemir et al. 2026, arXiv 2605.00850), Copernicus-FM (2025), Prithvi family (EO, WxC, Surya), AlphaEarth (2025)	ESFM: Aurora-style backbone with per-variable tokens over ERA5, CMIP6, sparse MODIS and station data with missingness handling. Copernicus-FM unifies Sentinel sensors incl. S-5P. Prithvi: separate sphere-specific FMs. AlphaEarth: embedding field with ERA5-Land as an input.	Heterogeneous sources in one embedding; missingness as a first-class token (ESFM) — same primitive as our masked / never-measured tokens	All atmosphere- or land-centric; none has the ocean interior or the ocean biosphere as a target
Ocean biosphere / carbon ML — SOM-FFN (Landschützer), CSIR-ML6 (Gregor 2019), pCO_2-Residual (Bennington 2022), CANYON-B (Sauzède 2017), Granite-Geospatial-Ocean (2025)	Data-driven surface-pCO_2 and carbon-sink maps from SST, SSS, chl, MLD; NN inference of nutrients/carbonate from T, S, O_2 for BGC-Argo; one ocean-colour FM.	Exactly the physics → biology couplings step 2 of our sequence would learn end-to-end	Task models with hand-chosen predictors; no shared representation with the physical state
Coupled data assimilation — Penny & Hamill 2017 (BAMS); Penny et al. 2019 (JAMES); ECMWF CERA	The canonical statement that atmosphere (days) and ocean (months–years) memories differ by orders of magnitude; strongly vs weakly coupled DA; production systems still weakly coupled.	The multi-timescale problem is known and hard; our cone-per-sphere is a representation-side answer to it	Assimilation, not learned representation
Platforms & twins — NVIDIA Earth-2, Destination Earth (ECMWF/CMCC/ESA), EU Digital Twin of the Ocean (EDITO)	Generative km-scale weather tooling (NVIDIA); physics-based coupled km-scale twins (DestinE); ocean data/model platform (EDITO).	Same four-sphere integration goal at production scale	Not learned embeddings; NVIDIA 'Earth-2' is a name collision we must disambiguate

Reading. The nearest lineage on the physics side is Samudra → SamudrACE (ocean interior first, atmosphere second — but emulating a simulator); on the representation side it is ESFM and Copernicus-FM (heterogeneous sources, missingness tokens — but no ocean). Nobody starts from the ocean interior as an embedding, and nobody joins the ocean biosphere to a physical-state representation. Two names to disambiguate in any write-up: NVIDIA's 'Earth-2' platform and Aurora's 'A foundation model for the Earth system'.

Related work II — the four-sphere plan vs. the literature

The four-sphere ambition against the literature, by family.

Coupled emulators: Samudra is a 3-D global ocean emulator (temperature, salinity, currents at depth) trained on reanalysis; SamudrACE (2026) couples it to Ai2's machine-learned atmosphere ACE2. This is the closest structural analogue to our order — ocean interior first, atmosphere joined later — but it emulates a simulator rather than embedding observations, and has no biosphere. ACE2-SOM and ACE2-NEMO couple an ML atmosphere to a slab or dynamical ocean.

ECMWF's 2026 AIFS surface-ocean paper forecasts atmosphere plus sea-surface temperature, salinity, height, currents, waves and sea ice in one "component-agnostic" model, and explicitly gives the slowly evolving ocean and ice fields larger loss weights — the first ML system we found that names and handles the cross-sphere timescale mismatch. Surface only, and a forecast model rather than a reusable embedding.

Aurora (Nature 2025, titled "A foundation model for the Earth system") is one pretrained backbone with sphere-specific fine-tunes; atmosphere-centric, no interior ocean.

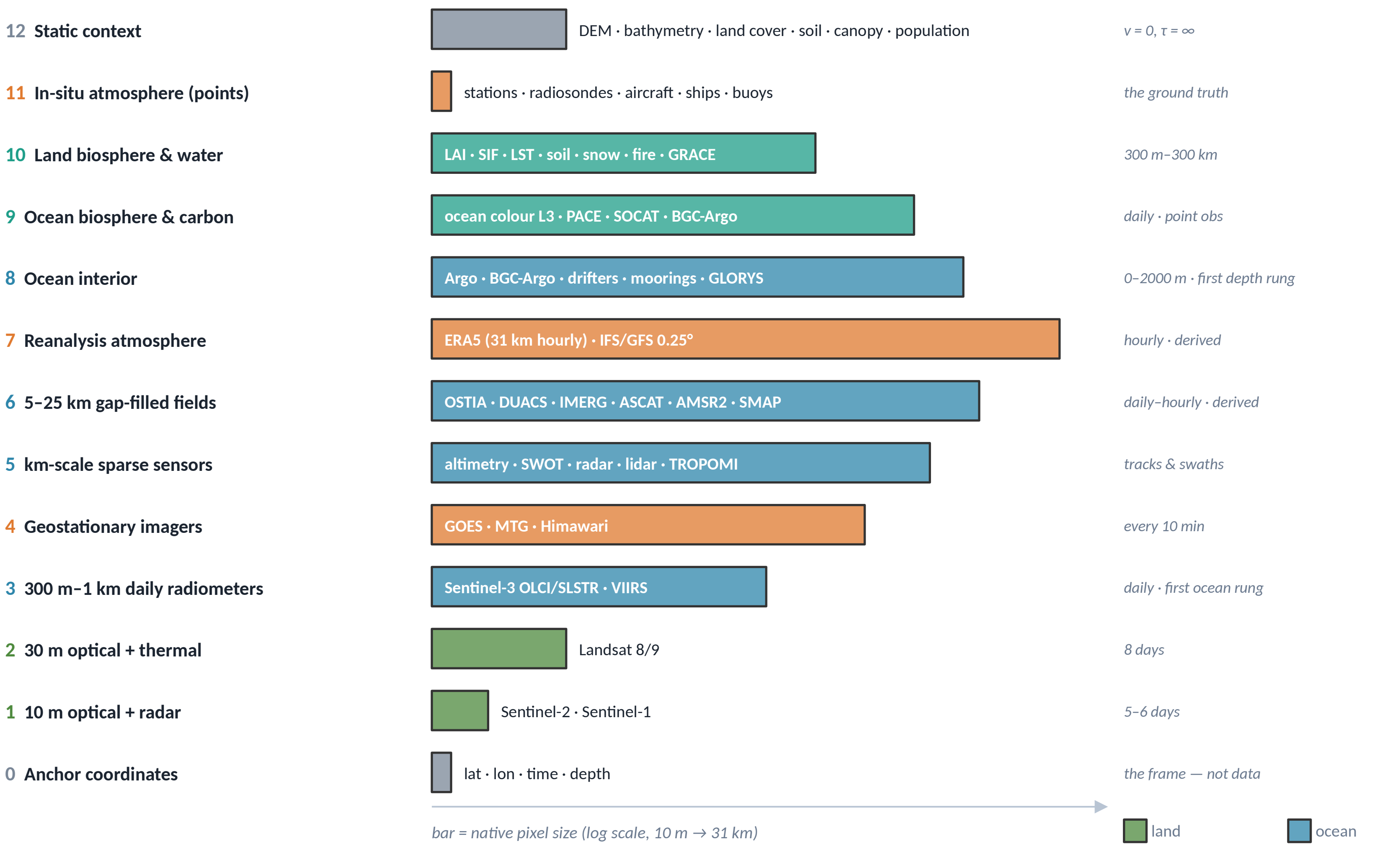
Observation-native learning: Aardvark Weather and GraphDOP learn directly from raw observations, skipping reanalysis; 4DVarNet learns to fuse sparse altimetry and temperature into gap-free surface fields. This is the recipe for taking Argo profiles, swaths and ship tracks natively — what "combine all relevant data" needs — but none of them has an Argo-depth encoder.

"Earth-system" foundation models: ESFM (2026) and Copernicus-FM unify heterogeneous sources with missingness tokens — the same primitive as our "masked" versus "never measured" tokens — but neither touches the ocean. The Prithvi family is a set of separate sphere-specific models; AlphaEarth uses ERA5-Land only as an input.

Ocean-biosphere and carbon ML (SOM-FFN, CSIR-ML6, CANYON-B, Granite) are task models with hand-chosen predictors. Coupled data assimilation (Penny & Hamill 2017) is where the timescale problem was first named. Platforms and twins (NVIDIA Earth-2, Destination Earth, EDITO) share the goal but are not learned embeddings — and NVIDIA's "Earth-2" is a name collision we must disambiguate, as is Aurora's title.

What should a generic Earth embedding see?

The rule: every signal that is relevant to the four spheres at a point and cannot be derived from the other inputs. Thirteen rungs, finest first.



atmosphere · biosphere · static / frame

Three tests for a derived product

Observations (a reflectance, a radar echo, a float's thermometer) are irreducible: always in. A derived product (a gap-filled map, a reanalysis, an ML reconstruction) is admitted only if:

1 it carries information we do not ingest raw — ERA5 holds radiosondes, aircraft, radiances; GLORYS holds a full ocean model around the floats;

2 it is flagged derived and dropped from the input more often, so the model must also work without it;

3 its effective time is shifted by its assimilation window, so it is not a peek into the future (DUACS: ±6 weeks; GLORYS: 7 days; ERA5: 12 h).

Dropped as derivable (targets only): NDVI and band ratios, solar geometry, slope/aspect, geostrophic currents from sea-level slope, per-scene class maps, Stokes drift.

What should a generic Earth embedding see?

This slide opens the last part of the deck: if we want one embedding that is generic — usable for the ocean, the atmosphere, the land and both biospheres — what should it be allowed to see?

The rule at the top is the whole answer in one sentence: every signal that is relevant to the state of the four spheres at a point, and that cannot be derived from the other inputs. "Cannot be derived" is the important half. A vegetation index is a formula applied to two satellite bands; if the bands are in, the index adds nothing. Sunlight angle follows from position and time. The slope of a hill follows from the elevation map. None of those belong in the input — they are fine as things to predict, which costs nothing and gives the model free supervision.

The ladder on the left lists thirteen rungs, finest at the bottom. The bar is the native pixel size on a logarithmic scale, so 10-metre Sentinel imagery is a short bar and the 31-kilometre reanalysis is a long one. Colours are the sphere the rung mostly serves. Two rungs deserve a mention: rung 3, the 300-metre-to-1-kilometre daily radiometers, is the first rung that covers the ocean — the fine imagery of rungs 1 and 2 only sees land and coast. And rung 8 is the first rung that looks below the sea surface.

The box on the right is about derived products — a "reanalysis" is a weather or ocean model that is continuously nudged towards observations, and a "gap-filled map" is an observation product with the clouds interpolated away. They are admitted on three conditions: they must carry information we do not otherwise ingest (ERA5 has weather balloons and aircraft inside it), they are flagged so the model knows they are second-hand and are hidden from the input more often, and their time stamp is shifted by the window of observations they were built from, so they cannot smuggle in the future. That last point gets its own slide.

Input ladder I — fine and fast (rungs 0–4)

Land at 10–30 m, the first ocean-covering rung at 300 m–1 km, and the atmosphere's 10-minute movie

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
0	**Anchor coordinates — latitude, longitude, time of year, time of day, depth**	Where and when we are looking. Not data, but the model needs it: 15 °C means something different in January at 60° N than in July at the equator.	—	The frame of reference; every other input is relative to it.
1	**Sentinel-2 L2A (13 bands, 10/20/60 m, 5-day revisit with two satellites; three in orbit to end-2026) · Sentinel-1 GRD (VV/VH radar, 10 m pixels, 6-day repeat; S1C + S1D)**	Sentinel-2: a colour-plus-infrared photograph of the sunlit surface, corrected for the atmosphere — crops, forests, soil, water, snow, buildings. Sentinel-1: a radar image, day or night, through cloud, of how rough and how wet the surface is — floods, sea-ice edges, ships, soil moisture.	O	Field-scale texture and the fine state of the land biosphere; radar sees through cloud and at night. Land and coast only.
2	**Landsat 8/9 Collection 2 (surface reflectance + surface temperature, 30 m, 8-day combined; archive to 1982)**	The same kind of photograph plus a thermal image — how warm the ground is — at field scale, with a forty-year archive.	O	Thermal at field scale; the long record for phenology and change.
3	**Sentinel-3 OLCI (21 bands, 300 m, < 2 days) + SLSTR (500 m / 1 km, sea- and land-surface temperature, daily) · VIIRS NOAA-20/21 (375/750 m daily; MODIS retiring late 2026/27)**	Daily whole-planet pictures in visible, infrared and thermal. Over the ocean the colour of the water says how much phytoplankton is in it; the thermal channels give the skin temperature of sea and land; hot spots are fires.	O	First rung that covers the ocean surface, and the first daily one: ocean colour (the biosphere), fires, snow, daily land temperature.
4	**GOES-19/18 ABI · Meteosat-12 FCI · Himawari-9 AHI (16 channels, 0.5–2 km, every 10 minutes; full disks to about ± 60–70° latitude)**	A continuous movie of clouds, water vapour and surface radiance from fixed points in the sky; clouds visibly form, move and dissolve between frames.	O	Atmospheric motion at minute scale — the only direct view of the wind field acting on clouds, and of the daily cycle. The fast end of the cone.

Input ladder I — fine and fast (rungs 0–4)

The first table walks up the fine end of the ladder.

Rung 0 is not data at all but the frame: where and when we are looking, and how deep. The model needs it because the same temperature means different things in January at sixty degrees north and in July at the equator.

Rung 1 is the 10-metre imagery: Sentinel-2, which is a colour-plus-infrared photograph of the sunlit surface with the atmosphere corrected out, and Sentinel-1, a radar that works day and night and through cloud, and that measures how rough and how wet a surface is. Between them they give field-scale texture — crops, forests, floods, sea-ice edges, ships. The constellation facts are current as of this week: Sentinel-2 has three satellites in orbit until the end of 2026, and Sentinel-1 is now the C and D satellites, since 1A was switched off at the end of June.

Rung 2 is Landsat, at 30 metres: the same kind of photograph plus a thermal image — how warm the ground is — and a forty-year archive.

Rung 3 is where the ocean appears. Sentinel-3 and VIIRS take a daily picture of the whole planet at 300 metres to a kilometre. Over the ocean the colour of the water tells you how much phytoplankton is in it — that is the ocean biosphere — and the thermal channels give the skin temperature of sea and land. MODIS, the older sensor everyone has used since 2000, is being retired at the end of this year, which is why VIIRS is named here.

Rung 4 is the geostationary satellites — the ones parked over one spot, like GOES over the Americas or Meteosat over Europe and Africa. They take a picture every ten minutes, so they are the only place where you can actually watch the atmosphere move. That is the fast end of the dependency cone.

The "O / D" column says whether each rung is observed or derived; the whole of this first table is observed.

Input ladder II — fields and ocean memory (rungs 5–8)

Sparse kilometre-scale sensors, the gap-filled daily maps, the reanalysis atmosphere, and the first rung below the surface

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
5	**SWOT (2 km swath sea level, 21-day) · nadir altimetry Sentinel-6/Jason-3/Sentinel-3 (~7 km along-track, ~10-day) · Sentinel-1 ocean winds (1 km on swaths) · ground weather radar (MRMS 1 km/2 min; OPERA 1 km/5 min) · GLM lightning (8 km) · GEDI & ICESat-2 lidar (25 m / 11 m footprints) · TROPOMI, OCO-2/3 (trace-gas columns)**	Sea-surface height to a few centimetres along the satellite's track — the ocean's pressure map, whose slopes drive currents; rain seen from the ground every few minutes; where lightning strikes; the height of trees and of sea ice; the gases in the air column.	O	Sea level (hence currents), storm-scale rain, 3-D structure of vegetation and ice, greenhouse-gas columns. All sparse in space or time — the case where a set of dots beats a grid.
6	**OSTIA SST (5 km daily; MUR 1 km) · DUACS sea level L4 (0.125° daily, with geostrophic u, v) · IMERG rain (0.1°, 30 min) · ASCAT winds (12.5/25 km, ≈ twice daily) · AMSR2 sea-ice concentration (10 / 6.25 km daily) · ice drift (62.5 km, 2-day) · SMAP soil moisture (36 km daily; 9 km 3-hourly model) · SSS L4 (weekly) · Copernicus waves (1/12°, 3-hourly, model)**	The cleaned-up daily maps: sea temperature with the clouds filled in, sea level everywhere and not only under the track, rain everywhere, wind over the whole ocean, how much of the sea is ice and how fast it drifts, how wet the top centimetres of soil are.	D*	Completeness and a usable initial state for the slow media; they merge sources we do not ingest raw (AVHRR, microwave SST, every altimeter). Flag, mask heavily, time-shift by their windows. (*ASCAT, SMAP L3 and AMSR2 are swath observations on a grid: O.)
7	**ERA5 (31 km native, 0.25° grid, hourly, 1940–, ~5-day latency): 2 m temperature, 10 m wind, sea-level pressure, precipitation, surface radiation, cloud cover, boundary-layer height, dewpoint, 37 pressure levels · real time: ECMWF IFS/AIFS open data or GFS (0.25°, 6-hourly)**	The best physically consistent estimate of the whole atmosphere at every hour — temperature, wind, pressure, humidity, radiation, clouds, at the surface and aloft — made by running a weather model and continuously nudging it towards millions of observations.	D	The forcing of everything else: heat and momentum into ocean and land; the vertical structure of the air. ERA5's SST and sea ice are its inputs, not outputs — never use them as ocean observations.
8	**Argo core (T, S profiles 0–2000 m every 10 days; 4,372 floats, ~150 k profiles/yr) · Deep Argo (to 4–6 km; 219) · BGC-Argo (O_2, NO_3, pH, chlorophyll, backscatter, light; 989) · moorings (TAO, OceanSITES) · drifters (1,317; velocity at 15 m + SST) · GLORYS12 reanalysis (1/12°, 50 levels, daily, 1993–) · Copernicus 1/12° analysis/forecast (real time)**	What the ocean does below the surface: how warm and salty it is at each depth (heat content, density, the mixed layer), how fast and where it flows, where heat and fresh water are stored. Floats are sparse points; the reanalysis is a model's gridded estimate around them.	O + D	First rung below the surface: heat capacity, density gradients (thermal wind), the AMOC — the slow memory of the system. Only dots do this justice; gridding Argo throws away its depth resolution.

Input ladder II — fields and ocean memory (rungs 5–8)

The second table covers the physical fields and the ocean's memory.

Rung 5 is a collection of kilometre-scale sensors that are sparse — they see along a line or a narrow strip rather than everywhere. The important one for the ocean is altimetry: a radar that measures the height of the sea surface to a few centimetres along the satellite's track. Sea-surface height is the ocean's pressure map; its slopes drive the currents. SWOT is the new altimeter that sees a 2-kilometre-resolution strip instead of a line. The same rung holds ground weather radar, lightning, the laser altimeters that measure tree height and sea-ice thickness, and the satellites that measure greenhouse gases in the air column. Everything here is sparse in space or time, which is exactly the case where a set of points beats a grid — a theme that returns in the proposal.

Rung 6 is the cleaned-up daily maps: sea temperature with the clouds filled in, sea level everywhere and not only under the track, rain everywhere, wind over the whole ocean, sea-ice cover and drift, and the wetness of the top centimetres of soil. Most of these are derived — they merge several sensors and interpolate the gaps — so they are flagged, masked more heavily, and time-shifted. A few in this rung are genuine observations put on a grid, like the ASCAT scatterometer winds.

Rung 7 is the reanalysis atmosphere, ERA5: the best physically consistent estimate of the whole atmosphere at every hour since 1940, made by running a weather model and nudging it towards millions of observations. It is what forces everything else — heat and momentum into the ocean and land. One warning that surprised us in verification: ERA5's sea-surface temperature and sea ice are inputs to ERA5, taken from other products, not things ERA5 estimates, so they must never be treated as ocean observations. For real time, where ERA5 lags five days, the free ECMWF or GFS analyses at a quarter degree fill in.

Rung 8 is the first look below the surface: Argo floats, which drift at depth and surface every ten days to report temperature and salinity down to 2,000 metres — there are about 4,400 of them right now; deep floats to 6,000 metres; biogeochemical floats that add oxygen, nitrate, acidity and chlorophyll; moorings; surface drifters, which give currents directly; and GLORYS, the ocean reanalysis, a model's gridded best guess around all of these. This rung holds the heat capacity, the density gradients and the AMOC — the slow memory of the whole system. The note in the last column matters for the proposal: only points do this justice; gridding Argo throws away its depth resolution.

Input ladder III — biospheres, truth, stage (rungs 9–13)

Ocean carbon, land carbon and water, the point observations the reanalyses are fitted to, and the static boundary conditions

Rung	Input (native resolution · cadence)	What it contains, in plain English	O / D	Adds — what nothing below it has
9	**Ocean colour L3 (OC-CCI / GlobColour, 4 km daily, cloud gaps kept) · PACE OCI (hyperspectral 1.2 km, 2-day, 2024–) · SOCAT surface CO_2 (44 M point obs) · BGC-Argo (rung 8) · Copernicus surface carbon (0.25° monthly, ML) · SeaFlux (1° monthly)**	How much plant life is in the surface ocean and of what kind; how much CO_2 the water holds relative to the air — whether the ocean is absorbing or releasing carbon; oxygen and nutrients at depth.	O + D	The ocean carbon sink and its drivers.
10	**LAI / FAPAR (300 m 10-day) · SIF fluorescence (TROPOMI ~7 km daily) · LST (1 km daily) · evapotranspiration (500 m 8-day; GLEAM 0.1° daily — models) · soil moisture (ASCAT 12.5 km, ~2 h) · snow (500 m daily) · fire (VIIRS 375 m, < 3 h) · water storage (GRACE-FO, ~300 km monthly) · rivers (GRDC gauges; GloFAS 0.05° daily model; Hydroweb heights) · flux towers (half-hourly CO_2, water, heat)**	How much leaf area there is and how actively it photosynthesises (fluorescence is the plants' own faint glow), how much water evaporates, how much snow and soil water is stored, where it burns, how much water sits in rivers, lakes and aquifers.	O + D	The land carbon and water cycles — the fourth sphere. Almost all of it is column-only: it does not move sideways.
11	**Surface stations (ISD > 14,000 active, hourly; GHCN-Daily > 100,000) · radiosondes (~800 in real time, 00/12 UTC) · aircraft (AMDAR) · ships & buoys (ICOADS, NDBC) · tide gauges (GLOSS)**	Thermometers, barometers and anemometers on the ground, balloons through the atmosphere, instruments on ships and buoys, sea-level gauges on coasts — the measurements rung 7 is fitted to.	O	Irreducible ground truth. Redundant while rung 7 is present; essential when it is masked — the model must rebuild the atmosphere from points.
12	**Copernicus DEM (30 m) · GEBCO bathymetry (~450 m) · land cover (WorldCover 10 m; CCI 300 m annual) · SoilGrids (250 m) · canopy height (10 m / 1 m) · glacier outlines · population (100 m) · night lights (500 m) · roads**	The stage: how high, how deep, what kind of surface and soil, how tall the trees, where the people are.	O / D	Boundary conditions: depth sets what currents can do, terrain sets rain and rivers, soil sets what water does. One dot each — $v = 0$, $\tau = \infty$.
13	**Optional: aerosol optical depth (MAIAC 1 km daily) · MERRA-2 aerosols (0.5°, hourly, 3-week latency) · trace-gas columns (rung 5)**	Dust, smoke and pollution in the air, and the greenhouse-gas columns above each point.	O + D	Radiative-forcing detail and the carbon-cycle link between land, ocean and air.

Input ladder III — biospheres, truth, stage (rungs 9–13)

The third table completes the ladder with the two biospheres, the ground truth, and the stage.

Rung 9 is the ocean biosphere and carbon: ocean colour as observed, with the cloud gaps left as gaps; PACE, the new hyperspectral colour sensor launched in 2024; and the carbon measurements — SOCAT, which is forty-four million ship measurements of the CO_2 content of surface water, telling you whether the ocean is absorbing or releasing carbon at that spot; plus the machine-learned monthly carbon maps built on top of it, which are derived and flagged as such.

Rung 10 is the land biosphere and land water: leaf area, fluorescence — plants give off a faint glow when they photosynthesise, and a satellite can measure it — surface temperature, evaporation, soil moisture, snow, fire, the total water stored underground as measured by the GRACE gravity satellites, rivers, and the flux towers that measure carbon and water exchange every half hour at a few hundred sites. Almost everything here stays where it is: soil, plants and snow do not move sideways. That is the column-only cone family on slide 39.

Rung 11 is the ground truth: weather stations, balloons, aircraft, ships, buoys, tide gauges. These are the measurements the reanalysis of rung 7 is fitted to, so they are redundant while the reanalysis is present — and essential when it is masked out, because the model must then be able to rebuild the atmosphere from points.

Rung 12 is the static context: elevation, sea-floor depth, land cover, soil, tree height, glaciers, people. Depth sets what currents can do; terrain sets rain and rivers; soil sets what water does. Each is a single dot with zero speed and infinite memory.

Rung 13 is optional in the first phase: aerosols and greenhouse-gas columns, which matter for radiative forcing and close the carbon loop between land, ocean and air.

If one had to start with one rung per sphere, it would be rung 7 for the atmosphere, rungs 6 and 8 for ocean physics, rung 9 for the ocean biosphere, and rungs 1 and 10 for land — which is the sequencing agreed earlier: ocean interior first, then ocean colour, then the atmosphere as forcing, then land.

Observed, derived, or derivable — and the leakage trap

Why the source of each dot is a token the model sees, and why delayed-time products must be time-shifted

Always in — observations

A number a sensor produced. Reflectances, radar echoes, radar ranges to the sea surface, float thermometers, gauges, towers, balloons. Irreducible: nothing else contains them.

- Sentinel-1/2/3, Landsat, VIIRS, geostationary imagers
- altimeter tracks, SWOT swaths, lidar footprints
- Argo / BGC-Argo profiles, drifters, moorings
- ocean-colour L3 (gaps stay gaps), SOCAT
- stations, radiosondes, ships, buoys, tide gauges
- swath products on a grid (ASCAT, SMAP L3, AMSR2)

Admitted derived — flagged

A number a program produced from observations plus a model or a statistical prior. Admitted when it carries information we do not ingest raw. Every dot carries a source token; derived sources are dropped from the input at ≥ 50 % of anchors, so the model must also work from observations alone.

- ERA5 / IFS / GFS (radiosondes, aircraft, radiances inside)
- GLORYS12, Copernicus 1/12° analysis (full ocean model around the floats)
- OSTIA, MUR, DUACS L4, IMERG, SSS L4 (multi-sensor merges)
- SMAP L4, GloFAS, GLEAM, MOD16 (land models)
- Copernicus carbon, SeaFlux (ML reconstructions)

Dropped — derivable from the ladder

A function of inputs already present. Not an input; an optional target (a masked-channel loss on it costs nothing and gives free supervision).

- NDVI / EVI and other band ratios (rungs 1–3)
- solar zenith, day length, Coriolis f (position + time)
- slope, aspect, distance to coast (rung 12)
- geostrophic u, v and OSCAR (SSH, wind, SST)
- Dynamic World and per-scene classifiers (Sentinel-2)
- Stokes drift (wave spectrum)

The leakage trap: delayed-time products look backwards and forwards. A gap-filled map or a reanalysis for day t is built from observations on both sides of t. If the codec is trained to predict the present or the future from the past, a derived dot at lag ℓ secretly contains information from lag ℓ − (half its window) — sometimes from after the anchor. Windows to respect:

DUACS sea level L4 — delayed-time grids use a centred ±6-week window of tracks (the NRT stream only the past 7 weeks) · GLORYS12 — 7-day assimilation cycle · ERA5 — 12-hour assimilation window · OSTIA / MUR — a window of recent days · ocean-colour L4 — space–time interpolation across cloud gaps (use L3)

Rule: shift each derived source's effective time by its window, never use a delayed-time product at lag 0, and train the forecasting part of the objective on the near-real-time streams, which only see the past (ERA5T, OSTIA NRT, DUACS NRT, the Copernicus analysis). Like ERA5T vs ERA5, this yields two embeddings per anchor: an NRT one (honest for forecasting) and a final one.

Observed, derived, or derivable — and the leakage trap

This slide sorts every input into three bins and then explains a trap that is easy to fall into.

Left, always in: observations — a number a sensor produced. A reflectance, a radar echo, a float's thermometer reading. Nothing else contains them, so they are irreducible.

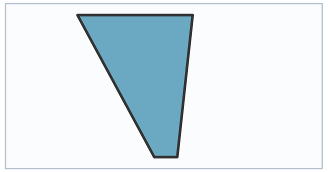
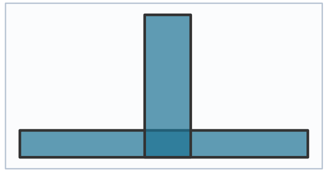
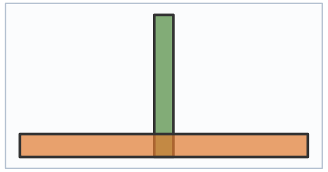
Middle, admitted but flagged: derived products — a number a program produced from observations plus a model or a statistical assumption. Reanalyses, gap-filled maps, machine-learned reconstructions. They come in because they carry information we do not ingest raw, but every dot carries a "source token" saying what kind of thing it is, and derived sources are removed from the input at least half the time. That way the model cannot get lazy and simply copy the reanalysis; it must also be able to work from observations alone.

Right, dropped: things that are pure functions of inputs already present — vegetation indices, sun angle, slope, the currents you can compute from the sea-level slope. They are not inputs. They are optional prediction targets.

The orange band is the trap. A gap-filled map or a reanalysis for a given day is built from observations on both sides of that day. The delayed-time sea-level maps, for example, use satellite tracks from up to six weeks before and after the day in question — only the near-real-time stream is built from the past alone; the ocean reanalysis works in seven-day chunks; ERA5 in twelve-hour chunks. If we train the codec to predict the present or the future from the past, then a derived dot labelled "ten days ago" secretly contains information from a few days ago — or even from after the anchor time. The model would learn to peek. The rule is simple: shift each derived source's effective time by its window, never use a delayed-time product at lag zero, and train the forecasting part of the objective on the near-real-time streams, which by construction only saw the past. This gives two embeddings per anchor — a near-real-time one and a final one — exactly the way ECMWF publishes ERA5T first and ERA5 later.

Five cone families — one shape per rung

Apply $\Delta x \le v\cdot(\Delta t+\ell)$, $\Delta t+\ell \le \tau$ to every input: the shape of the context each channel needs differs by orders of magnitude

	Family	Carrier — what moves the information	v (order of magnitude)	τ — memory at a fixed point	Reach after 1 day · 1 week · 1 month	Ladder rungs	
A	Fast, wide, short	wind, synoptic weather	10 m/s ≈ 860 km/day	3–10 days (rain: hours to 1 day)	860 km · 6,000 km · capped at 10,000 km	4 · 5 (radar, lightning) · 6 (IMERG, ASCAT) · 7 (wind, pressure, cloud)	
B	Slow, narrow, long	ocean currents and eddies; Rossby waves for sea level; Kelvin / coastal waves along boundaries	currents 0.1–0.2 m/s ≈ 13 km/day · Rossby 0.03 m/s ≈ 2.6 km/day westward · coastal 2.5 m/s	eddies, sea level 2–6 months · interior T, S 1–10 yr (thermocline), decades (deep)	13 · 90 · 400 km (currents) · 3 · 18 · 80 km (Rossby)	5 (altimetry, SWOT) · 6 (DUACS, SSS) · 8 (Argo, GLORYS, drifters)	
C	L-shaped union	fast atmospheric forcing on a slow medium: the forcing arm is A, the memory arm is B	A at short lags; B at long lags	SST, SSS 3–6 months (mixed-layer inertia) · sea ice seasonal + multi-year thickness · chlorophyll 1–4 weeks	forcing arm 860 km at lag 0; memory arm 100 km + 13 km/day	3 (SST) · 6 (OSTIA, MUR, ice) · 9 (chlorophyll)	
D	Column-only (land)	nothing horizontal — soil, plants, snow stay put; forced from above by A; rivers carry water downstream (~1 m/s along the network); sea ice drifts 5–10 km/day	0 horizontally (river network ≈ 90 km/day along channels)	surface soil moisture 1–3 weeks, root zone months · vegetation weeks to seasons · snow seasonal · water storage months to years · fire days	L_{corr} ≈ 1–10 km at every lag, plus the atmosphere disc at lags ≤ τ_{atm}	1 · 2 · 10 · part of 12	
E	Static	none	0	∞	one dot	12	

Glyphs: Δx sideways, lag upwards, present at the bottom. Depth is a sixth axis, not a family: the mixed layer (top 10–100 m, deeper in winter) is stirred within hours and belongs to the surface cone; below it memory grows with depth — months at 100–300 m, years at 500–1,500 m, decades below 2,000 m — while vertical speeds are ~1 m/day. Depth dots are log-spaced (0, 10, 30, 60, 100, 200, 400, 700, 1000, 1500, 2000 m), dense near the surface, each band with its own τ; the atmosphere mirrors this at 1000, 850, 500, 250 hPa. Consequence: fine rungs want a small dense patch and a long local history; coarse physical rungs a large sparse far field and a short (A) or thin-and-long (B) history; the interior depth and years. No fixed grid × window serves all three — a per-channel dot sampler does.

Five cone families — one shape per rung

Now we apply the dependency cone from the earlier slides to every rung of the ladder. The recap: a process that travels at speed v can only influence a point from within a distance v times the time elapsed, and only lags shorter than the process's memory tau are worth reading. When we do that for all thirteen rungs, five shapes come out.

Family A, fast, wide and short, is the atmosphere: wind carries information at about ten metres per second, which is 860 kilometres a day, but the atmosphere forgets in three to ten days. So the support is a wide flat disc — you need dots from far away but only from the last few days.

Family B, slow, narrow and long, is the ocean interior and sea level: currents move at a tenth of a metre per second, 13 kilometres a day; sea-level anomalies drift westward as Rossby waves at 3 kilometres a day. But the memory is months for eddies and years to decades for the interior. The support is a thin tall column, tilted upstream.

Family C is the L-shaped union we met before: sea-surface temperature and salinity, sea ice, surface chlorophyll. They are pushed around by the fast atmosphere and remembered by the slow ocean, so the support is wide at short lags and narrow at long lags.

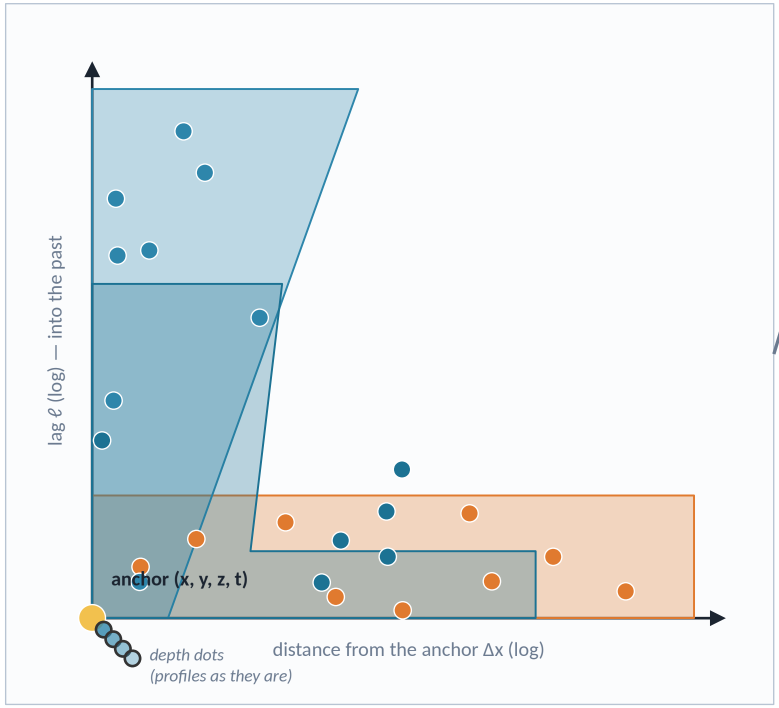
Family D is land. Soil, plants and snow do not move sideways at all; they are forced from above by the atmosphere and remember locally for weeks to seasons. The support is a narrow column with a wide flat hat on top — the atmosphere disc at the shortest lags. Rivers and sea ice are the exceptions that move along a path.

Family E is static: one dot, zero speed, infinite memory.

The paragraph at the bottom adds depth as a sixth axis. The top ten to a hundred metres of the ocean, the mixed layer, are stirred within hours and belong to the surface cone; below it memory grows with depth — months at a few hundred metres, decades below two thousand. So depth dots are spaced logarithmically, dense near the surface, each depth band with its own memory. The consequence for sampling is the point of the slide: fine rungs want a small dense patch and a long local history; coarse physical rungs want a large sparse far field and a short history; the interior wants depth and years. No single grid and window serves all three. A per-channel dot sampler does.

Proposal — a cone-native codec: dots in, embedding out

The codec reads (Δx, Δy, depth, lag, channel, value) samples from each channel's cone and predicts the dots it does not see — including future ones



wind (A) · currents, SSH (B) · SST (C) — each channel samples its own cone

1 Dot sampler (fixed by physics)
Per channel group: stratified log-polar × log-lag draws inside the cone, slot budget from the family. Point, profile and track channels contribute their actual observations. Fine imagery arrives as tokens from the local codec. Missing dots are simply absent.

2 Dot → token
value embedding + Fourier features of (Δx, Δy in km, z in m, ℓ in days — all log) + channel embedding + source flag (observed swath / point / L4 / reanalysis / model / local token).

3 Encoder (Perceiver-style)
K = 128–256 latents cross-attend to N ≈ 2,000–4,000 dots, then 4–8 self-attention blocks over the latents. Cost O(N·K), a few GFLOPs per anchor — below one ViT-Base image forward (≈ 17 GFLOPs).

4 Pool → e_dyn (256-D, int8)
the dynamic embedding: the state at the anchor, with its tendency and its context.

5 Objective: masked dots, including the future

A held-out dot is a query token (coordinates + channel, no value); a light decoder cross-attends from queries to the latents and predicts a distribution (Gaussian or bins; NLL / CRPS). Masking schemes mixed per batch:

- channel drop — remove a whole channel group (derived ones more often)
- lag-band drop — hide the recent past, predict it from the older past = forecasting inside pretraining
- future dots — queries at negative lags; the embedding becomes state + tendency
- sector drop — hide upstream; depth-band drop — predict interior from surface and back
- anchor reconstruction — always predict the anchor's own channels

Never-measured vs masked survives on the target side: no loss where nothing was ever observed. Slow fields up-weighted (AIFS-ocean practice).

Why a set encoder and not a 4-D transformer over (x, y, z, t) grids: the input is ragged — a profile here, a swath there, a 10 m tile in the corner — and the cone is not a box. A set of dots handles both without padding, and retires the 'never-measured vs masked' distinction on the input side. Two products per anchor: an NRT embedding (observations + near-real-time streams) and a final one, exactly as ERA5T and ERA5; because the cone only looks backwards, an embedding is final once its sources have arrived and never needs recomputing.

Proposal — a cone-native codec: dots in, embedding out

This is the proposal itself: a cone-native codec — dots in, embedding out.

Until now the codec embedded one pixel-month, and a second stage assembled a stencil of those embeddings across space and time. Here the codec itself reads a set of dots — each dot is a sample of one channel at some offset in space, some depth, and some lag into the past — drawn from that channel's cone around an anchor point. The picture on the left shows it: the anchor is the yellow dot at the origin; the orange wind dots spread wide but stay near the present; the blue current and sea-level dots stay close in space but reach far back in time; the teal temperature dots fill the L. Depth dots are the small stack at the anchor — a float profile enters as it is, not regridded.

Step 1, the sampler, is fixed by physics. For gridded channels we draw dots in log-polar space and log lag, denser near the anchor and near the present, with the slot budget from the family. For point, profile and track channels — Argo, drifters, stations, altimeter tracks — the dots are the actual observations that fall inside the cone. Nothing is interpolated or invented. Fine imagery arrives as tokens from a small local codec, which the hybrid slide explains. Missing dots are simply absent: a set has no holes to fill.

Step 2 turns each dot into a token: the value, the coordinates as Fourier features on a log scale so that ten metres and ten thousand kilometres fit in the same code, a channel embedding, and the source flag from the previous slide.

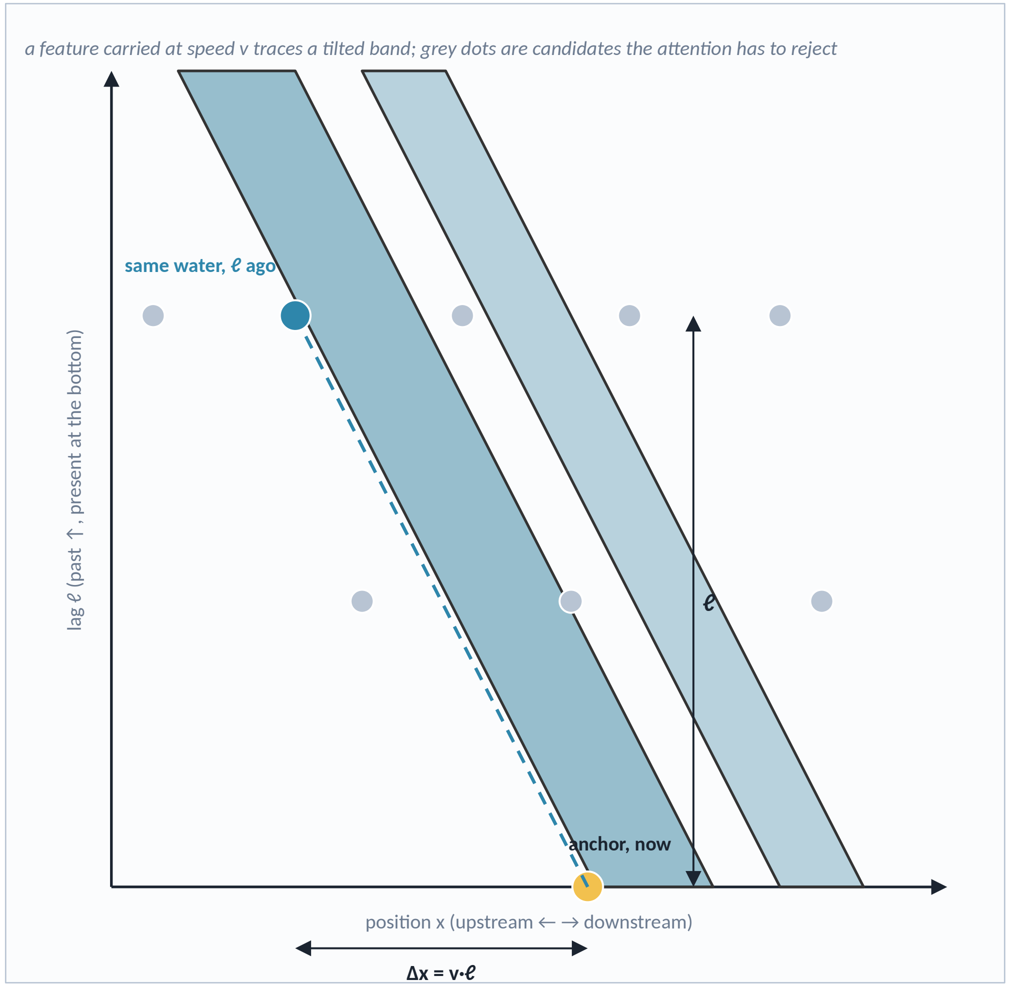
Step 3 is the encoder, Perceiver-style: a couple of hundred learned latent vectors cross-attend to the two to four thousand dots, then a few self-attention blocks over the latents. The cost grows linearly with the number of dots, and per anchor it is a few billion operations — less than running a standard vision transformer on one image.

Step 4 pools the latents into the dynamic embedding, 256 numbers stored as 8-bit integers.

Step 5 is the objective. A held-out dot becomes a query — coordinates and channel without the value — and a light decoder predicts its value as a distribution. We mix several ways of hiding dots: whole channels, as before; the recent past, so that the model must forecast from the older past; dots from the future, so that the embedding learns tendency; an upstream sector; a depth band; and always the anchor's own channels, so the embedding still means "the state at this point". The distinction between "never measured" and "hidden" survives on the target side: there is no loss where nothing was ever observed.

Where velocity comes from — two dots of the same field at two lags

A photograph does not show speed; two photographs a known time apart do. Attention over dots with relative coordinates can implement feature tracking.



The argument

If a field is carried along, $f(x, t + \ell) \approx f(x - v \cdot \ell, t)$: the best predictor of the anchor's value is the dot at displacement $-v \cdot \ell$ at lag ℓ. The cross-correlation between the anchor's neighbourhood now and the field ℓ ago peaks at $\Delta x = v \cdot \ell$. That is the maximum-cross-correlation method oceanographers use to track features in SST images (Emery et al. 1986). A query attending over dots with relative $(\Delta x, \ell)$ encodings can implement exactly this search — and the winning ratio $\Delta x / \ell$ is a velocity readout. Tendency (two lags at one place) and convergence (dots around the anchor) follow the same way. A per-pixel snapshot codec cannot do any of this; it has one lag.

Making sure the capacity is used, not merely available

- Flow as maskable channels — drifter velocities, sea-ice drift, ASCAT / ERA5 wind, river discharge, and (flagged) OSCAR / GLORYS u, v are inputs, so the masked objective asks the embedding to produce velocity whenever the velocity channel is hidden.
- Multiple lags of the same field by construction — the B and C budgets always contain lags > 0, and the cone guarantees the upstream dots are in the set.
- Optional advected-sampling prior — draw a fraction of the dots along $x - \hat{v} \cdot \ell$ using a first-guess velocity (geostrophic u, v; ERA5 wind): PARADIS's semi-Lagrangian gather used as a sampler, not a layer.

The test: a velocity probe

A ridge head from the frozen embedding to drifter velocity or GLORYS u, v at the anchor, with derived channels masked at test time. Prediction: the snapshot embedding scores near climatology ($R^2 \approx 0$); the cone codec does not. This is the cheapest decisive experiment on the list.

Where velocity comes from — two dots of the same field at two lags

This slide answers the question in the request directly: can an embedding capture momentum features — velocity, water flow — if it sees several points in space and time? Yes, and here is why.

A single photograph does not show speed. Two photographs a known time apart do: whatever moved, moved by speed times time. In the diagram, position runs left to right and lag runs upward, the present at the bottom. A feature carried along by the flow — a warm patch, an eddy, a cloud — traces a tilted band. The yellow dot is the anchor now; the blue dot is the same water some time ago, upstream by exactly speed times lag. The grey dots are candidates at the same lag that the model has to reject.

The argument in prose: if a field is carried along, its value here now is best predicted by its value upstream a while ago. The correlation between the anchor's neighbourhood now and the field some time ago peaks at a displacement equal to speed times lag. Oceanographers have used exactly this — the maximum cross-correlation method — since the 1980s to compute currents from pairs of satellite temperature images. Attention over dots that carry their relative position and lag can implement that search, and the winning ratio of displacement to lag is a velocity. The rate of change at one place comes from two lags at that place; convergence comes from dots around the anchor. A per-pixel snapshot codec has one lag and cannot do any of this.

Having the capacity is not the same as using it, so three measures make sure it is used. First, flow enters as maskable channels — drifter velocities, sea-ice drift, wind, river discharge, and the flagged reanalysis currents — so the masked objective directly asks the embedding to produce velocity whenever the velocity channel is hidden. Second, the slow families always contain several lags of the same field, by construction of the cone budgets. Third, optionally, a fraction of the dots can be drawn along the path a first-guess velocity would predict — the semi-Lagrangian idea from the PARADIS paper used as a sampler rather than as a layer.

The test is a velocity probe: a simple linear head from the frozen embedding to drifter velocity or reanalysis currents at the anchor, with the derived channels hidden at test time. The snapshot embedding should score near zero; the cone codec should not. It is the cheapest decisive experiment in the plan.

Is it zero-sum? Thin codec + thick stage 2 vs cone codec + thin stage 2

A: embed one pixel-month, let a stage-2 stencil connect embeddings (today's Earth 2 and every GeoFM in the survey). B: embed the cone, keep stage 2 thin.

	A — thin codec, thick stage 2	B — thick (cone) codec, thin stage 2
Cost per embedding	Minimal: one pixel-month (27 tokens for 3 × 3 × 3 months)	30–100 × more input per anchor (≈ 3,000 dots) — still ≈ one ViT-B image forward
Reusability, caching	Embedding = this place, this time; computed once, never changes	Embedding = this place, this time and its past context; final once its sources have arrived (NRT and final versions)
Interpretability	Clear provenance: this pixel	Provenance is a set of dots; attention maps recover it, with more work
Dynamics: velocity, tendency, transport	Not representable from a snapshot; stage 2 must rebuild it from compressed codes	Representable in the codec (previous slide); supervised by flow channels
Information bottleneck	Codec compresses before it can know what stage 2 needs; sub-pixel phase and small gradients that carry velocity may be dropped as noise — what the code drops, stage 2 cannot recover	Compression happens after the dynamics are visible; the objective (predict future and upstream dots) tells the codec what to keep
Sparse channels: Argo, tracks, stations	A lone profile with no context yields a near-empty code; gridding loses depth resolution	Points enter natively as dots and are contextualised at encode time
Cone geometry	Re-learned per task and per channel by stage 2 (24 × 89 × channels tokens, each time)	Fixed once by physics, shared by all tasks; 7 % of the cylinder's slots
Sample efficiency of probes	Attention head must learn the geometry from labels (AMOC: ~20 years of monthly labels)	Geometry already inside; ridge / linear probes should suffice more often
Static and mapping tasks	Ideal: snapshot semantics (land cover, LCZ, canopy)	Neutral to slightly worse — context can blur a static class; expose the local code alongside
Forecasting in pretraining	Not available: no time axis in the codec	Native: future dots as targets
Training complexity, shortcut risk	Simple per-pixel pipeline; nothing to copy	Heterogeneous dot sampler, bucketing, leakage control; derived channels can be copied — needs source flags and drop rates
Comparability with published GeoFMs	Direct	Indirect: a different unit of embedding

Green = advantage, red = cost. A wins on cost, caching, provenance, simplicity; B on everything that involves time. Next slide: why this is not a symmetric trade.

Is it zero-sum? Thin codec + thick stage 2 vs cone codec + thin stage 2

This is the slide the request asked for explicitly: is the architecture zero-sum? Either we keep the embedding fine-grained and local and let stage 2 make the spatial and temporal connections, or we put that work into the embedding — and does one side's gain equal the other's loss?

Column A is the thin codec with a thick stage 2: embed one pixel-month, then a second-stage model assembles stencils of those embeddings. That is today's Earth 2, and it is what every model in the survey does — they embed images, and the user connects them. Column B is the cone codec from the previous slides with a thin stage 2.

Reading down the rows, green is an advantage and red a cost. A wins on cost per embedding, on caching — an embedding is this place at this time and never changes — on provenance, on simplicity, and on direct comparability with published models. B wins on everything that involves time: dynamics, which a snapshot cannot represent at all; the information bottleneck, because a per-pixel codec has to compress before it can know what the dynamics need, and whatever it throws away as noise — the tiny sub-pixel shifts and gradients that carry velocity — is gone for good; sparse observations, because a lone Argo profile with no context is nearly empty but a profile among its neighbours is not; the cone geometry, which A has to re-learn for every task from labels and B fixes once from physics; sample efficiency of the probes, which matters when the AMOC has only twenty years of monthly labels; and forecasting, which B gets for free inside pretraining.

Two rows are honest costs of B: it is a more complex training pipeline, and there is a real risk that the model copies the derived channels instead of learning — which is what the source flags and drop rates are for. And for purely static tasks like land cover, A's snapshot semantics are actually better.

So the table is not one-sided, but it is also not symmetric — which is the argument of the next slide.

Not zero-sum — the asymmetry, and the split that follows from it

The trade is about where the bottleneck sits relative to the dynamics, not about how much work there is in total

Two facts break the symmetry

1 B contains A. A cone codec that ignores every dot but the anchor's own is a per-pixel codec. Whatever A can represent, B can; the reverse fails because of the bottleneck: information the per-pixel code discards before the dynamics are visible is gone for stage 2 (data-processing inequality). In capability the trade is one-sided; the price is compute and the cleanliness of the embedding's meaning.

2 Compute is paid in different places. A pays little per embedding but re-pays the context assembly in every stage-2 model, for every task, every time. B pays once per anchor and amortises across tasks — provided the cone is generic, which is exactly what a physics-set support promises: it depends on the channel, not on the task.

The principled split

Into the embedding: everything whose relevance is decided by physics — the cone (v, τ per channel), depth coupling, the local past. Identical for every task.

Into stage 2: everything whose relevance is decided by the task — which section to integrate over (AMOC), which basin to sum (carbon), which teleconnection matters (ENSO's Pacific state, 10,000 km and six months away, outside any single cone), which label.

Where A stays right: purely static targets (land cover, soil, canopy height, LCZ), pixel-level anomaly detection, and very-high-volume or on-board inference (Prithvi-EO's in-orbit deployment). Keep the local code for these.

What existing evidence says: nothing yet. The attribution matrix's 3 × 3 codec-vs-raw contrast (0.672 vs 0.659) is a single-seed probe number inside the probe noise band, so it is consistent with parity; and every rolled-skill number from before 2 Sep 2026 was withdrawn with the paper reset (heads trained under the endpoint pool had seen the held-out years). The v8 paper's null ladder — a Linear Inverse Model in pixel space against one in embedding space — is the first measurement of the bottleneck row, and Phase 0 is built to run after it.

The hybrid: split by scale, not by fiat

Raw observations and admitted derived products
the thirteen rungs; point observations stay points; every dot carries a source flag

ladder
rungs 0–13

↓

Local codec — per tile-date, cached, snapshot semantics
the existing per-pixel / 3 × 3 MAE codec generalised to a tile (64 × 64 px at 10 m, STP-style pyramid or FlexiViT tokens); masked-pixel objective; a few 64-D tokens per tile-date reused by every anchor whose cone contains the tile

e_loc 64-D
mapping tasks

↓

Cone codec — per anchor, dynamic semantics
reads local tokens + coarse dots + native point observations inside each channel's cone; masked-dot objective with future dots; the momentum features live here

e_dyn 256-D
prediction tasks

↓

Stage 2 — thin
teleconnections beyond the cone (attention over distant anchors' e_dyn), aggregation along sections and over basins, task heads; the probe ladder stays (ridge → MLP → attention) with the expectation that more tasks stop at ridge

AMOC, carbon,
warming, biosphere

Not zero-sum — the asymmetry, and the split that follows from it

The answer to "is it zero-sum" is no, and the reason is an asymmetry.

First: B contains A. A cone codec that ignores every dot except the anchor's own is a per-pixel codec. Whatever A can represent, B can. The reverse is false, because of the bottleneck: information that the per-pixel code discards before the dynamics are visible cannot be recovered by stage 2 — this is the data-processing inequality, the formal statement that you cannot get information back by processing. So in capability the trade is one-sided; what B pays is compute and a less tidy meaning of "the embedding of this place".

Second: the compute is paid in different places. A pays little per embedding but pays the context assembly again inside every stage-2 model, for every task. B pays once per anchor and amortises it across tasks — provided the cone is generic. And that is exactly what a physics-set support promises: the shape depends on the channel, not on the task.

That gives a principled split rather than a compromise. Into the embedding goes everything whose relevance is decided by physics — the cone, depth coupling, the local past — because that is identical for every task. Into stage 2 goes everything whose relevance is decided by the task: which section to integrate for the AMOC, which basin to sum for carbon, which teleconnection matters — El Niño's Pacific state is ten thousand kilometres and six months away, outside any single cone — and which label.

The right side shows the resulting hierarchy. Raw observations at the bottom. A local codec, per tile and date, cached, with snapshot semantics — the existing per-pixel codec generalised to a tile — producing a small local code that mapping tasks use directly. A cone codec per anchor, which reads local tokens plus coarse dots plus native point observations inside each channel's cone, and produces the dynamic code where the momentum features live. And a thin stage 2 for teleconnections, aggregation and task heads.

What does existing evidence say? Nothing decisive yet, and it is worth being precise about why. The attribution matrix's contrast between a 3-by-3 codec and the same 3-by-3 raw input — 0.672 against 0.659 — is a single-seed probe number, and our own record shows probe numbers moving by far more than that between seeds, so the honest reading is parity. And on 2 September the paper was reset: every rolled forecast number from before that date was withdrawn, because the second-stage heads had been trained under a pool that had seen the held-out years. The first clean measurement of the bottleneck question is the one the reset paper puts first on its list — a Linear Inverse Model fitted in pixel space against one fitted in the codec's embedding space. If the embedding-space model is worse, the codec has discarded predictable signal, which is exactly column A's weakness. Phase 0, next slide, is built to run after that measurement exists.

Phase 0 first — the decisive experiment fits the data we already have

Then one sphere per phase, in the agreed order: ocean surface + atmosphere as forcing, interior, ocean biosphere, land, fast atmosphere

Phase 0 — the A/B on today's tokens (weeks). Same tokens the stage-2 pool reads today: Earth 2 ocean pixel-months, 3 × 3 codec, sunflower-89 × 24 months. Train a cone codec over those tokens with channel drop + lag-band drop + anchor reconstruction. Compare at equal total context under the corrected window-scope pool (paper v8, 2 Sep 2026), after its null ladder exists, with $n \geq 3$ seeds per arm:

A current codec + stage-2 attention head · B cone codec + ridge probe · B´ cone codec + attention head

Verdict metric: rolled skill — MSSS vs climatology and vs damped persistence per lead, trained and held-out longitudes separately, block-bootstrap intervals, the LIM null beside every number. Diagnostics (not verdicts): a velocity probe (GLORYS u, v at the anchor; drifters where available; derived channels masked at test) and the RAPID transport probe.

Predictions: B ≥ A on rolled skill with a linear head, and B beats the embedding-space LIM where A does not; velocity probe B ≫ A; static probes equal. If B needs B´ to match A, the 'work in the embedding' argument loses on sample efficiency and the hybrid keeps a thick stage 2. Effects inside the tier's replicate band are written as consistencies, never levels.

Phase	Sphere	What enters	New probes
1	Ocean surface + atmosphere as forcing	rungs 6 + 7 as gridded dots (OSTIA, DUACS, ASCAT, IMERG, AMSR2 ice; ERA5 / IFS open data); L-shaped and A-family cones; pentad anchors at 25 km	SST and SSH forecast skill at 1–4 weeks vs persistence and vs the Copernicus 1/12° forecast; sea-ice edge
2	Interior	Argo core / Deep / BGC and moorings as native dots; GLORYS flagged derived, dropped 50 %; depth-band drop objective	AMOC again; mixed-layer depth; heat content 0–700 m
3	Ocean biosphere	ocean-colour L3 (gaps stay gaps), PACE, SOCAT, BGC-Argo variables	carbon-flux reconstruction vs SeaFlux / Copernicus carbon (held out, not input)
4	Land	local codec on Sentinel-2/-1 and Landsat tiles (rungs 1–2); rung-10 fields as dots; column cones with the atmosphere hat	LAI / SIF and soil-moisture forecasts, fire risk; standard GeoFM benchmarks on e_loc for comparability
5	Fast atmosphere	geostationary imagers and ground radar at 10-min cadence through a second local codec (spatio-temporal tubelets); family-A disc at hourly Δt	nowcasting skill; wind-driven SST and ice response

Ablations that run in every phase

1 cone-in vs cone-on-top at equal context
2 drop lag bands
3 drop the far field
4 drop depth
5 derived channels masked at test
6 cone × 2 vs cone × ½ on v and τ

Engineering first: a pre-materialised cone index per channel family (which tiles, dates and profiles fall in which anchors' cones). The sampler, not the network, is the bottleneck — the network is under an hour per year of 25 km pentad ocean embeddings at 100 TFLOP/s, about a day for 1 km monthly land.

Phase 0 first — the decisive experiment fits the data we already have

The plan starts with an experiment that needs no new data.

Phase 0 uses exactly the tokens the second stage already reads: ocean pixel-months, the 3-by-3 codec, and the sunflower stencil over 24 months. We train a cone codec over those same tokens with the masked-dot objective, and compare three things at equal total context: the current codec with its stage-2 attention head; the cone codec with a plain ridge regression on top; and the cone codec with an attention head. Three things about the protocol are non-negotiable after the 2 September paper reset. The verdict is rolled skill under the corrected window-scope pool — mean skill against climatology and against damped persistence at each lead, trained and held-out longitudes reported separately, with block-bootstrap intervals and the Linear-Inverse-Model null beside every number. Probes — the velocity probe and the RAPID transport probe — are diagnostics of what the embedding encodes, never verdicts. And at least three seeds per arm, because anything probe-scored is unreadable at one seed. The predictions are written down so they can be wrong: the cone codec should match or beat the current setup on rolled skill with only a linear head, and beat the embedding-space LIM where the current codec does not; it should win by a wide margin on the velocity probe; and it should tie on static probes. If the cone codec only matches when it also gets an attention head, then the "work in the embedding" argument loses on sample efficiency and the hybrid keeps a thick stage 2.

After that, one sphere per phase, in the order agreed earlier. Phase 1 adds the ocean surface and the atmosphere as forcing — the gap-filled surface maps and ERA5 — and is scored on forecasting sea temperature and sea level one to four weeks ahead against persistence and against the operational Copernicus forecast. Phase 2 adds the interior — Argo as native dots, GLORYS as a flagged, heavily masked prior — and is scored on the AMOC again, on mixed-layer depth, and on upper-ocean heat content. Phase 3 adds the ocean biosphere and is scored on reconstructing the carbon flux with the carbon maps held out. Phase 4 adds land, with the local codec on Sentinel and Landsat tiles, and is scored on vegetation and soil-moisture forecasts as well as the standard benchmarks for comparability. Phase 5 adds the fast atmosphere at ten-minute cadence through a second local codec.

Six ablations run in every phase; the first — cone inside versus cone on top at equal context — is the one that keeps answering the zero-sum question as the inputs grow. The engineering note at the bottom is a warning: the network is cheap, under an hour per year of ocean embeddings on a single modern accelerator; the sampler that finds which tiles, dates and profiles fall into which cones is the thing to build first.

Data continuity 2026–27 — build the archive sensor-agnostic

What the verification pass found switching off or changing (agency notices, 2 Sep 2026). Channel-drop training turns a missing sensor into a masked channel, not a failure.

Source	What changes	Consequence for the ladder
Sentinel-1	S1B failed Dec 2021; S1C launched 5 Dec 2024, S1D 4 Nov 2025 (data open 17 Apr 2026); S1A terminated 29 Jun 2026	Constellation is S1C + S1D at 6-day repeat, timeline shifted one day; radar record has a 2022–24 single-satellite gap
Sentinel-2	Three satellites (S2A/B/C) only until end-2026; S2C is prime	Revisit drops back to 5 days in 2027 — do not tune the local codec to the 2025–26 cadence
MODIS → VIIRS	Terra/Aqua shut down late 2026 / early 2027; Suomi-NPP data delivery ends 1 Nov 2026	Rung 3 continues on NOAA-21 (primary) and NOAA-20; the 2000–2026 MODIS archive is training data with a hard end
Passive microwave	DMSP SSMIS retiring ~Sep 2026; NSIDC NRT SSMIS sea ice retired 18 Jun 2026; NOAA AMSR2 NRT ends Sep 2026 for AMSR3 (GOSAT-GW, launched 28 Jun 2025)	NRT sea-ice concentration and drift now depend on one 2012 satellite (AMSR2) until AMSR3 products are operational
Altimetry	Sentinel-6B launched 17 Nov 2025, reference hand-over date not yet published; DUACS L4 now 0.125°; SWOT L4 only experimental	Sea-level rung stable; treat the 0.25° legacy grid and the new 0.125° grid as one channel with a resolution flag
Climate records that stop	OC-CCI v6 ends 31 Dec 2024; ESA CCI soil moisture and snow are annual retrospective releases; OSCAR final ends Aug 2022; FLUXNET2015 frozen (Feb 2020)	Use the NRT twins for 2025+ (GlobColour L3, ASCAT / SMAP, OSCAR NRT, ICOS / AmeriFlux); expect version seams
Licences to watch	EN4 non-commercial; GRDC research-only, no redistribution; OPERA radar for EUMETNET members; GLEAM commercial use by approval; ECMWF 9 km IFS licensed (0.25° open data free)	Keep licensed sources out of any embedding meant for release; they can still supervise probes

Also verified: ERA5's SST and sea ice are prescribed inputs (HadISST2, OSI SAF, OSTIA), so they must never be scored as ocean observations; ERA5-Land has no data assimilation at all — a land model replayed on ERA5 forcing; Copernicus waves are MFWAM, not WAVEWATCH; Copernicus surface carbon is 0.25°, not 1°; the OPERA composite is now 1 km / 5 min (CIRRUS), not the retired 2 km / 15 min ODYSSEY. Float counts are the live OceanOPS figures of 2 Sep 2026 and move daily.

Data continuity 2026–27 — build the archive sensor-agnostic

The verification pass this week turned up an unusual amount of change in the satellite fleet, and it affects which channels are safe to build on.

Sentinel-1 lost its A satellite at the end of June; the constellation is now C and D, and there is a two-year single-satellite gap in the radar record. Sentinel-2 has three satellites only until the end of this year. MODIS, the sensor behind most 500-metre products since the year 2000, is being switched off around the turn of the year, and Suomi-NPP, the first VIIRS satellite, stops delivering data on the first of November; the newer NOAA-20 and NOAA-21 continue. The passive-microwave sea-ice record is in the middle of a hand-over from the retiring military DMSP satellites to Japan's AMSR2 and, eventually, AMSR3 — which means that for a while near-real-time sea ice depends on a single satellite launched in 2012. Sea level is stable; Sentinel-6B is up but not yet the reference. Several climate-quality records have hard ends — the ocean-colour climate record at the end of 2024, the FLUXNET collection frozen in 2020 — and their near-real-time twins must be used for later years.

The architectural answer is the one the whole deck has been building towards: with channel-drop training, a sensor that disappears is a masked channel, not a failure. But the training archive has to be assembled sensor-agnostic, with the sensor identity in the source flag, so that the model learns "thermal radiometer at 1 km" rather than "MODIS".

The licence line matters for anything meant to be published: the EN4 ocean analysis is non-commercial, the river-gauge archive is research-only and cannot be redistributed, the European radar composite is for members, and ECMWF's native 9-kilometre analysis is licensed while the quarter-degree open data is free.

The paragraph at the bottom lists smaller corrections that came out of the same pass — worth knowing because they are commonly assumed the other way: ERA5's sea-surface temperature is an input, not a product; ERA5-Land has no data assimilation; the Copernicus wave model is MFWAM, not WAVEWATCH; and the European radar composite is now 1 kilometre every 5 minutes.

Sources

Papers, model cards and repositories consulted (accessed 31 Aug – 2 Sep 2026)

Input ladder & dataset specs (2 Sep 2026) Verified against agency product pages (Copernicus Data Space / Sentinel Online, USGS, NASA Earthdata / LP DAAC / PO.DAAC / NSIDC, NOAA NESDIS / NCEI, WMO OSCAR, OSI SAF, EUMETNET, ECMWF / CDS, Copernicus Marine, Argo + OceanOPS API, AVISO, ESA CCI, SOCAT, GEBCO, Met Office, ISRIC, Copernicus Land, FLUXNET / ICOS / AmeriFlux, GRDC, GloFAS, Hydroweb, WorldPop, GHSL) — full spec sheet with URLs in generic-earth-embedding-inputs-proposal.md. Method refs: Emery et al. 1986 (JGR 91, 12865) · Perceiver IO arxiv.org/abs/2107.14795 · PARADIS arxiv.org/abs/2601.21151.

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TESSERA arxiv.org/abs/2506.20380 · github.com/ucam-eo/tessera · github.com/ucam-eo/geotessera · geotessera.org · CVPR 2026 open-access page

OlmoEarth arxiv.org/abs/2511.13655 · allenai.org/blog/olmoearth-models · github.com/allenai/olmoearth_pretrain · huggingface.co/blog/allenai/olmoearth-v1-1 · allenai.org/papers/olmoearth-v1-2

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Ocean-including models OceanSAR-1: huggingface.co/galeio-research/OceanSAR-1 · arxiv.org/abs/2504.06962 · OceanSAR-2: arxiv.org/abs/2601.07392 · Copernicus-FM: arxiv.org/abs/2503.11849 · huggingface.co/datasets/wangyi111/Copernicus-Pretrain · github.com/zhu-xlab/Copernicus-FM · Aurora: microsoft.github.io/aurora · wekeo.copernicus.eu (ocean FM workspace)

Benchmarks & head-to-head tables OlmoEarth Tables 2, 3, 7 (arxiv.org/abs/2511.13655) · TerraMind Table 6 / PANGAEA (arxiv.org/abs/2504.11171; arxiv.org/abs/2412.04204) · Prithvi-EO 2.0 Tables A1–A4 / GEO-Bench (arxiv.org/abs/2412.02732; arxiv.org/abs/2306.03831) · TESSERA Table 1 (arxiv.org/abs/2506.20380) · GEO-Bench-2 ranks (arxiv.org/abs/2511.15658) · AlphaEarth & TESSERA for LCZ mapping (arxiv.org/abs/2606.20034) · Granite-Ocean Table 1 (arxiv.org/abs/2509.21273) · OceanSAR-1 Tables 1–3 (arxiv.org/abs/2504.06962)

Related work OceanNet arxiv.org/abs/2310.00813 · PARADIS arxiv.org/abs/2601.21151 · ClimODE arxiv.org/abs/2404.10024 · Neural-LAM arxiv.org/abs/2309.17370 · stretched grid arxiv.org/abs/2409.02891 · Keisler arxiv.org/abs/2202.07575 · GraphCast arxiv.org/abs/2212.12794 · Pangu arxiv.org/abs/2211.02556 · FuXi arxiv.org/abs/2306.12873 · Samudra arxiv.org/abs/2412.03795 · SamudrACE doi 10.1029/2025GL119340 · ACE2-SOM arxiv.org/abs/2412.04418 · ACE2-NEMO arxiv.org/abs/2603.28704 · AIFS surface ocean arxiv.org/abs/2604.25559 · Aurora doi 10.1038/s41586-025-09005-y · Aardvark doi 10.1038/s41586-025-08897-0 · GraphDOP arxiv.org/abs/2412.15687 · 4DVarNet doi 10.1029/2023MS003609 · ESFM arxiv.org/abs/2605.00850 · Penny & Hamill 2017 BAMS · CSIR-ML6 doi 10.5194/gmd-12-5113-2019 · CANYON-B doi 10.3389/fmars.2017.00128

Also checked Prithvi WxC (arxiv.org/abs/2409.13598) — atmosphere-only, no ocean downstream tasks found; NASA marine-debris work — not Prithvi-based. Earth 2 numbers quoted from paper/paper.tex (attribution matrix, probe ladder).

Sources

Every paper, model card and repository the deck draws on, grouped by model and topic, as short URLs. A few notes on how they were used.

For the six model slides, numbers were taken from the papers and, where possible, verified against the published configuration files: Prithvi-EO 2.0's token sizes come from its Hugging Face config.json (16-pixel patches for the 300 M model, 14 for the 600 M), and Granite-Geospatial-Ocean's architecture from the config and model file in the ibm-granite GitHub repository (42-pixel tiles, 17 channels, embedding 512, 12 layers, patch size 2 by 2, one frame).

The head-to-head tables were transcribed from the papers' own tables: OlmoEarth Tables 2, 3 and 7; TerraMind Table 6; TESSERA Table 1; Granite Table 1; OceanSAR-1 Tables 1–3; GEO-Bench-2 for independent ranks; and an independent Local Climate Zone study for AlphaEarth versus TESSERA.

For the ocean-including models and related work, sources were checked in August 2026 and include several 2026 papers (AIFS surface ocean, ESFM, SamudrACE, PARADIS, ACE2-NEMO). Earth 2 numbers — the attribution matrix and probe ladder — are quoted from the project's own paper draft.

Items listed as "also checked" are things we looked at and excluded, with the reason, so that nobody re-chases them.